Mathematics Grade 11 - Solved Exercises (BTBB)

Balochistan Textbook Board (BTBB)

Unit 1: Complex Numbers

Exercise 1.1

Q1. Separate the real and imaginary parts in the following: (i) 5 + 8i; (ii) 3/2 - (7/3)i; (iii) (2 + 3i)i; (iv) (7 - 8i)(1 + i); (v) sqrt(2) - 18i; (vi) 3.

(i) 5 + 8i is already in standard form: real part 5, imaginary part 8.

(ii) 3/2 - (7/3)i: real part 3/2, imaginary part -7/3.

(iii) (2 + 3i)i = 2i + 3i^2 = -3 + 2i: real part -3, imaginary part 2.

(iv) (7 - 8i)(1 + i) = 7 + 7i - 8i - 8i^2 = 15 - i: real part 15, imaginary part -1.

(v) sqrt(2) - 18i: real part sqrt(2), imaginary part -18.

(vi) 3 = 3 + 0i: real part 3, imaginary part 0.

(i) Re = 5, Im = 8; (ii) Re = 3/2, Im = -7/3; (iii) Re = -3, Im = 2; (iv) Re = 15, Im = -1; (v) Re = sqrt(2), Im = -18; (vi) Re = 3, Im = 0

Q2. Which of the following complex numbers are equal? (i) z1 = 4 + 4i, z2 = sqrt(16) + 4i; (ii) z1 = 7 - 5i, z2 = 5i - 7; (iii) z1 = 15 + sqrt(-9), z2 = 15 + 3i; (iv) z1 = 8i - sqrt(-25) i, z2 = 5 + 8i.

(i) sqrt(16) = 4, so z2 = 4 + 4i = z1: equal.

(ii) z2 = 5i - 7 = -7 + 5i, which differs from 7 - 5i in both parts: not equal. (The book prints 'z2' for both numbers; the second is the one compared.)

(iii) sqrt(-9) = 3i, so z1 = 15 + 3i = z2: equal.

(iv) sqrt(-25) = 5i, so sqrt(-25) i = 5i^2 = -5, and z1 = 8i - (-5) = 5 + 8i = z2: equal. (The notation is unusual; read as 8i minus the product sqrt(-25) times i.)

(i) Equal; (ii) Not equal; (iii) Equal; (iv) Equal

Q3. Simplify the following complex numbers: (i) (8 + 2i) + (4 - 6i); (ii) 3(2 + 9i) + (1 - 3i); (iii) (1/7 - (2/3)i) + (3/5 - (4/7)i); (iv) (12 + 7i) + (8i + 9); (v) (sqrt(-3) + 2) + (13 + sqrt(3) i); (vi) (8 + 15i) + (16 - 13i).

(i) Add real parts and imaginary parts separately: (8 + 4) + (2 - 6)i = 12 - 4i.

(ii) 3(2 + 9i) = 6 + 27i; add (1 - 3i): (6 + 1) + (27 - 3)i = 7 + 24i.

(iii) Real parts: 1/7 + 3/5 = 26/35. Imaginary parts: -2/3 - 4/7 = -26/21. Result: 26/35 - (26/21)i.

(iv) (12 + 9) + (7 + 8)i = 21 + 15i.

(v) sqrt(-3) = i sqrt(3). So (2 + i sqrt(3)) + (13 + sqrt(3) i) = 15 + 2

sqrt(3) i.

(vi) (8 + 16) + (15 - 13)i = 24 + 2i.

(i) 12 - 4i; (ii) 7 + 24i; (iii) 26/35 - (26/21)i; (iv) 21 + 15i; (v) 15 + 2 sqrt(3) i; (vi) 24 + 2i

Q4. Simplify the following: (i) (12 + 6i) - (4 + 5i); (ii) (5 - 6i) - (7 + 9i); (iii) (15 + 13i) - (12 - 10i); (iv) (8 - 17i) - (-9 - 15i); (v) (sqrt(3) - sqrt(-16)) - (sqrt(27) - sqrt(-25)); (vi) (19 + 15i) - (17 - 14i).
(i) (12 - 4) + (6 - 5)i = 8 + i.

(ii) (5 - 7) + (-6 - 9)i = -2 - 15i.

(iii) (15 - 12) + (13 + 10)i = 3 + 23i.

(iv) (8 + 9) + (-17 + 15)i = 17 - 2i.

(v) sqrt(-16) = 4i and sqrt(-25) = 5i, so (sqrt(3) - 4i) - (3 sqrt(3) - 5i) = (1 - 3) sqrt(3) + (-4 + 5)i = -2 sqrt(3) + i.

(vi) (19 - 17) + (15 + 14)i = 2 + 29i.

(i) 8 + i; (ii) -2 - 15i; (iii) 3 + 23i; (iv) 17 - 2i; (v) -2 sqrt(3) + i; (vi) 2 + 29i

Q5. Simplify the following: (i) (2 - i)(3 + i); (ii) (3 + 2i)(3 - 2i); (iii) (2 sqrt(-9) - 3)(3 sqrt(-16) - 1); (iv) (3 - i)(4 + 3i)(5 - 2i); (v) (4 - 7i)^2;

(vi) (i + 5)(7 - 6i).

(i) (2 - i)(3 + i) = 6 + 2i - 3i - i^2 = 7 - i.

(ii) (3 + 2i)(3 - 2i) = 9 - (2i)^2 = 9 + 4 = 13.

(iii) 2 sqrt(-9) = 6i and 3 sqrt(-16) = 12i, so (6i - 3)(12i - 1) = 72i^2 - 6i - 36i + 3 = -69 - 42i.

(iv) First (3 - i)(4 + 3i) = 12 + 9i - 4i - 3i^2 = 15 + 5i. Then (15 + 5i)(5 - 2i) = 75 - 30i + 25i - 10i^2 = 85 - 5i.

(v) (4 - 7i)^2 = 16 - 56i + 49i^2 = -33 - 56i.

(vi) (i + 5)(7 - 6i) = 7i - 6i^2 + 35 - 30i = 41 - 23i.

(i) 7 - i; (ii) 13; (iii) -69 - 42i; (iv) 85 - 5i; (v) -33 - 56i; (vi) 41 - 23i

Q6. Simplify the following: (i) (3 - 5i)/(3 - 2i); (ii) (5 - sqrt(-4))/(4 + 9i); (iii) (7 + 8i)/(9 - 2i); (iv) (sqrt(9) - sqrt(-16))/(2 + i); (v) (17 + 11i)/(6 + 5i); (vi) (6 + 7i)/(5 + 4i).

(i) Multiply numerator and denominator by the conjugate (3 + 2i): (3 - 5i)(3 + 2i)/13 = (19 - 9i)/13.

(ii) sqrt(-4) = 2i. (5 - 2i)/(4 + 9i) = (5 - 2i)(4 - 9i)/97 = (2 - 53i)/97.

(iii) (7 + 8i)/(9 - 2i) = (7 + 8i)(9 + 2i)/85 = (47 + 86i)/85.

(iv) sqrt(9) = 3, sqrt(-16) = 4i. (3 - 4i)/(2 + i) = (3 - 4i)(2 - i)/5 = (2 -

11i)/5.

(v) (17 + 11i)/(6 + 5i) = (17 + 11i)(6 - 5i)/61 = (157 - 19i)/61.

(vi) (6 + 7i)/(5 + 4i) = (6 + 7i)(5 - 4i)/41 = (58 + 11i)/41.

(i) (19 - 9i)/13; (ii) (2 - 53i)/97; (iii) (47 + 86i)/85; (iv) (2 - 11i)/5; (v) (157 - 19i)/61; (vi) (58 + 11i)/41

Q7. Find the conjugate of the following: (i) 2 + 7i; (ii) (5 - 7i)^2; (iii) (3 - 2i)(2 + 3i); (iv) 4 - sqrt(-8); (v) 8/3 + sqrt(-25/16); (vi) (5i/3) - (7/4).

The conjugate changes the sign of the imaginary part.

(i) Conjugate of 2 + 7i is 2 - 7i.

(ii) (5 - 7i)^2 = 25 - 70i + 49i^2 = -24 - 70i; its conjugate is -24 + 70i.

(iii) (3 - 2i)(2 + 3i) = 6 + 9i - 4i - 6i^2 = 12 + 5i; its conjugate is 12 - 5i.

(iv) 4 - sqrt(-8) = 4 - 2 sqrt(2) i; its conjugate is 4 + 2 sqrt(2) i.

(v) sqrt(-25/16) = (5/4)i, so the number is 8/3 + (5/4)i; its conjugate is 8/3 - (5/4)i.

(vi) (5i/3) - (7/4) = -7/4 + (5/3)i; its conjugate is -7/4 - (5/3)i.

(i) 2 - 7i; (ii) -24 + 70i; (iii) 12 - 5i; (iv) 4 + 2 sqrt(2) i; (v) 8/3 - (5/4)i; (vi) -7/4 - (5/3)i

Q8. Find the absolute value of the following: (i) z = 5 + 13i; (ii) z = 13 - 18i; (iii) z = (5 - 12i) + (17 + 10i); (iv) z = (3 - 7i)(5 + 11i); (v) z = (7 + 8i)/(11 - 9i); (vi) z = (sqrt(2) + sqrt(-9)) - (sqrt(8) - 4i).

(i) |5 + 13i| = sqrt(25 + 169) = sqrt(194).

(ii) |13 - 18i| = sqrt(169 + 324) = sqrt(493).

(iii) z = 22 - 2i, so |z| = sqrt(484 + 4) = sqrt(488) = 2 sqrt(122).

(iv) z = 15 + 33i - 35i - 77i^2 = 92 - 2i, so |z| = sqrt(8464 + 4) = sqrt(8468) = 2 sqrt(2117).

(v) |z| = |7 + 8i|/|11 - 9i| = sqrt(113)/sqrt(202).

(vi) z = (sqrt(2) + 3i) - (2 sqrt(2) - 4i) = -sqrt(2) + 7i, so |z| = sqrt(2 + 49) = sqrt(51).

(i) sqrt(194); (ii) sqrt(493); (iii) 2 sqrt(122); (iv) 2 sqrt(2117); (v) sqrt(113)/sqrt(202); (vi) sqrt(51)

Exercise 1.2

Q1(i). Solve the simultaneous linear equations with complex co-efficients: 2z + 3iw = 1 + i; (1 - i)z + w = 2 + i.

From the second equation, w = 2 + i - (1 - i)z.

Substitute in the first equation: 2z + 3i(2 + i - (1 - i)z) = 1 + i.

Expand: 2z + 6i + 3i^2 - 3i(1 - i)z = 1 + i, i.e. 2z + 6i - 3 - (3i + 3)z = 1 + i.

Collect z: z(2 - 3 - 3i) + (-3 + 6i) = 1 + i, so z(-1 - 3i) = 4 - 5i.

Therefore z = (4 - 5i)/(-1 - 3i) = (4 - 5i)(-1 + 3i)/10 = (11 + 17i)/10.

Then w = 2 + i - (1 - i)(11/10 + 17i/10) = 2 + i - (28 + 6i)/10 = -4/5 + (2/5)i.

Check: 2z + 3iw = 11/5 + 17i/5 - 12i/5 - 6/5 = 1 + i, as required.

z = 11/10 + (17/10)i, w = -4/5 + (2/5)i

Q1(ii). Solve: z + (z + i)w = 3i; 3iz - 2iw = 3 + 2i. [Note: the first equation contains a zw term, so this is a quadratic system, not a linear one; the solution below was verified by substitution.]
From the second equation, i(3z - 2w) = 3 + 2i, so 3z - 2w = (3 + 2i)/i = 2 - 3i, giving w = (3z - 2 + 3i)/2.

Substitute in the first equation: z + (z + i)(3z - 2 + 3i)/2 = 3i.
Multiply by 2: 2z + (z + i)(3z - 2 + 3i) = 6i.

Expand (z + i)(3z - 2 + 3i) = 3z^2 + z(-2 + 6i) - 2i - 3. Hence 3z^2 + 6iz - 3 - 8i = 0, i.e. z^2 + 2iz - 1 - (8/3)i = 0.

Discriminant D = (2i)^2 - 4(-1 - (8/3)i) = -4 + 4 + (32/3)i = (32/3)i. Its square root is (4/sqrt(3))(1 + i), since ((4/sqrt(3))(1 + i))^2 = (16/3)(2i) = (32/3)i.

Therefore z = (-2i +- (4/sqrt(3))(1 + i))/2 = -i +- (2/sqrt(3))(1 + i).

Then w = (3z - 2 + 3i)/2 gives w = (sqrt(3) - 1) + i sqrt(3) for the first root, and w = (-sqrt(3) - 1) - i sqrt(3) for the second root.

Check (first pair): z + (z + i)w = 3i, verified by direct substitution.

z = 2/sqrt(3) + i(2/sqrt(3) - 1) or z = -2/sqrt(3) - i(2/sqrt(3) + 1), with matching w = (sqrt(3) - 1) + i sqrt(3) or w = (-sqrt(3) - 1) - i sqrt(3)

Q1(iii). Solve: (3 - i)z + 2iw = 5i; z + iw = 7i.
From the second equation, z = 7i - iw.

Substitute: (3 - i)(7i - iw) + 2iw = 5i. Expand: (3 - i)(7i) - (3 - i)iw + 2iw = 5i.

(3 - i)(7i) = 21i - 7i^2 = 7 + 21i, and -(3 - i)iw + 2iw = -iw(1 - i). So 7 + 21i - iw(1 - i) = 5i.

Hence iw(1 - i) = 7 + 16i, so w = (7 + 16i)/(i(1 - i)) = (7 + 16i)/(1 + i) = (7 + 16i)(1 - i)/2 = (23 + 9i)/2.

Then z = 7i - i(23/2 + 9i/2) = 7i - 23i/2 + 9/2 = 9/2 - (9/2)i.

Check: (3 - i)z + 2iw = (9/2)(2 - 4i) + 23i - 9 = 5i, as required.

z = 9/2 - (9/2)i, w = 23/2 + (9/2)i

Q2. Factorize the polynomials P(z) into linear factors.

(i) z^2 + 9 = z^2 - (3i)^2 = (z + 3i)(z - 3i).

(ii) z^2 + 121 = z^2 - (11i)^2 = (z + 11i)(z - 11i).

(iii) z^2 + 3 = z^2 - (i sqrt(3))^2 = (z + i sqrt(3))(z - i sqrt(3)).

(iv) z^2 + 196 = z^2 - (14i)^2 = (z + 14i)(z - 14i).

(v) z^3 + 3z^2 + 4z + 2: z = -1 is a root (-1 + 3 - 4 + 2 = 0), so divide: (z + 1)(z^2 + 2z + 2). And z^2 + 2z + 2 = (z + 1 - i)(z + 1 + i).

(vi) z^3 + 2z^2 - 2z - 12: z = 2 is a root (8 + 8 - 4 - 12 = 0), so divide: (z - 2)(z^2 + 4z + 6). And z^2 + 4z + 6 = (z + 2 - i sqrt(2))(z + 2 + i sqrt(2)).

(vii) z^3 - 7z^2 + 19z - 13: z = 1 is a root (1 - 7 + 19 - 13 = 0), so divide: (z - 1)(z^2 - 6z + 13). And z^2 - 6z + 13 = (z - 3 - 2i)(z - 3 + 2i).

(viii) RESOLVED from the book's Answers section (book p360): as

printed, z^3 - z^2 + 17z + 15 has no rational root (P(-1) = -4, P(1) = 32, P(3) = 84, P(-3) = -72, P(5) = 200, P(-5) = -220), so it cannot be factorised as printed. The Answers section gives (z + 1)(z + 1 - i sqrt(14))(z + 1 + i sqrt(14)), which expands to z^3 + 3z^2 + 17z + 15. Hence the printed '-z^2' is a misprint for '+3z^2'. With the intended polynomial: z = -1 is a root, divide to get (z + 1)(z^2 + 2z + 15), and z^2 + 2z + 15 = (z + 1 - i sqrt(14))(z + 1 + i sqrt(14)).

(i) (z + 3i)(z - 3i); (ii) (z + 11i)(z - 11i); (iii) (z + i sqrt(3))(z - i sqrt(3)); (iv) (z + 14i)(z - 14i); (v) (z + 1)(z + 1 - i)(z + 1 + i); (vi) (z - 2)(z + 2 - i sqrt(2))(z + 2 + i sqrt(2)); (vii) (z - 1)(z - 3 - 2i)(z - 3 + 2i); (viii) (z + 1)(z + 1 - i sqrt(14))(z + 1 + i sqrt(14))

Q3. Solve the quadratic equations.

(i) z^2 - 4z + 13 = 0: (z - 2)^2 = 4 - 13 = -9, so z - 2 = +- 3i, z = 2 +- 3i.

(ii) 13z^2 - 6z + 1 = 0: discriminant = 36 - 52 = -16, so z = (6 +- 4i)/26 = (3 +- 2i)/13.

(iii) 4z^2 - 4z + 17 = 0: discriminant = 16 - 272 = -256, so z = (4 +- 16i)/8 = (1 +- 4i)/2.

(iv) 2z^2 = 2z - 13, i.e. 2z^2 - 2z + 13 = 0: discriminant = 4 - 104 = -100, so z = (2 +- 10i)/4 = (1 +- 5i)/2.

(v) z^2 - 3iz - 4 = 0: discriminant = (-3i)^2 - 4(1)(-4) = -9 + 16 = 7, so z = (3i +- sqrt(7))/2.

(vi) z^2 - 2z + 15 = 0: (z - 1)^2 = 1 - 15 = -14, so z = 1 +- i sqrt(14).

(i) z = 2 +- 3i; (ii) z = (3 +- 2i)/13; (iii) z = (1 +- 4i)/2; (iv) z = (1 +- 5i)/2; (v) z = (3i +- sqrt(7))/2; (vi) z = 1 +- i sqrt(14)

Q4. Solve the equation in polar form: (i) 10z - 8i = 1 - 5iz; (ii) 5z + 3i = 2 - 5iz.

(i) 10z + 5iz = 1 + 8i, so z(10 + 5i) = 1 + 8i. Hence z = (1 + 8i)/(10 + 5i) = (1 + 8i)(10 - 5i)/125 = (50 + 75i)/125 = 2/5 + (3/5)i.

Modulus r = sqrt(4/25 + 9/25) = sqrt(13)/5; argument theta = tan^-1(3/2) (first quadrant). Polar form: (sqrt(13)/5)(cos theta + i sin theta).

(ii) 5z + 5iz = 2 - 3i, so z(5 + 5i) = 2 - 3i. Hence z = (2 - 3i)/(5 + 5i) = (2 - 3i)(5 - 5i)/50 = (-5 - 25i)/50 = -1/10 - (1/2)i.

Modulus r = sqrt(1/100 + 1/4) = sqrt(26)/10; the point is in the third quadrant, so theta = pi + tan^-1(5). Polar form: (sqrt(26)/10)(cos theta + i sin theta).

(i) z = 2/5 + (3/5)i, polar form (sqrt(13)/5)(cos theta + i sin theta) with theta = tan^-1(3/2); (ii) z = -1/10 - (1/2)i, polar form (sqrt(26)/10)(cos theta + i sin theta) with theta = pi + tan^-1(5)

Q5. Represent the complex numbers in polar form: (i) -1 - i; (ii) 1 + i

sqrt(3); (iii) 2 + i 2 sqrt(3); (iv) sqrt(6) + i sqrt(2).

(i) -1 - i: r = sqrt(1 + 1) = sqrt(2); third quadrant, theta = 5pi/4. Form: sqrt(2)(cos(5pi/4) + i sin(5pi/4)).

(ii) 1 + i sqrt(3): r = sqrt(1 + 3) = 2; first quadrant, theta = pi/3. Form: 2(cos(pi/3) + i sin(pi/3)).

(iii) 2 + i 2 sqrt(3): r = sqrt(4 + 12) = 4; first quadrant, theta = pi/3. Form: 4(cos(pi/3) + i sin(pi/3)).

(iv) sqrt(6) + i sqrt(2): r = sqrt(6 + 2) = sqrt(8) = 2 sqrt(2); first quadrant, tan theta = sqrt(2)/sqrt(6) = 1/sqrt(3), theta = pi/6. Form: 2 sqrt(2)(cos(pi/6) + i sin(pi/6)).

(i) sqrt(2)(cos(5pi/4) + i sin(5pi/4)); (ii) 2(cos(pi/3) + i sin(pi/3)); (iii) 4(cos(pi/3) + i sin(pi/3)); (iv) 2 sqrt(2)(cos(pi/6) + i sin(pi/6))

Q6. If z1 = 5 sqrt(3) + 5i and z2 = 3 + 3 sqrt(3) i then find z1 z2 in polar form.

z1 z2 = (5 sqrt(3) + 5i)(3 + 3 sqrt(3) i). Multiply: 15 sqrt(3) + 45i + 15i + 15 sqrt(3) i^2.

Since i^2 = -1, the real parts cancel: 15 sqrt(3) - 15 sqrt(3) = 0, and the imaginary parts give 60i.

So z1 z2 = 60i, whose modulus is 60 and argument is pi/2. Polar form: 60(cos(pi/2) + i sin(pi/2)).

z1 z2 = 60i = 60(cos(pi/2) + i sin(pi/2))

Q7. Solve the equations for z: (i) 10z - 8i = 1 - 5iz; (ii) (z - 3)/(z + i) = i; (iii) 4z - 3i = 1 + 3iz; (iv) (3i - 5z)/(4 - i) = z/2.

(i) This is the same equation as Q4(i): z = 2/5 + (3/5)i (see Q4(i) for steps).

(ii) (z - 3) = i(z + i) = iz - 1, so z - iz = 2, z(1 - i) = 2, z = 2/(1 - i) = 2(1 + i)/2 = 1 + i.

(iii) 4z - 3iz = 1 + 3i, so z(4 - 3i) = 1 + 3i. Hence z = (1 + 3i)/(4 - 3i) = (1 + 3i)(4 + 3i)/25 = (-5 + 15i)/25 = -1/5 + (3/5)i.

(iv) 2(3i - 5z) = z(4 - i), so 6i - 10z = 4z - iz, 6i = z(14 - i). Hence z = 6i/(14 - i) = 6i(14 + i)/197 = (-6 + 84i)/197.

(i) z = 2/5 + (3/5)i [same equation as Q4(i)]; (ii) z = 1 + i; (iii) z = -1/5 + (3/5)i; (iv) z = -6/197 + (84/197)i

Q8. Solve the following in-equalities (Polar form), where z is a complex number: (i) |z + 2i| < 3; (ii) |z|^2 + |z - 2i|^2 <= 10; (iii) |z^2 - 4| < 1; (iv) |z - 2| + |z + 2| <= 6.

(i) |z + 2i| < 3: the set of points whose distance from (0, -2) is less than 3, i.e. the interior of the circle with centre (0, -2) and radius 3.

(ii) Put z = x + iy: |z|^2 = x^2 + y^2 and |z - 2i|^2 = x^2 + (y - 2)^2. Their sum is 2x^2 + 2y^2 - 4y + 4 <= 10, i.e. x^2 + y^2 - 2y <= 3, i.e. x^2 + (y - 1)^2 <= 4: the closed disc with centre (0,

1) and radius 2.

(iii) |z^2 - 4| = |z - 2||z + 2| < 1: the set of points whose product of distances from 2 and -2 is less than 1. Since the product at 0 is 4 > 1, the set has two separate lobes, one around 2 and one around -2 (a Cassini oval).

(iv) |z - 2| + |z + 2| <= 6: the set of points whose sum of distances from the foci +-2 is at most 6, i.e. the interior of the ellipse with major axis 6 and minor axis 2 sqrt(9 - 4) = 2 sqrt(5): x^2/9 + y^2/5 <= 1.

(i) Interior of the circle centre (0, -2), radius 3; (ii) disc centre (0, 1), radius 2: x^2 + (y - 1)^2 <= 4; (iii) set of points whose product of distances from 2 and -2 is less than 1 (two small lobes around 2 and -2); (iv) interior of the ellipse x^2/9 + y^2/5 <= 1 with foci +-2

Q9. Two waves are represented by A = 3e^{i pi/6} and B = 2e^{-i pi/4}. Find the resulting wave amplitude after interference.
Write in rectangular form: A = 3(cos(pi/6) + i sin(pi/6)) = 3(sqrt(3)/2 + i/2); B = 2(cos(pi/4) - i sin(pi/4)) = 2(sqrt(2)/2 - i sqrt(2)/2).

Add: A + B = (3 sqrt(3)/2 + sqrt(2)) + i(3/2 - sqrt(2)).

Amplitude = sqrt((3 sqrt(3)/2 + sqrt(2))^2 + (3/2 - sqrt(2))^2) = sqrt(13 + 3 sqrt(6) - 3 sqrt(2)).

Numerically: 13 + 7.348 - 4.243 = 16.105, so the amplitude is

sqrt(16.105) = 4.01, approximately 4.0.

Resultant amplitude approximately 4.0

Q10. In an AC circuit, the voltage is represented by V = 20e^{i pi/3}. Find the real and imaginary parts of the voltage.
V = 20(cos(pi/3) + i sin(pi/3)).

cos(pi/3) = 1/2, so the real part is 20 x 1/2 = 10.

sin(pi/3) = sqrt(3)/2, so the imaginary part is 20 x sqrt(3)/2 = 10 sqrt(3).

Real part = 10, imaginary part = 10 sqrt(3)

Review Exercise 1

Q1(i). Conjugate of (1 + 2i)(1 - i) is:

- ✓ 3 - i
- 3 + i
- -3 + i
- -3 - i

Q1(ii). |1 + i| / |1 + i| is equal to:

- sqrt(2)
- 1/sqrt(2)
- ✓ 1
- 2/sqrt(2)

Q1(iii). Multiplicative inverse of a + ib is:

a/(a^2+b^2) + b/(a^2+b^2) i

✓ a/(a^2+b^2) - b/(a^2+b^2) i

-a/(a^2+b^2) + b/(a^2+b^2) i

-a/(a^2+b^2) - b/(a^2+b^2) i

Q1(iv). Factors of z^2 + 256 are:

(z + 16i)(z + 16i)

(z - 16i)(z - 16i)

✓ (z + 16i)(z - 16i)

(z - 16i)(z - 16)

Q1(v). Conjugate of (1 + i)/(1 - i) is:

1

i

✓ -i

1 + i

Q1(vi). Absolute value of (2 + i)(2 - i) + 1 is:

✓ 6

-36

36

36 + i

Q1(vii). (3 + i)/(2 + i) is equal to:

✓ 7/5 - (1/5)i

7 - i

3 + i

(7 + i)/5

Q1(viii). If z1 = 4 - 3i, and z2 = 5 + 7i then Im(z1 + z2) is:

✓ 4

4i

-4

-4i

Q1(ix). Factors of z^2 + 169 are:

(z + 13)(z - 13)

(z + 14)(z - 14i)

(z - 13)(z + 13i)

✓ (z - 13i)(z + 13i)

Q1(x). Standard form of (2 + 3i)/(2 - i) is:

✓ 1/5 + (8/5)i

1/5 - 8i

15 - (8/5)i

5 + 8i

Q2. Find the conjugate of (3 - 2i)(1 + 3i) / ((1 + i)(2 - i)).

Numerator: (3 - 2i)(1 + 3i) = 3 + 9i - 2i - 6i^2 = 9 + 7i.

Denominator: (1 + i)(2 - i) = 2 - i + 2i - i^2 = 3 + i.

Quotient: (9 + 7i)/(3 + i) = (9 + 7i)(3 - i)/10 = (34 + 12i)/10 = 17/5 + (6/5)i.

The conjugate changes the sign of the imaginary part: 17/5 - (6/5)i.

17/5 - (6/5)i

Q3. Evaluate [i^20 + (1/i)^25]^3.

i^20 = (i^4)^5 = 1^5 = 1.

1/i = -i, so (1/i)^25 = (-i)^25 = (-i)^24 x (-i) = 1 x (-i) = -i.

The bracket is 1 - i. Its cube: (1 - i)^3 = 1 - 3i + 3i^2 - i^3 = 1 - 3i - 3 + i = -2 - 2i.

-2 - 2i

Q4. Reduce into standard form (1/(1 - 4i) - 2/(1 + i))((3 - 4i)/(5 + i)).

1/(1 - 4i) = (1 + 4i)/17. 2/(1 + i) = 2(1 - i)/2 = 1 - i.

Difference: (1 + 4i)/17 - (1 - i) = (1 + 4i - 17 + 17i)/17 = (-16 + 21i)/17.

(3 - 4i)/(5 + i) = (3 - 4i)(5 - i)/26 = (11 - 23i)/26.

Product: (-16 + 21i)(11 - 23i)/(17 x 26) = (307 + 599i)/442 = 307/442 + (599/442)i.

307/442 + (599/442)i

Q5. Solve by completing squares: (i) 3z^2 - 15z = 21; (ii) 2z^2 =

-3z - 15.

(i) Divide by 3: z^2 - 5z = 7. Complete the square: (z - 5/2)^2 = 7 + 25/4 = 53/4. So z = 5/2 +- sqrt(53)/2.

(ii) Divide by 2: z^2 + (3/2)z = -15/2. Complete the square: (z + 3/4)^2 = -15/2 + 9/16 = -111/16. So z = -3/4 +- i sqrt(111)/4.

(i) z = 5/2 +- sqrt(53)/2; (ii) z = -3/4 +- i sqrt(111)/4

Q6. If z1 = 1 + 2i, z2 = 1 - i then find |(z1 + z2 + 1)/(z1 - z2 + 1)|.

z1 + z2 + 1 = (1 + 2i) + (1 - i) + 1 = 3 + i.

z1 - z2 + 1 = (1 + 2i) - (1 - i) + 1 = 1 + 3i.

The quotient has modulus |3 + i|/|1 + 3i| = sqrt(10)/sqrt(10) = 1.

1

Q7. If z1 = 1 + i, z2 = 2 - i then find: (i) Re(z1 z2 / conj(z1)); (ii) Im(conj(z2)/(z1 z2)).

z1 z2 = (1 + i)(2 - i) = 3 + i.

(i) conj(z1) = 1 - i. (3 + i)/(1 - i) = (3 + i)(1 + i)/2 = (2 + 4i)/2 = 1 + 2i. Real part = 1.

(ii) conj(z2) = 2 + i. (2 + i)/(3 + i) = (2 + i)(3 - i)/10 = (7 + i)/10. Imaginary part = 1/10.

(i) 1; (ii) 1/10

Q8. Express each of the complex numbers in polar form: (i) -sqrt(5) - sqrt(5) i; (ii) -1/2 - (sqrt(3)/2) i.

(i) -sqrt(5) - sqrt(5) i: r = sqrt(5 + 5) = sqrt(10); third quadrant, theta = 5pi/4. Form: sqrt(10)(cos(5pi/4) + i sin(5pi/4)).

(ii) -1/2 - (sqrt(3)/2) i: r = sqrt(1/4 + 3/4) = 1; third quadrant, theta = 4pi/3. Form: cos(4pi/3) + i sin(4pi/3).

(i) sqrt(10)(cos(5pi/4) + i sin(5pi/4)); (ii) cos(4pi/3) + i sin(4pi/3)

Unit 2: Matrices and Determinants

Exercise 2.1

Q1. If A = [[1, 3, 7], [1, 4, 5], [2, 3, 6]] and B = [[3, -1, 1], [1, 1, 3], [2, 4, 5]], then find (i) 3A - 4B (ii) A + 3(B - A).

(i) 3A = [[3, 9, 21], [3, 12, 15], [6, 9, 18]] and 4B = [[12, -4, 4], [4, 4, 12], [8, 16, 20]]. Subtracting: 3A - 4B = [[-9, 13, 17], [-1, 8, 3], [-2, -7, -2]].

(ii) B - A = [[2, -4, -6], [0, -3, -2], [0, 1, -1]], so 3(B - A) = [[6, -12, -18], [0, -9, -6], [0, 3, -3]]. Adding A: A + 3(B - A) = [[7, -9, -11], [1, -5, -1], [2, 6, 3]].

(i) [[-9, 13, 17], [-1, 8, 3], [-2, -7, -2]]; (ii) [[7, -9, -11], [1, -5, -1], [2, 6, 3]]

Q2. If A = [[1, 4], [3, 5]], B = [[-3, 2], [1, 4]] and C = [[1, 0], [3, 2]], then find A^2 + BC.

A^2 = [[1 + 12, 4 + 20], [3 + 15, 12 + 25]] = [[13, 24], [18, 37]].

BC = [[-3 + 6, 0 + 4], [1 + 12, 0 + 8]] = [[3, 4], [13, 8]].

A^2 + BC = [[16, 28], [31, 45]].

[[16, 28], [31, 45]]

Q3. If A = [[4, -2], [-3, 6]], B = [[0, 2], [1, 3]] and C = [[-2, 3], [1, 4]], then find (i) A^2 (ii) BC (iii) B^2 - AB.

(i) A^2 = [[16 + 6, -8 - 12], [-12 - 18, 6 + 36]] = [[22, -20], [-30, 42]].

(ii) BC = [[0 + 2, 0 + 8], [-2 + 3, 3 + 12]] = [[2, 8], [1, 15]].

(iii) B^2 = [[2, 6], [3, 11]] and AB = [[-2, 2], [6, 12]], so B^2 - AB = [[4, 4], [-3, -1]].

(i) [[22, -20], [-30, 42]]; (ii) [[2, 8], [1, 15]]; (iii) [[4, 4], [-3, -1]]

Q4. If A = [[4, 2i, 3], [i, 4, 1], [3, -5, 6]] and B = [[3i, 2, 1], [-2, 7i, 3], [2i, 3, 8]], then find (i) AB (ii) A^2 - AB.

(i) Row 1 of AB: [12i - 4i + 6i, 8 - 14 + 9, 4 + 6i + 24] = [14i, 3, 28 + 6i]. Row 2: [-3 - 8 + 2i, 2i + 28i + 3, i + 12 + 8] = [-11 + 2i, 3 + 30i, 20 + i]. Row 3: [9i + 10 + 12i, 6 - 35i + 18, 3 - 15 + 48] = [10 + 21i, 24 - 35i, 36].

(ii) A^2 = [[23, -15 + 16i, 30 + 2i], [3 + 8i, 9, 10 + 3i], [30 - 5i, -50 + 6i, 40]]. Subtracting AB gives [[23 - 14i, -18 + 16i, 2 - 4i], [14 + 6i, 6 - 30i, -10 + 2i], [20 - 26i, -74 + 41i, 4]].

(i) AB = [[14i, 3, 28 + 6i], [-11 + 2i, 3 + 30i, 20 + i], [10 + 21i, 24 - 35i, 36]]; (ii) A^2 - AB = [[23 - 14i, -18 + 16i, 2 - 4i], [14 + 6i, 6 - 30i, -10 + 2i], [20 - 26i, -74 + 41i, 4]]

Q5. If A = [[5, 3, 1], [-2, 1, 4], [1, 3, -2]], then show that A I_3 = A.

Multiplying by the identity leaves every row unchanged: A I_3 = [[5, 3, 1], [-2, 1, 4], [1, 3, -2]] = A.

A I_3 = A is shown

Q6. If A = [[3, 1, 2], [2, 1, 6], [4, 2, 1]], B = [[3, -1, 2], [2, 1, 4], [1, 5, 4]], then show that; (i) (A + B)^t = A^t + B^t (ii) (A - B)^t = A^t - B^t.

(i) A + B = [[6, 0, 4], [4, 2, 10], [5, 7, 5]], so (A + B)^t = [[6, 4, 5], [0, 2, 7], [4, 10, 5]]. Also A^t + B^t = [[3, 2, 4], [1, 1, 2], [2, 6, 1]] + [[3, 2, 1], [-1, 1, 5], [2, 4, 4]] = [[6, 4, 5], [0, 2, 7], [4, 10, 5]]. Equal.

(ii) A - B = [[0, 2, 0], [0, 0, 2], [3, -3, -3]], so (A - B)^t = [[0, 0, 3], [2, 0, -3], [0, 2, -3]]. Also A^t - B^t = [[0, 0, 3], [2, 0, -3], [0, 2, -3]]. Equal.

Both transpose laws are shown

Q7. If C = [[5, -3], [2, 7]], D = [[2, -3/2], [1, 3]] then show that; C - 2D = I_2.

2D = [[4, -3], [2, 6]]. Hence C - 2D = [[5 - 4, -3 + 3], [2 - 2, 7 - 6]] = [[1, 0], [0, 1]] = I_2.

C - 2D = I_2 is shown

Q8. If A = [[3, 1, 2], [2, 1, 5], [4, 1, 6]], B = [[1, 0, -2], [3, 1, 0], [0, 2, 1]] then show that; (i) (A^t)^t = A (ii) (AB)^t = B^t A^t.

(i) Transposing twice returns the original matrix: (A^t)^t = A.

(ii) AB = [[6, 5, -4], [5, 11, 1], [7, 13, -2]], so (AB)^t = [[6, 5, 7], [5, 11, 13], [-4, 1, -2]]. Also B^t A^t = [[1, 3, 0], [0, 1, 2], [-2, 0, 1]] x [[3, 2, 4], [1, 1, 1], [2, 5, 6]] = [[6, 5, 7], [5, 11, 13], [-4, 1, -2]]. Equal.

Both laws are shown

Q9. If A = [[sin^2(theta), -cos(theta)], [sin(theta), cos^2(theta)]], B = [[cos^2(theta), cos(theta)], [sin(theta), sin^2(theta)]] then show that; A + B = I_2. (Flag: as printed this is FALSE. A + B = [[1, 0], [2sin(theta), 1]], which equals I_2 only when sin(theta) = 0. The (2,1) entries add to 2sin(theta), not 0.)
Adding entry by entry: (1,1): sin^2(theta) + cos^2(theta) = 1. (1,2): -cos(theta) + cos(theta) = 0. (2,1): sin(theta) + sin(theta) = 2sin(theta). (2,2): cos^2(theta) + sin^2(theta) = 1.

Hence A + B = [[1, 0], [2sin(theta), 1]]. This is I_2 only if sin(theta) = 0; the book's claim as printed is incorrect.

A + B = [[1, 0], [2sin(theta), 1]] — NOT I_2 in general (see flag)

Exercise 2.2

Q1. Evaluate determinant of the following matrices using cofactors. (i) [[7, 5, -9], [3, 6, -1], [2, 4, -6]]; (ii) [[5, 2, 1], [-3, 1, 2], [-1, 2, 5]]; (iii) [[4, 2, 2], [7, 3, 4], [1, -2, 7]]; (iv) [[2, -1, 7], [3, 1, 8], [4, -5, 13]]; (v) [[1, 3, -5], [4, 7, 2], [-6, 0, 1]].

(i) Expanding by row 1: 7(6 x -6 - (-1) x 4) - 5(3 x -6 - (-1) x 2) + (-9)(3 x 4 - 6 x 2) = 7(-32) - 5(-16) - 9(0) = -224 + 80 = -144.

(ii) Expanding by row 1: 5(1 x 5 - 2 x 2) - 2(-3 x 5 - 2 x -1) + 1(-3 x 2 - 1 x -1) = 5(1) - 2(-13) + 1(-5) = 5 + 26 - 5 = 26.

(iii) Expanding by row 1: 4(3 x 7 - 4 x -2) - 2(7 x 7 - 4 x 1) + 2(7 x -2 - 3 x 1) = 4(29) - 2(45) + 2(-17) = 116 - 90 - 34 = -8.

(iv) Expanding by row 1: 2(1 x 13 - 8 x -5) + 1(3 x 13 - 8 x 4) + 7(3 x -5 - 1 x 4) = 2(53) + 1(7) + 7(-19) = 106 + 7 - 133 = -20.

(v) Expanding by row 1: 1(7 x 1 - 2 x 0) - 3(4 x 1 - 2 x -6) + (-5)(4 x 0 - 7 x -6) = 7 - 3(16) - 5(42) = 7 - 48 - 210 = -251.

(i) -144; (ii) 26; (iii) -8; (iv) -20; (v) -251

Q2. Using the properties of determinants, prove the following without expanding. (i) |x, a, x+a; y, b, y+b; z, c, z+c| = 0; (ii) |0, a, -b; -a, 0, -c; b, c, 0| = 0; (iii) |2, 7, 65; 3, 8, 75; 5, 9, 86| = 0; (iv) |a-b, b-c, c-a; b-c, c-a, a-b; c-a, a-b, b-c| = 0; (v) |sin^2(alpha), cos^2(alpha), cos^2(alpha); sin^2(beta), cos^2(beta), cos^2(beta); sin^2(gamma), cos^2(gamma), cos^2(gamma)| = 0; (vi) |(a+b)^2, a^2+b^2, ab; (c+d)^2, c^2+d^2, cd; (g+h)^2, g^2+h^2, gh| = 0; (vii) |a, a^2, a/(bc); b, b^2, b/(ca); c, c^2, c/(ab)| = 0.

(i) Column 3 = column 1 + column 2, so the determinant is 0.

(ii) The matrix is skew-symmetric (A^t = -A), so its determinant is 0.

(iii) Column 3 = column 1 + 9 x column 2 (check: 2 + 63 = 65, 3 + 72 = 75, 5 + 81 = 86), so the determinant is 0.

(iv) Column 1 + column 2 + column 3 = 0 in every row, so the determinant is 0.

(v) Column 2 and column 3 are identical, so the determinant is 0.

(vi) Column 1 = column 2 + 2 x column 3, since (a+b)^2 = a^2 + b^2 + 2ab, and similarly for the other rows, so the determinant is 0.

(vii) Column 3 = column 2 divided by abc, since a/(bc) = a^2/(abc), and similarly for the other rows, so the determinant is 0.

All seven determinants are 0, by the column relations shown

Q3. Using the properties of determinants, prove the following determinants. (i) |-a^2, ab, ac; ba, -b^2, bc; ca, cb, -c^2| = 4a^2 b^2 c^2; (ii) |1, 1, 1; a, b, c; a^3, b^3, c^3| = (a-b)(b-c)(c-a)(a+b+c); (iii) |x, x^2, yz; y, y^2, zx; z, z^2, xy| = (x-y)(y-z)(z-x)(xy+yz+zx); (iv) |x+4, 2x, 2x; 2x, x+4, 2x; 2x, 2x, x+4| = (5x+4)(4-x)^2; (v) |a-b-c, 2a, 2a; 2b, b-c-a, 2b; 2c, 2c, c-a-b| = (a+b+c)^3.

(i) Take a, b, c common from rows 1, 2, 3: determinant = abc x |-a,

b, c; a, -b, c; a, b, -c|. Expanding the second determinant: -a(bc - bc) - b(-ac - ac) + c(ab + ab) = 4abc. Total: 4a^2 b^2 c^2.

(ii) Apply C2 - C1 and C3 - C1, take (b - a) and (c - a) common, then the remaining 2x2 determinant is (c - a)(a + b + c) - (b - a)(a + b + c) up to sign; simplifying gives (a - b)(b - c)(c - a)(a + b + c).

(iii) The determinant is 0 when x = y, y = z or z = x (two rows equal), so (x - y)(y - z)(z - x) divides it. It is of degree 4, so the remaining factor is a symmetric polynomial of degree 1, namely xy + yz + zx; comparing the coefficient of x^2 y z gives 1.

(iv) Adding columns 2 and 3 to column 1 gives 5x + 4 in every row of column 1; take it out, then R2 - R1 and R3 - R1 leave a triangular determinant (4 - x)^2. Total: (5x + 4)(4 - x)^2.

(v) Adding rows 2 and 3 to row 1 gives a + b + c in every column of row 1; take it out, then C2 - C1 and C3 - C1 leave a triangular determinant (a + b + c)^2. Total: (a + b + c)^3.

(i) 4a^2 b^2 c^2; (ii) (a-b)(b-c)(c-a)(a+b+c); (iii) (x-y)(y-z)(z-x)(xy+yz+zx); (iv) (5x+4)(4-x)^2; (v) (a+b+c)^3

Exercise 2.3

Q1. Reduce the following matrices into echelon form. (i) [[1, 2, -1], [2, 4, 1], [3, 6, 2]]; (ii) [[2, 3, -1], [4, 5, 3], [6, 2, 1]]; (iii) [[1, -3, 2], [4, 5, 3], [2, 1, 5]]; (iv) [[1, -1, 3], [0, 3, 2], [2, -4, 1]]. (Note: echelon form is not unique — row scalings of these forms are equally valid.)

(i) R2 - 2R1 = [0, 0, 3]; R3 - 3R1 = [0, 0, 5]; R3 - (5/3)R2 = [0, 0, 0]. Echelon form: [[1, 2, -1], [0, 0, 3], [0, 0, 0]].

(ii) R2 - 2R1 = [0, -1, 5]; R3 - 3R1 = [0, -7, 4]; R3 - 7R2 = [0, 0, -31]. Echelon form: [[2, 3, -1], [0, -1, 5], [0, 0, -31]].

(iii) R2 - 4R1 = [0, 17, -5]; R3 - 2R1 = [0, 7, 1]; R3 - (7/17)R2 = [0, 0, 52/17]. Echelon form: [[1, -3, 2], [0, 17, -5], [0, 0, 52/17]].

(iv) R3 - 2R1 = [0, -2, -5]; R3 + (2/3)R2 = [0, 0, -11/3]. Echelon form: [[1, -1, 3], [0, 3, 2], [0, 0, -11/3]].

(i) [[1, 2, -1], [0, 0, 3], [0, 0, 0]]; (ii) [[2, 3, -1], [0, -1, 5], [0, 0, -31]]; (iii) [[1, -3, 2], [0, 17, -5], [0, 0, 52/17]]; (iv) [[1, -1, 3], [0, 3, 2], [0, 0, -11/3]]

Q2. Reduce the following matrices into reduced echelon form. (i) [[1, 3, -2], [1, -4, 3], [3, 2, 1]]; (ii) [[1, -2, -3], [3, -1, 4], [4, 7, 3]]; (iii) [[3, 5, 4], [4, 1, 5], [7, 6, 3]]; (iv) [[0, 3, 2], [2, -4, 6], [2, 3, -1]].

(i) R2 - R1 = [0, -7, 5]; R3 - 3R1 = [0, -7, 7]; R3 - R2 = [0, 0, 2]. Pivots in all three columns; scaling and back-substitution give I_3.

(ii) R2 - 3R1 = [0, 5, 13]; R3 - 4R1 = [0, 15, 15]; R3 - 3R2 = [0, 0, -24]. Three pivots; reduced form is I_3.

(iii) R2 - (4/3)R1 and R3 - (7/3)R1, then elimination, give three pivots; reduced form is I_3.

(iv) Swap R1 and R2, then eliminate below and above; three pivots; reduced form is I_3.

All four reduce to I_3

Q3. Find rank by using row operations. (i) [[1, 2, 3], [2, 5, 4], [3, 0, 5]]; (ii) [[4, 2, 5], [-2, -3, 3], [-1, 3, 2]]; (iii) [[2, 1, 3], [0, 2, 1], [5, -3, 2]]; (iv) [[3, 6, 15], [0, 6, 2], [1, 2, 5]].

(i) Echelon form [[1, 2, 3], [0, 1, -2], [0, 0, -16]] has 3 non-zero rows: rank 3.

(ii) Echelon form [[4, 2, 5], [0, -2, 11/2], [0, 0, 103/8]] has 3 non-zero rows: rank 3.

(iii) Echelon form [[2, 1, 3], [0, 2, 1], [0, 0, -11/4]] has 3 non-zero rows: rank 3.

(iv) Echelon form [[3, 6, 15], [0, 6, 2], [0, 0, 0]] has 2 non-zero rows: rank 2.

(i) 3; (ii) 3; (iii) 3; (iv) 2

Q4. Find the inverse by using row operations and by adjoint method. (i) [[1, 2, -3], [0, 3, 0], [3, -2, 2]]; (ii) [[3, 5, 7], [4, 2, 1], [2, 0, 3]]; (iii) [[5, -2, -1], [0, 3, 2], [2, 1, 1]]; (iv) [[0, 1, 5], [2, 4, 3], [-3, -1, 6]].

(i) Determinant 33. Adjoint method: cofactors give adj = [[6, 2, 9], [0, 11, 0], [-9, 8, 3]]; dividing by 33 gives [[2/11, 2/33, 3/11], [0,

1/3, 0], [-3/11, 8/33, 1/11]]. Row reduction of [A | I] gives the same matrix.

(ii) Determinant -60. Adjoint divided by -60 gives [[-1/10, 1/4, 3/20], [1/6, 1/12, -5/12], [1/15, -1/6, 7/30]]; row reduction of [A | I] agrees.

(iii) Determinant 3. Adjoint divided by 3 gives [[1/3, 1/3, -1/3], [4/3, 7/3, -10/3], [-2, -3, 5]]; row reduction of [A | I] agrees.

(iv) Determinant 29. Adjoint divided by 29 gives [[27/29, -11/29, -17/29], [-21/29, 15/29, 10/29], [10/29, -3/29, -2/29]]; row reduction of [A | I] agrees.

(i) [[2/11, 2/33, 3/11], [0, 1/3, 0], [-3/11, 8/33, 1/11]]; (ii) [[-1/10, 1/4, 3/20], [1/6, 1/12, -5/12], [1/15, -1/6, 7/30]]; (iii) [[1/3, 1/3, -1/3], [4/3, 7/3, -10/3], [-2, -3, 5]]; (iv) [[27/29, -11/29, -17/29], [-21/29, 15/29, 10/29], [10/29, -3/29, -2/29]]

Exercise 2.4

Q1. Solve a system of homogeneous linear equations. (i) 2x + 4y + 6z = 0; 4x + 5y + 6z = 0; 3x + y - 2z = 0. (ii) 3x_1 - 4x_2 + 2x_3 = 0; -9x_1 + 12x_2 - 6x_3 = 0; -6x_1 + 8x_2 - 4x_3 = 0. (Flag for (ii): the book's Answers section says 'No Solution', which is impossible for a homogeneous system — it always has the trivial solution — and here all three equations are multiples of one another, so there are infinitely many solutions.)

(i) The coefficient matrix has rank 3 (determinant non-zero), so only the trivial solution x = 0, y = 0, z = 0 exists.

(ii) Row 2 = -3 x row 1 and row 3 = -2 x row 1, so the system is just 3x_1 - 4x_2 + 2x_3 = 0: infinitely many solutions, e.g. x_1 = (4x_2 - 2x_3)/3 with x_2, x_3 free (including the trivial solution).

(i) x = 0, y = 0, z = 0 only; (ii) infinitely many solutions: 3x_1 - 4x_2 + 2x_3 = 0 (flag: book's Answers 'No Solution' is wrong — see steps)

Q2. Solve the system of linear equation and also state that they are consistent or inconsistent system. (i) x + 2y - 3z = 2; 6x + 3y - 9z = 6; 7x + 14y - 21z = 13. (ii) 2x_1 - x_2 = -1; x_1 + 2x_2 - x_3 = -2; x_2 + x_3 = -2. (iii) 4y + z = 2; 2x + 6y - 2z = 3; 4x + 8y - 5z = 4.
(i) Row 3 = 7 x row 1 in the coefficients, but 7 x 2 = 14 is not 13: the augmented matrix reduces to a row [0, 0, 0 | 1]. Inconsistent: no solution.

(ii) Solving: from the third equation x_3 = -2 - x_2; substituting gives x_1 = -1, x_2 = -1, x_3 = -1. Consistent with a unique solution.

(iii) Reduction gives x - (7/4)z = 0 and y + (1/4)z = 1/2, with z free: x = 7t/4, y = -t/4 + 1/2, z = t. Consistent with infinitely many solutions.

(i) No solution (inconsistent); (ii) x_1 = -1, x_2 = -1, x_3 = -1 (consistent); (iii) infinitely many: x = 7t/4, y = -t/4 + 1/2, z = t (consistent)

Q3. Solve the non-homogeneous linear equations using matrix inversion method. (i) 2x + y - z = 8; -3x - y + 2z = -11; -2x + y + 2z = -3. (ii) x_1 + 3x_2 + x_3 = 9; x_1 + x_2 - x_3 = 1; 3x_1 + 11x_2 + 5x_3 = 35. (iii) x + y - z = 1; -x + y - z = -1; x + y + z = 0. (Flags: for (ii) the book's Answers 'No solution' is wrong — equation 3 is satisfied identically by the family given (check: 3(-3 + 2t) + 11(4 - t) + 5t = 35). For (iii) the book's Answers (1/2, 1, -1/2) fails equation 1 (1/2 + 1 + 1/2 = 2, not 1) and equation 3 (1/2 + 1 - 1/2 = 1, not 0); the correct solution is (1, -1/2, -1/2).)
(i) The inverse of the coefficient matrix times (8, -11, -3) gives (2, 3, -1). Check: 4 + 3 + 1 = 8; -6 - 3 - 2 = -11; -4 + 3 - 2 = -3.

(ii) Subtracting equations 1 and 2 gives x_2 + x_3 = 4; then x_1 = -3 + 2x_3 and x_2 = 4 - x_3, with x_3 free; equation 3 is then satisfied identically. Infinitely many solutions.

(iii) Adding equations 1 and 2 gives y = z; equation 1 then gives x = 1; equation 3 gives 1 + 2y = 0, so y = z = -1/2. Solution (1, -1/2, -1/2).

(i) x = 2, y = 3, z = -1; (ii) infinitely many: x_1 = -3 + 2x_3, x_2 = 4 - x_3 (flag: book's Answers 'No solution' is wrong — see steps); (iii) x = 1, y = -1/2, z = -1/2 (flag: book's Answers (1/2, 1, -1/2) fails the equations — see steps)

Q4. Solve by Cramer's rule if possible. (i) 2x + 3y + 3z = 5; x - 2y + z = -4; 3x - y - 2z = 3. (ii) x - y + z = 4; 2x + y - 3z = 0; x + y + z =

2. (iii) 9x_1 + 3x_2 + 4x_3 = 7; 4x_1 + 3x_2 + 4x_3 = 8; x_1 + x_2 + x_3 = 3.
(i) D = 40; D_x = 40, D_y = 80, D_z = -40, so x = 1, y = 2, z = -1.

(ii) D = 10; D_x = 20, D_y = -10, D_z = 10, so x = 2, y = -1, z = 1.

(iii) D = -5; D_x = 1, D_y = -20, D_z = 4, so x_1 = -1/5, x_2 = 4, x_3 = -4/5.

(i) x = 1, y = 2, z = -1 (D = 40); (ii) x = 2, y = -1, z = 1 (D = 10); (iii) x_1 = -1/5, x_2 = 4, x_3 = -4/5 (D = -5)

Q5. Solve by Gaussian elimination method (echelon form). (i) x - 2y + z = -4; 2x - 3y + 2z = -6; 2x + 2y + z = 5. (ii) x_1 + 3x_2 + 2x_3 = 3; 4x_1 + 5x_2 - 3x_3 = -3; x_1 - 2x_2 + 17x_3 = 42. (iii) 3x_1 + x_2 - x_3 = 4; x_1 + x_2 - 2x_3 = -4; -x_1 + 2x_2 - x_3 = 1. (Flags: the book's Answers for all three parts fail the printed equations when substituted, as noted in each answer; the solutions given here were verified by substitution.)
(i) Echelon form gives z = -1, then y = 2, then x = 1. Check: 1 - 4 - 1 = -4; 2 - 6 - 2 = -6; 2 + 4 - 1 = 5.

(ii) Echelon form gives x_3 = 87/40, then x_2 = -51/40, then x_1 = 99/40. Check: 99/40 - 153/40 + 174/40 = 3; 396/40 - 255/40 - 261/40 = -3; 99/40 + 102/40 + 1479/40 = 42.

(iii) Echelon form gives x_3 = 14/3, then x_2 = 11/3, then x_1 = 5/3. Check: 5 + 11/3 - 14/3 = 4; 5/3 + 11/3 - 28/3 = -4; -5/3 + 22/3

- 14/3 = 1.

(i) x = 1, y = 2, z = -1 (flag: book's Answers (1, 2, 1) fails equation 1); (ii) x_1 = 99/40, x_2 = -51/40, x_3 = 87/40 (flag: book's Answers (2, -1, 2) fails equation 3); (iii) x_1 = 5/3, x_2 = 11/3, x_3 = 14/3 (flag: book's Answers (-1, 1, 2) fails equation 1)

Q6. Solve by Gauss Jordon method (reduced echelon form). (i) x - 2y + z = -1; 3x + y - 2z = 4; y - z = 1. (ii) 2x_1 + x_2 + 3x_3 = 3; x_1 + x_2 - 2x_3 = 0; -3x_1 - x_2 + 2x_3 = -4. (iii) x + y = 2; 2x - z = 1; 2y - 3z = -1.

(i) Reduced echelon form is I with right side (1, 1, 0): x = 1, y = 1, z = 0.

(ii) Reduced echelon form is I with right side (2, -8/5, 1/5): x_1 = 2, x_2 = -8/5, x_3 = 1/5.

(iii) Reduced echelon form is I with right side (1, 1, 1): x = 1, y = 1, z = 1.

(i) x = 1, y = 1, z = 0; (ii) x_1 = 2, x_2 = -8/5, x_3 = 1/5; (iii) x = 1, y = 1, z = 1

Exercise 2.5

Q1. Decode this cipher text that has been encoded using the matrix A = [[2, 1], [1, 2]]: DPOOOHZZVOQOVYYVQQ. (Flag: the book never states the reduction step, but it is forced by the data — plain integer decoding gives fractions, while decoding modulo 26 (with 0

read as Z) gives a meaningful message, and re-encoding that message with A modulo 26 reproduces the cipher letter for letter. The final W is a padding letter, since the message has 17 letters and the code works in pairs.)
Convert letters to numbers (A = 1, ..., Z = 26) and pair them: (4, 16), (15, 15), (15, 8), (26, 26), (22, 15), (17, 15), (22, 25), (25, 22), (17, 17).

The inverse of A modulo 26 is [[18, 17], [17, 18]] (since det(A) = 3 and 3 x 9 = 27 = 1 mod 26). Multiply each pair by it, modulo 26: (6, 18), (5, 5), (16, 9), (26, 26), (1, 20), (15, 13), (15, 18), (18, 15), (23, 23).

Convert back to letters: F, R, E, E, P, I, Z, Z, A, T, O, M, O, R, R, O, W, W — i.e. 'FREE PIZZA TOMORROW' with a final padding W.

FREE PIZZA TOMORROW (see flag for the method and the padding letter)

Q2. Using matrix B = [[1, 1, -1], [1, 0, 1], [2, 1, 1]], encode the message: ATTACH NOW. (Flag: 'ATTACH NOW' has 10 characters including the space, but a 3x3 code works in triples; the space is omitted here, giving the 9 letters ATTACHNOW in 3 triples. No answers for Exercise 2.5 are printed in the book's Answers section.)
Letter numbers (A = 1, ..., Z = 26): ATTACHNOW = (1, 20, 20), (1, 3, 8), (14, 15, 23).

B x (1, 20, 20) = (1 + 20 - 20, 1 + 0 + 20, 2 + 20 + 20) = (1, 21,

42).

B x (1, 3, 8) = (1 + 3 - 8, 1 + 0 + 8, 2 + 3 + 8) = (-4, 9, 13).

B x (14, 15, 23) = (14 + 15 - 23, 14 + 0 + 23, 28 + 15 + 23) = (6, 37, 66).

Coded triples: (1, 21, 42), (-4, 9, 13), (6, 37, 66) (see flag for the space)

Q3. Decode the following message that was encoded using matrix B = [[1, 1, -1], [1, 0, 1], [2, 1, 1]]: [11; 20; 43], [25; 10; 41], [22; 14; 41].

The inverse of B is [[1, 2, -1], [-1, -3, 2], [-1, -1, 1]] (det(B) = -1).

B^(-1) x (11, 20, 43) = (11 + 40 - 43, -11 - 60 + 86, -11 - 20 + 43) = (8, 15, 12) = H, O, L.

B^(-1) x (25, 10, 41) = (25 + 20 - 41, -25 - 30 + 82, -25 - 10 + 41) = (4, 27, 6) = D, space, F (27 = space, as in Example 2.25).

B^(-1) x (22, 14, 41) = (22 + 28 - 41, -22 - 42 + 82, -22 - 14 + 41) = (9, 18, 5) = I, R, E. Message: 'HOLD FIRE'.

HOLD FIRE

Q4. Find the image of triangle PQR with vertices P(2, -3), Q(0, 1) and R(-1, -2) under the translation vector [5; -1]. Sketch the original

and image figures. (Flag: a sketch question — the image triangle is given exactly; sketch both triangles from their vertices.)
Add (5, -1) to each vertex: P' = (2 + 5, -3 - 1) = (7, -4); Q' = (0 + 5, 1 - 1) = (5, 0); R' = (-1 + 5, -2 - 1) = (4, -3).

Sketch: original triangle (2, -3), (0, 1), (-1, -2) and image triangle (7, -4), (5, 0), (4, -3), the image being the original shifted 5 right and 1 down.

P'(7, -4), Q'(5, 0), R'(4, -3) (sketch described)

Q5. Find the images of the vertices of a unit square ABCD under the transformation given by L = [[1, 2], [-1, 1]]. The square has D(0, 0), C(1, 0), B(1, 1), A(0, 1).
L x D = (0, 0), so D' = (0, 0). L x C = (1, -1), so C' = (1, -1). L x B = (1 + 2, -1 + 1) = (3, 0), so B' = (3, 0). L x A = (2, 1), so A' = (2, 1). The image is the parallelogram (0, 0), (1, -1), (3, 0), (2, 1).

A'(2, 1), B'(3, 0), C'(1, -1), D'(0, 0)

Q6. An image is represented by a 2x2 matrix of pixel brightness values: I = [[120, 180], [200, 240]]. Find the new image matrix (Hint: find 0.5I).
0.5I = [[60, 90], [100, 120]].

[[60, 90], [100, 120]]

Q7. A two-floor building has a stiffness matrix: K = [[10, -2], [-2, 5]]

and F = [18; 11]. Find displacement solving the system of equations. (Hint: Find KF). (Flag: the hint 'Find KF' is a misprint — displacement u solves K u = F, so u = K^(-1) F, not KF. KF = (158, 19) is not a displacement.)
det(K) = 50 - 4 = 46, so K^(-1) = (1/46)[[5, 2], [2, 10]].

u = K^(-1) F = (1/46)(5 x 18 + 2 x 11, 2 x 18 + 10 x 11) = (1/46)(112, 146) = (56/23, 73/23), about (2.43, 3.17).

Displacement (56/23, 73/23), about (2.43, 3.17) (see flag for the hint)

Q8. Point B = (1, 3) is scaled by 3 in x-direction and 2 in y-direction. Find new coordinate using S = [[3, 0], [0, 2]]. (Hint: Find SB).
S B = (3 x 1, 2 x 3) = (3, 6).

(3, 6)

Review Exercise 2

Q1(i). A matrix having 1-by-n order is called;

- ✓ row matrix
- column matrix
- square matrix
- diagonal matrix

Q1(ii). A square matrix with a_ij = 1 for i = j and a_ij = 0 for i not equal to j is called;

- diagonal matrix

scalar matrix

✓ unit matrix

none of these

Q1(iii). Value of |1, 2, 3; 4, 5, 6; 1, 2, 3| is; (rows 1 and 3 are identical, so the value is 0.)

-1

✓ 0

1

2

Q1(iv). Inverse of [[3, 4], [4, 5]] is; (determinant 15 - 16 = -1, so the inverse is -[[5, -4], [-4, 3]] = [[-5, 4], [4, -3]].)

[[3, 4], [-4, -5]]

[[-3, -4], [4, 5]]

[[5, 4], [4, 3]]

✓ [[-5, 4], [4, -3]]

Q1(v). Adjoint of [[2, 3], [-4, 5]] is (flag: the book's answer key says (d), but the adjoint is the transpose of the cofactor matrix [[5, 4], [-3, 2]], i.e. [[5, -3], [4, 2]] = (c).)

[[2, -3], [-4, 5]]

[[5, 3], [-4, 2]]

✓ [[5, -3], [4, 2]]

[[5, -3], [4, -2]]

Q1(vi). In a row vector number of rows are

✓ 1

2

n

m

Q1(vii). Inverse of [[1, 2], [1, 3]] is

[[1, 1], [2, 3]]

[[1, 2], [3, 1]]

[[3, 2], [1, 1]]

✓ [[3, -2], [-1, 1]]

Q1(viii). Minor of an element of a square matrix is denoted by

M

✓ M_ij

M_i

M_j

Q1(ix). Solution of system of linear equations is; x - 2y + z = 2; 2x + 2y - z = 4; x + y = 0. (From the third equation y = -x; the second gives z = -4; the first gives x = 2, y = -2.)

(2, 2, 4)

✓ (2, -2, -4)

(-2, 2, -4)

(2, -2, 4)

Q1(x). System of equations is called; a_1 x + b_1 y = K_1; a_2 x + b_2 y = K_2 where K_1 not equal to 0, K_2 not equal to 0

✓ non-homogeneous

homogeneous

quadratic equations

cubic equations

Q2. Find the values of x and y from 2[[x, 5], [7, y - 3]] + [[3, -4], [1, 2]] = [[7, 6], [15, 14]].

Entry (1,1): 2x + 3 = 7, so x = 2. Entry (1,2): 10 - 4 = 6, which checks. Entry (2,1): 14 + 1 = 15, which checks. Entry (2,2): 2(y - 3) + 2 = 14, so y - 3 = 6 and y = 9.

x = 2, y = 9 (flag: book's answer key says y = 10, but 2(y - 3) + 2 = 14 gives y = 9)

Q3. Calculate AC, BC and (A + B)C. Also verify that (A + B)C = AC + BC for A = [[0, 6, 7], [-6, 0, 8], [7, -8, 0]], B = [[0, 1, 1], [1, 0, 2], [1, 2, 0]], C = [2; -2; 3].

AC = (0 - 12 + 21, -12 + 0 + 24, 14 + 16 + 0) = (9, 12, 30).

BC = (0 - 2 + 3, 2 + 0 + 6, 2 - 4 + 0) = (1, 8, -2).

A + B = [[0, 7, 8], [-5, 0, 10], [8, -6, 0]], so (A + B)C = (0 - 14 + 24, -10 + 0 + 30, 16 + 12 + 0) = (10, 20, 28).

AC + BC = (9 + 1, 12 + 8, 30 - 2) = (10, 20, 28). Verified.

AC = (9, 12, 30); BC = (1, 8, -2); (A + B)C = (10, 20, 28) = AC + BC

Q4. If A = [[3, 1], [-1, 2]] show that A^2 - 5A + 7I = 0.

A^2 = [[9 - 1, 3 + 2], [-3 - 2, -1 + 4]] = [[8, 5], [-5, 3]].

5A = [[15, 5], [-5, 10]] and 7I = [[7, 0], [0, 7]].

A^2 - 5A + 7I = [[8 - 15 + 7, 5 - 5], [-5 + 5, 3 - 10 + 7]] = [[0, 0], [0, 0]].

A^2 - 5A + 7I = 0 is shown

Q5. Show that |a, b, c; a + 2x, b + 2y, c + 2z; x, y, z| = 0.
Row 2 = row 1 + 2 x row 3, so the rows are linearly dependent and the determinant is 0.

0 is shown

Q6. Show that |x + y + 2z, x, y; z, y + z + 2x, y; z, x, z + x + 2y| = 2(x + y + z)^3.
Every row sums to 2(x + y + z). Add columns 2 and 3 to column 1 and take 2(x + y + z) out; then row and column operations reduce the remaining determinant to (x + y + z)^2. Total: 2(x + y + z)^3.

2(x + y + z)^3 is shown

Q7. Show that |sin(alpha), cos(alpha), cos(alpha + delta); sin(beta), cos(beta), cos(beta + delta); sin(gamma), cos(gamma), cos(gamma + delta)| = 0.
cos(alpha + delta) = cos(alpha)cos(delta) - sin(alpha)sin(delta), and similarly for beta and gamma. Hence column 3 = cos(delta) x column 2 - sin(delta) x column 1, so the determinant is 0.

0 is shown

Q8. Solve the system of linear equations using matrix inversion method. 2x + 3y + 3z = 5; x - 2y + z = -4; 3x - y - 2z = 3.
This is the same system as Exercise 2.4 Q4(i). Inverting the coefficient matrix and multiplying by (5, -4, 3) gives (1, 2, -1). Check: 2 + 6 - 3 = 5; 1 - 4 - 1 = -4; 3 - 2 + 2 = 3.

x = 1, y = 2, z = -1

Q9. Use Cramer's rule to solve the system: x + y + z = 4; 2x + y - z = 3; 3x + 3y + 2z = 10. (Flag: the first equation is printed as 'z + y + z = 4'; it is read as x + y + z = 4, matching the book's Answers (3, -1, 2).)
D = |1, 1, 1; 2, 1, -1; 3, 3, 2| = 1(2 + 3) - 1(4 + 3) + 1(6 - 3) = 1.

D_x = |4, 1, 1; 3, 1, -1; 10, 3, 2| = 4(2 + 3) - 1(6 + 10) + 1(9 - 10) = 3.

D_y = |1, 4, 1; 2, 3, -1; 3, 10, 2| = 1(6 + 10) - 4(4 + 3) + 1(20 - 9) = -1.

D_z = |1, 1, 4; 2, 1, 3; 3, 3, 10| = 1(10 - 9) - 1(20 - 9) + 4(6 - 3) = 2.

Hence x = 3, y = -1, z = 2.

x = 3, y = -1, z = 2 (flag: the first equation is printed as 'z + y + z

= 4', read as x + y + z = 4)

Q10. Find the image of triangle ABC with vertices A(3, 2), B(0, 5) and C(-3, -2) under the translation vector [2; -1], sketch the original and image figure. (Flag: a sketch question — the image triangle is given exactly. The book's answer key prints A'(15, 1); A(3, 2) + (2, -1) = (5, 1).)

Add (2, -1) to each vertex: A' = (3 + 2, 2 - 1) = (5, 1); B' = (0 + 2, 5 - 1) = (2, 4); C' = (-3 + 2, -2 - 1) = (-1, -3). Sketch both triangles from their vertices.

A'(5, 1), B'(2, 4), C'(-1, -3) (flag: book's answer key prints A'(15, 1) — a misprint for (5, 1))

Unit 3: Sequences and Series

Exercise 3.1

Q1. Find the indicated terms of the following sequences. (i) 1, 5, 9, ..., a15; (ii) -15, -9, -3, ..., a10; (iii) -5, 4, 13, ..., a30; (iv) 23, 26, 29, ..., a14.

Use $a_n = a_1 + (n - 1)d$, where a_1 is the first term and d is the common difference.

(i) $a_1 = 1$, $d = 5 - 1 = 4$. $a_{15} = 1 + (15 - 1)(4) = 1 + 56 = 57$.

(ii) $a_1 = -15$, $d = -9 - (-15) = 6$. $a_{10} = -15 + (10 - 1)(6) = -15 + 54 = 39$.

(iii) $a_1 = -5$, $d = 4 - (-5) = 9$. $a_{30} = -5 + (30 - 1)(9) = -5 + 261 = 256$.

(iv) $a_1 = 23$, $d = 26 - 23 = 3$. $a_{14} = 23 + (14 - 1)(3) = 23 + 39 = 62$.

(i) 57; (ii) 39; (iii) 256; (iv) 62

Q2. Find the first four terms of the following arithmetic sequences, if (i) $a_n = 2n - 5$; (ii) $a_{n-2} = 3n - 4$; (iii) $a_n - a_{n-1} = 15$, $a_1 = 4$; (iv) $a_3 = 18$ and other three consecutive terms are 54, 60, 66.

(i) Put n = 1, 2, 3, 4 in a_n = 2n - 5: a_1 = -3, a_2 = -1, a_3 = 1, a_4 = 3.

(ii) a_{n-2} = 3n - 4. Put n = 3, 4, 5, 6 to get a_1, a_2, a_3, a_4: a_1 = 9 - 4 = 5, a_2 = 12 - 4 = 8, a_3 = 15 - 4 = 11, a_4 = 18 - 4 = 14.

(iii) a_n - a_{n-1} = 15 means the common difference d = 15. With a_1 = 4: a_2 = 19, a_3 = 34, a_4 = 49.

(iv) The consecutive terms 54, 60, 66 give d = 60 - 54 = 6. From a_3 = 18: a_1 = 18 - 2d = 18 - 12 = 6, a_2 = 12, a_4 = 24. (The wording is compressed; it is read as: 54, 60, 66 are three consecutive terms of the same A.P., which fixes d = 6.)

(i) -3, -1, 1, 3; (ii) 5, 8, 11, 14; (iii) 4, 19, 34, 49; (iv) 6, 12, 18, 24

Q3. If a_{n-3} = 9n - 4, find n-th term of the sequence.

Let m = n - 3, so n = m + 3.

Then a_m = 9(m + 3) - 4 = 9m + 27 - 4 = 9m + 23.

Replacing m by n, the n-th term is a_n = 9n + 23.

a_n = 9n + 23

Q4. Find 25th term of the sequence 1, 2 - y, 3 - 2y, ...

First term a_1 = 1. Common difference d = (2 - y) - 1 = 1 - y. Check: (3 - 2y) - (2 - y) = 1 - y, consistent.

$a_{25} = a_1 + 24d = 1 + 24(1 - y) = 1 + 24 - 24y = 25 - 24y$.

25 - 24y

Q5. Find the 18th term of the A.P. if its 6th term is 11 and the 15th term is 47.

Let the first term be a and the common difference be d.

$a_6 = a + 5d = 11$ ... (1). $a_{15} = a + 14d = 47$ ... (2).

Subtract (1) from (2): $9d = 36$, so $d = 4$.

Put $d = 4$ in (1): $a + 20 = 11$, so $a = -9$.

$a_{18} = a + 17d = -9 + 17(4) = -9 + 68 = 59$.

59

Q6. Which term of the sequence -15, -9, -3, ... is 75?

First term $a = -15$, common difference $d = -9 - (-15) = 6$.

Let the n-th term be 75: $a + (n - 1)d = 75$.

$-15 + (n - 1)(6) = 75$, so $6(n - 1) = 90$, $n - 1 = 15$, $n = 16$.

75 is the 16th term

Q7. Which term of the sequence 5/7, 3/7, 1/7, ... is -105/7?

First term a = 5/7, common difference d = 3/7 - 5/7 = -2/7.

Let the n-th term be -105/7: 5/7 + (n - 1)(-2/7) = -105/7.

Multiply by 7: 5 - 2(n - 1) = -105, so 2(n - 1) = 110, n - 1 = 55, n = 56.

-105/7 is the 56th term

Q8. Find the n-th term of the sequence (6/7)^2, (11/7)^2, (16/7)^2, ...

The bases form the sequence 6/7, 11/7, 16/7, ..., an A.P. with first term 6/7 and common difference 5/7.

The n-th base = 6/7 + (n - 1)(5/7) = (6 + 5n - 5)/7 = (5n + 1)/7.

Therefore the n-th term of the given sequence is the square of the n-th base.

a_n = ((5n + 1)/7)^2

Q9. If a, b, c are the u-th, v-th and w-th terms of an A.P., show that: (i) a(v - w) + b(w - u) + c(u - v) = 0; (ii) u(b - c) + v(c - a) + w(a - b) = 0.

Let A be the first term and d the common difference. Then a = A + (u - 1)d, b = A + (v - 1)d, c = A + (w - 1)d.

(i) a(v - w) + b(w - u) + c(u - v) = A[(v - w) + (w - u) + (u - v)] + d[(u - 1)(v - w) + (v - 1)(w - u) + (w - 1)(u - v)].

The coefficient of A is v - w + w - u + u - v = 0.

The coefficient of d expands to (uv - uw - v + w) + (vw - vu - w + u) + (wu - wv - u + v) = 0. Hence the whole expression is 0.

(ii) b - c = (v - w)d, c - a = (w - u)d, a - b = (u - v)d.

So u(b - c) + v(c - a) + w(a - b) = d[u(v - w) + v(w - u) + w(u - v)] = d[uv - uw + vw - vu + wu - wv] = 0.

Both identities are proved (each side reduces to 0)

Q10. If 1/c, 1/d, 1/e are in A.P., show that the common difference is (c - e)/(2ce). (Note: the book writes the common difference with the same letter d as the middle denominator; here the common difference is called D to avoid confusion.)
Since 1/c, 1/d, 1/e are in A.P., the difference of consecutive terms is constant: let it be D.

Then 1/d - 1/c = D and 1/e - 1/d = D.

Subtracting: (1/e - 1/d) - (1/d - 1/c) = 0 is not needed; instead add the two gaps: (1/e - 1/c) = 2D.

But 1/e - 1/c = (c - e)/(ce). So 2D = (c - e)/(ce), giving D = (c -

e)/(2ce).

Common difference = (c - e)/(2ce)

Exercise 3.2

Q1. Find A.M. between. (i) 9 and 16; (ii) 6 + 2 sqrt(7) and 4 - 2 sqrt(7); (iii) 3x + 2y and 5x - 6y; (iv) 3 + 4 sqrt(3) i and 9 - 4 sqrt(3) i; (v) 3x^2 - 5x + 6 and -x^2 + 7x - 4; (vi) 3x - 5 and 5x + 3.

The arithmetic mean (A.M.) between a and b is (a + b)/2.

(i) (9 + 16)/2 = 25/2.

(ii) (6 + 2 sqrt(7) + 4 - 2 sqrt(7))/2 = 10/2 = 5.

(iii) (3x + 2y + 5x - 6y)/2 = (8x - 4y)/2 = 4x - 2y.

(iv) (3 + 4 sqrt(3) i + 9 - 4 sqrt(3) i)/2 = 12/2 = 6.

(v) (3x^2 - 5x + 6 - x^2 + 7x - 4)/2 = (2x^2 + 2x + 2)/2 = x^2 + x + 1.

(vi) (3x - 5 + 5x + 3)/2 = (8x - 2)/2 = 4x - 1.

(i) 25/2; (ii) 5; (iii) 4x - 2y; (iv) 6; (v) x^2 + x + 1; (vi) 4x - 1

Q2. If 3 and 8 are two A.Ms. between a and b, find a and b.

If 3 and 8 are A.Ms. between a and b, then a, 3, 8, b are in A.P.

The common difference is d = 8 - 3 = 5.

Then a = 3 - d = 3 - 5 = -2, and b = 8 + d = 8 + 5 = 13.

a = -2, b = 13

Q3. Insert three A.Ms. between 4 and 17.

Inserting three means gives 5 terms in A.P.: first term 4, fifth term 17.

Let the common difference be d: 4 + 4d = 17, so 4d = 13, d = 13/4.

The three means are: 4 + 13/4 = 29/4; 4 + 2(13/4) = 4 + 13/2 = 21/2; 4 + 3(13/4) = 55/4.

29/4, 21/2, 55/4

Q4. Insert four A.Ms. between sqrt(3) and 12/sqrt(3).

Note 12/sqrt(3) = 4 sqrt(3). Inserting four means gives 6 terms in A.P.: first term sqrt(3), sixth term 4 sqrt(3).

Let the common difference be d: sqrt(3) + 5d = 4 sqrt(3), so 5d = 3 sqrt(3), d = (3 sqrt(3))/5.

The four means are: sqrt(3) + (3 sqrt(3))/5 = (8 sqrt(3))/5; (11 sqrt(3))/5; (14 sqrt(3))/5; (17 sqrt(3))/5.

(8 sqrt(3))/5, (11 sqrt(3))/5, (14 sqrt(3))/5, (17 sqrt(3))/5

Q5. Find six A.Ms. between -15 and 15.

Inserting six means gives 8 terms in A.P.: first term -15, eighth term 15.

Let the common difference be d: -15 + 7d = 15, so 7d = 30, d = 30/7.

The six means are: -15 + 30/7 = -75/7; -45/7; -15/7; 15/7; 45/7; 75/7.

-75/7, -45/7, -15/7, 15/7, 45/7, 75/7

Q6. Find n so that (x^{n+4} + y^{n+4})/(x^{n+3} + y^{n+3}) may be the A.M. between x and y.

The A.M. between x and y is (x + y)/2. Set the given expression equal to it.

2(x^{n+4} + y^{n+4}) = (x + y)(x^{n+3} + y^{n+3}).

Expand the right side: x^{n+4} + x y^{n+3} + y x^{n+3} + y^{n+4}.

Bring all terms to one side: x^{n+4} + y^{n+4} - x y^{n+3} - y x^{n+3} = 0.

Factor: x(x^{n+3} - y^{n+3}) - y(x^{n+3} - y^{n+3}) = (x -

y)(x^{n+3} - y^{n+3}) = 0.

For x different from y this holds identically only when x^{n+3} = y^{n+3} for all x, y, i.e. when n + 3 = 0. Hence n = -3.

n = -3

Q7. Find the values of x if |1 0; 0 4| is the A.M. between |x 1; 1 x+2| and |1 0; 0 4|. (Determinants written row-wise.)
Evaluate the determinants: |1 0; 0 4| = 4; |x 1; 1 x+2| = x(x + 2) - 1 = x^2 + 2x - 1; the third determinant is again 4.

The A.M. condition gives: 4 = ((x^2 + 2x - 1) + 4)/2.

So x^2 + 2x + 3 = 8, i.e. x^2 + 2x - 5 = 0.

By the quadratic formula: x = (-2 +- sqrt(4 + 20))/2 = -1 +- sqrt(6).

x = -1 + sqrt(6) or x = -1 - sqrt(6)

Exercise 3.3

Q1. Sum the series. (i) 3 + 7 + 11 + ... + a15; (ii) 4.57 + 4.87 + 5.17 + ... + a24; (iii) 3/sqrt(2) + 2 sqrt(2) + 5/sqrt(2) + ... to n terms; (iv) (y - 3b) + (y - b) + (y + b) + ... to n terms; (v) 1/(1 - sqrt(x)) + 1/(1 - x) + 1/(1 + sqrt(x)) + ... to n terms; (vi) 2/sqrt(3) + sqrt(3) + 4/sqrt(3) + ... + a20.
Use S_n = (n/2)[2a + (n - 1)d].

(i) a = 3, d = 4, n = 15: S_15 = (15/2)[6 + 14(4)] = (15/2)(62) = 465.

(ii) a = 4.57, d = 0.30, n = 24: S_24 = 12[9.14 + 23(0.30)] = 12(16.04) = 192.48.

(iii) a = 3/sqrt(2), d = 2 sqrt(2) - 3/sqrt(2) = 1/sqrt(2). S_n = (n/2)[6/sqrt(2) + (n - 1)/sqrt(2)] = n(n + 5)/(2 sqrt(2)) = n(n + 5) sqrt(2)/4.

(iv) a = y - 3b, d = 2b. S_n = (n/2)[2(y - 3b) + (n - 1)(2b)] = (n/2)[2y + 2b(n - 4)] = ny + nb(n - 4).

(v) Let t = sqrt(x). The terms are 1/(1 - t), 1/(1 - t^2), 1/(1 + t); consecutive differences are both -t/(1 - t^2), so d = -sqrt(x)/(1 - x) and a = 1/(1 - sqrt(x)). S_n = (n/2)[2/(1 - t) + (n - 1)(-t/(1 - t^2))] = n[2 + (3 - n) sqrt(x)]/[2(1 - x)].

(vi) a = 2/sqrt(3), d = sqrt(3) - 2/sqrt(3) = 1/sqrt(3). a_20 = 2/sqrt(3) + 19/sqrt(3) = 21/sqrt(3) = 7 sqrt(3). S_20 = 10(2/sqrt(3) + 7 sqrt(3)) = 10(23/sqrt(3)) = 230/sqrt(3) = 230 sqrt(3)/3.

(i) 465; (ii) 192.48; (iii) n(n + 5) sqrt(2)/4; (iv) ny + nb(n - 4); (v) n[2 + (3 - n) sqrt(x)]/[2(1 - x)]; (vi) 230 sqrt(3)/3

Q2. Find the number of terms of the following. (i) -9 + (-6) + (-3) + ... amount to 3225? (ii) 1/(1 + sqrt(x)) + 1/(1 - x) + 1/(1 - sqrt(x)) + ... amount to 10(2 + 17 sqrt(x))/(1 - x)?

(i) a = -9, d = 3. S_n = (n/2)[-18 + 3(n - 1)] = n(3n - 21)/2 = 3225.

So 3n^2 - 21n - 6450 = 0, i.e. n^2 - 7n - 2150 = 0. Discriminant = 49 + 8600 = 8649 = 93^2. n = (7 + 93)/2 = 50 (the negative root is rejected).

(ii) Let t = sqrt(x). Then a = 1/(1 + t), d = t/(1 - t^2), and S_n = n[2 + (n - 3)t]/[2(1 - t^2)].

Set S_n = 10(2 + 17t)/(1 - t^2): n[2 + (n - 3)t]/2 = 10(2 + 17t), i.e. n[2 + (n - 3)t] = 40 + 340t.

Comparing the constant parts: 2n = 40, so n = 20; check the t-coefficients: n(n - 3) = 20(17) = 340, consistent. Hence 20 terms.

(i) 50 terms; (ii) 20 terms

Q3. Sum the series. (i) 3 + 5 - 7 + 10 + 12 - 14 + 17 + 19 - 21 + ... to 3n terms. (ii) 9 + 12 - 15 + 18 + 21 - 24 + 27 + 30 - 33 + ... to 3n terms.

(i) Group in threes: (3, 5, -7), (10, 12, -14), ... The k-th group is (7k - 4) + (7k - 2) + (-7k) = 7k - 6. Check: k = 1 gives 1 = 3 + 5 - 7; k = 2 gives 8 = 10 + 12 - 14.

Sum over k = 1 to n: 7[n(n + 1)/2] - 6n = (7n^2 - 5n)/2 = n(7n - 5)/2.

(ii) Group in threes: (9, 12, -15), (18, 21, -24), ... The k-th group is

9k + (9k + 3) - (9k + 6) = 9k - 3. Check: k = 1 gives 6 = 9 + 12 - 15; k = 2 gives 15 = 18 + 21 - 24.

Sum over k = 1 to n: 9[n(n + 1)/2] - 3n = (9n^2 + 3n)/2 = 3n(3n + 1)/2.

(i) n(7n - 5)/2; (ii) 3n(3n + 1)/2

Q4. If S_n = n(2n + 1), then find the series.

The n-th term is a_n = S_n - S_{n-1} for n >= 2, and a_1 = S_1.

S_1 = 1(3) = 3, so a_1 = 3.

S_n - S_{n-1} = (2n^2 + n) - [2(n - 1)^2 + (n - 1)] = (2n^2 + n) - (2n^2 - 3n + 1) = 4n - 1.

Hence a_n = 4n - 1: the series is 3 + 7 + 11 + 15 + ... Check: S_3 = 3(7) = 21 = 3 + 7 + 11.

The series is 3 + 7 + 11 + 15 + ... (a_n = 4n - 1)

Q5. Find the sum of first 600 natural numbers which are neither divisible by 2 nor by 5.

Among 1 to 600, the numbers neither divisible by 2 nor by 5 are the odd numbers not divisible by 5.

Odd numbers from 1 to 600: 300 numbers, their sum = 300^2 = 90000 (sum of the first 300 odd numbers).

Odd multiples of 5 among them: 5, 15, ..., 595, which are 60 numbers; their sum = 5(1 + 3 + ... + 119) = 5(60^2) = 18000.

Required sum = 90000 - 18000 = 72000. Cross-check by inclusion-exclusion: 180300 - 90300 - 36300 + 18300 = 72000.

72000

Q6. If S_{3n}, S_{5n}, S_{8n} are the sums of 3n, 5n and 8n terms of a_n in A.P., show that S_{8n} = 4(S_{5n} - S_{3n}).
Let the A.P. have first term a and common difference d. S_k = (k/2)[2a + (k - 1)d].

S_{5n} - S_{3n} = (5n/2)[2a + (5n - 1)d] - (3n/2)[2a + (3n - 1)d].

= (n/2)[10a + 25nd - 5d - 6a - 9nd + 3d] = (n/2)[4a + 16nd - 2d] = n[2a + 8nd - d].

Multiply by 4: 4(S_{5n} - S_{3n}) = 4n[2a + 8nd - d] = (8n/2)[2a + (8n - 1)d] = S_{8n}.

S_{8n} = 4(S_{5n} - S_{3n}) is proved

Q7. The sums of n terms of two arithmetic series are in the ratio of 4n - 3 : 2n + 1. Find the ratio of their 7th terms.
For an A.P., the sum of the first (2m - 1) terms equals (2m - 1) times the m-th term, so the ratio of the m-th terms equals the ratio of the

sums of (2m - 1) terms.

For m = 7: 2m - 1 = 13. Ratio of 7th terms = [4(13) - 3] : [2(13) + 1] = 49 : 27.

49 : 27

Exercise 3.4

Q1. Find the indicated terms of the following G.P. (i) 2, 6, 18, ..., a7; (ii) 1 + i, 2(1 - i), ..., a10; (iii) 4, 2 sqrt(2), 2, ..., a15; (iv) a^2 - b^2, a + b, (a + b)/(a - b), ..., a9.

Use a_n = a_1 r^{n-1}.

(i) a_1 = 2, r = 3. a_7 = 2(3^6) = 2(729) = 1458.

(ii) a_1 = 1 + i, r = 2(1 - i)/(1 + i) = (1 - i)^2 = -2i. (-2i)^9 = (-2)^9 i^9 = -512i. So a_10 = (1 + i)(-512i) = -512i + 512 = 512 - 512i.

(iii) a_1 = 4, r = (2 sqrt(2))/4 = 1/sqrt(2). a_15 = 4(1/sqrt(2))^{14} = 4/2^7 = 4/128 = 1/32.

(iv) a_1 = a^2 - b^2, r = (a + b)/(a^2 - b^2) = 1/(a - b). a_9 = (a^2 - b^2)(1/(a - b))^8 = (a - b)(a + b)/(a - b)^8 = (a + b)/(a - b)^7.

(i) 1458; (ii) 512 - 512i; (iii) 1/32; (iv) (a + b)/(a - b)^7

Q2. In a G.P., a_1 = 5/9, a_6 = 15625/9. Find its n-th term.

a_6 = a_1 r^5, so r^5 = a_6/a_1 = (15625/9)/(5/9) = 3125 = 5^5. Hence r = 5.

a_n = a_1 r^{n-1} = (5/9)(5^{n-1}) = 5^n/9.

a_n = 5^n/9

Q3. Find the n-th term of the geometric sequence if a_7/a_9 = 16/9 and a_2 = 9/4.
a_7/a_9 = 1/r^2 = 16/9, so r^2 = 9/16 and r = 3/4 or r = -3/4.

a_2 = a_1 r = 9/4. If r = 3/4, then a_1 = 3 and a_n = 3(3/4)^{n-1}. If r = -3/4, then a_1 = -3 and a_n = -3(-3/4)^{n-1}. Both satisfy the given data.

a_n = 3(3/4)^{n-1} or a_n = -3(-3/4)^{n-1}

Q4. Show that the reciprocals of the terms of the G.P. 1/a_1, 1/(a_1 r^2), 1/(a_1 r^4), ... form another G.P. Also find the common ratio of the new G.P.
The reciprocals of the terms are: 1/(1/a_1) = a_1; 1/(1/(a_1 r^2)) = a_1 r^2; 1/(1/(a_1 r^4)) = a_1 r^4; and so on.

Each term divided by the previous one gives (a_1 r^2)/a_1 = r^2 and (a_1 r^4)/(a_1 r^2) = r^2, a constant.

Hence the reciprocals form a G.P. with first term a_1 and common ratio r^2.

The reciprocals form a G.P. with common ratio r^2

Q5. How many terms are in G.P. 1/3, 1/12, 1/48, ..., 1/196608?

First term a = 1/3, common ratio r = (1/12)/(1/3) = 1/4.

Let the last term be the n-th term: (1/3)(1/4)^{n-1} = 1/196608.

Then (1/4)^{n-1} = 3/196608 = 1/65536 = (1/4)^8, so n - 1 = 8 and n = 9.

9 terms

Q6. If 1/(xy), 1/(yz), 1/(zx) are in G.P., show that x = +- sqrt(yz).

In a G.P., the square of the middle term equals the product of the neighbouring terms: (1/(yz))^2 = (1/(xy))(1/(zx)).

So 1/(y^2 z^2) = 1/(x^2 y z). Cross-multiplying: x^2 y z = y^2 z^2.

Divide by yz (non-zero): x^2 = yz, so x = +- sqrt(yz).

x = +- sqrt(yz) is proved

Exercise 3.5

Q1. Find G.M. between: (i) 4 and 9; (ii) 4 sqrt(5) and 5 sqrt(5); (iii) 5 + i and 5 - i; (iv) 16/9 and 9/25.

The geometric mean between a and b is G = +- sqrt(ab).

(i) +- sqrt(4 x 9) = +- sqrt(36) = +-6.

(ii) +- sqrt(4 sqrt(5) x 5 sqrt(5)) = +- sqrt(100) = +-10.

(iii) +- sqrt((5 + i)(5 - i)) = +- sqrt(26).

(iv) +- sqrt((16/9)(9/25)) = +- sqrt(16/25) = +-4/5.

(i) +-6; (ii) +-10; (iii) +-sqrt(26); (iv) +-4/5

Q2. Insert two G.Ms. between 3 and 81.

Inserting two means gives 4 terms in G.P.: 3, 3r, 3r^2, 81.

So 3r^3 = 81, r^3 = 27, r = 3 (real root).

The two means are 3(3) = 9 and 3(3^2) = 27.

9, 27

Q3. Insert five G.Ms. between 16 and 1024.

Inserting five means gives 7 terms in G.P.: first term 16, seventh term 1024.

So 16r^6 = 1024, r^6 = 64, r = 2 or r = -2.

With r = 2 the means are 32, 64, 128, 256, 512. With r = -2 they are -32, 64, -128, 256, -512. (The book's worked example uses the

positive root.)

With r = 2: 32, 64, 128, 256, 512; with r = -2: -32, 64, -128, 256, -512

Q4. Insert six G.Ms. between 9 and 19683.

Inserting six means gives 8 terms in G.P.: first term 9, eighth term 19683.

So 9r^7 = 19683, r^7 = 2187 = 3^7, r = 3.

The six means are 27, 81, 243, 729, 2187, 6561.

27, 81, 243, 729, 2187, 6561

Q5. The A.M. between two numbers is 17 and the geometric mean is 15, find the numbers.

Let the numbers be a and b. Then (a + b)/2 = 17, so a + b = 34; and sqrt(ab) = 15, so ab = 225.

Thus a and b are the roots of t^2 - 34t + 225 = 0.

Discriminant = 34^2 - 4(225) = 1156 - 900 = 256 = 16^2. So t = (34 +- 16)/2 = 25 or 9.

The numbers are 25 and 9.

25 and 9

Q6. For what value of n, is (x^{n-2} + y^{n-2})/(x^{n-3} + y^{n-3}) the G.M. between x and y?

The G.M. between x and y is sqrt(xy). Put m = n - 3, so the expression is (x^{m+1} + y^{m+1})/(x^m + y^m).

Set it equal to sqrt(xy) and cross-multiply: x^{m+1} + y^{m+1} = x^{m+1/2} y^{1/2} + x^{1/2} y^{m+1/2}.

Rearrange: x^{m+1/2}(x^{1/2} - y^{1/2}) - y^{m+1/2}(x^{1/2} - y^{1/2}) = 0, i.e. (x^{1/2} - y^{1/2})(x^{m+1/2} - y^{m+1/2}) = 0.

For x different from y this holds identically only when m + 1/2 = 0, i.e. m = -1/2. Hence n - 3 = -1/2, so n = 5/2.

Check: for n = 5/2 the expression is (sqrt(x) + sqrt(y))/(1/sqrt(x) + 1/sqrt(y)) = sqrt(xy).

n = 5/2

Q7. Show that the n-th root of the product of n geometric means between a and b is the geometric mean between a and b.

Let G_1, ..., G_n be the n geometric means between a and b. Then a, G_1, ..., G_n, b are in G.P. with n + 2 terms.

Let r be the common ratio: b = a r^{n+1}, so r^{n+1} = b/a, and G_k = a r^k.

The product G_1 ... G_n = (a r)(a r^2)...(a r^n) = a^n r^{1+2+...+n} = a^n r^{n(n+1)/2}.

Its n-th root is a r^{(n+1)/2} = a (r^{n+1})^{1/2} = a (b/a)^{1/2} = sqrt(ab), which is the geometric mean between a and b.

The n-th root of the product is sqrt(ab), the G.M. between a and b

Exercise 3.6

Q1. Compute the sum. (i) 1 + 1/2 + 1/4 + ... + 1/512; (ii) 2^6 + 2^7 + 2^8 + ... + 2^13; (iii) 1/u + 1/u^2 + 1/u^3 + ... + 1/u^8; (iv) 9 + 3 + 1 + ... + 1/2187; (v) 0.9 + 0.99 + 0.999 + ... to n terms; (vi) 7 + 77 + 777 + ... to n terms.

Use S_n = a(1 - r^n)/(1 - r) for r different from 1.

(i) a = 1, r = 1/2, n = 10 (since 1/512 = (1/2)^9 is the 10th term). S = (1 - (1/2)^10)/(1/2) = 2(1 - 1/1024) = 1023/512.

(ii) a = 64, r = 2, n = 8. S = 64(2^8 - 1)/(2 - 1) = 64(255) = 16320.

(iii) a = 1/u, r = 1/u, n = 8. S = (1/u)(1 - u^{-8})/(1 - 1/u) = (u^8 - 1)/(u^8 (u - 1)).

(iv) a = 9, r = 1/3. The last term 9(1/3)^{n-1} = 1/2187 = 3^{-7} gives n = 10. S = 9(1 - (1/3)^10)/(2/3) = (27/2)(59048/59049) = 29524/2187.

(v) The k-th term is 1 - (1/10)^k. Sum = n - (1/10)(1 - (1/10)^n)/(9/10) = n - (1/9)(1 - 10^{-n}) = n - 1/9 + 10^{-n}/9.

(vi) The k-th term is 7(10^k - 1)/9. Sum = (7/9)[10(10^n - 1)/9 - n] = 70(10^n - 1)/81 - 7n/9.

(i) 1023/512; (ii) 16320; (iii) (u^8 - 1)/(u^8 (u - 1)); (iv) 29524/2187; (v) n - 1/9 + 10^{-n}/9; (vi) 70(10^n - 1)/81 - 7n/9

Q2. Sum to n terms, the series. (i) 1 + (x + y) + (x^2 + x y + y^2) + (x^3 + x^2 y + x y^2 + y^3) + ... (ii) r + (1 + v) r^2 + (1 + v + v^2) r^3 + ...

(i) The k-th term (k = 0, 1, ...) is (x^{k+1} - y^{k+1})/(x - y). Summing to n terms: S_n = (1/(x - y))[x(1 + x + ... + x^{n-1}) - y(1 + y + ... + y^{n-1})] = [x(x^n - 1)/(x - 1) - y(y^n - 1)/(y - 1)]/(x - y).

(ii) The k-th term (k = 1, 2, ...) is (1 + v + ... + v^{k-1}) r^k = ((1 - v^k)/(1 - v)) r^k. Summing to n terms: S_n = (1/(1 - v))[r(1 - r^n)/(1 - r) - v r(1 - (v r)^n)/(1 - v r)].

(i) [x(x^n - 1)/(x - 1) - y(y^n - 1)/(y - 1)]/(x - y); (ii) [r(1 - r^n)/(1 - r) - v r(1 - (v r)^n)/(1 - v r)]/(1 - v)

Q3. If a_n = (1/4)^n then find its sum up to n terms.

The series is 1/4 + (1/4)^2 + ... + (1/4)^n, a G.P. with a = 1/4, r = 1/4.

Sum = (1/4)(1 - (1/4)^n)/(1 - 1/4) = (1/4)(1 - 4^{-n})/(3/4) = (1/3)(1 - 4^{-n}).

(1/3)(1 - 4^{-n})

Q4. Find the sums of the following infinite geometric series. (i) 15 + sqrt(15) + 1 + ... (ii) 1 + 3/2 + 9/4 + ... (iii) a_1 = 729, r = 1/3; (iv) a_1 = 1/128, r = -1/2.
Use S_inf = a/(1 - r), valid only when |r| < 1.

(i) a = 15, r = sqrt(15)/15 = 1/sqrt(15). S_inf = 15/(1 - 1/sqrt(15)) = 15 sqrt(15)/(sqrt(15) - 1) = 15(15 + sqrt(15))/14.

(ii) r = 3/2, and |r| >= 1, so the infinite series diverges; it has no finite sum.

(iii) S_inf = 729/(1 - 1/3) = 729/(2/3) = 2187/2.

(iv) S_inf = (1/128)/(1 + 1/2) = (1/128)/(3/2) = 1/192.

(i) 15(15 + sqrt(15))/14; (ii) Divergent (no finite sum); (iii) 2187/2; (iv) 1/192

Q5. Find the vulgar fraction equivalent to the following recurring decimals (dots verified at 300 dpi: the repetend is the block under the dots). (i) 1.5454... (ii) 0.267267... (iii) 1.14848... (iv) 1.6666... (v) 3.26767... (vi) 1.45757...

Write each recurring decimal as a geometric series and sum it, or use the standard rule: repetend over as many 9s, with non-repeating part adjusted.

(i) 1.5454... = 1 + 54/99 = 1 + 6/11 = 17/11.

(ii) 0.267267... = 267/999 = 89/333.

(iii) 1.14848... = 1.1 + 48/990 = 11/10 + 8/165 = 379/330.

(iv) 1.6666... = 1 + 6/9 = 5/3.

(v) 3.26767... = 3.2 + 67/990 = 16/5 + 67/990 = 647/198.

(vi) 1.45757... = 1.4 + 57/990 = 7/5 + 57/990 = 481/330.

(i) 17/11; (ii) 89/333; (iii) 379/330; (iv) 5/3; (v) 647/198; (vi) 481/330

Q6. If x = y/2 + y^2/4 + y^3/8 + ... if 0 < y < 2, prove that y = 2x/(1 + x).

The series is geometric with first term y/2 and ratio y/2. Since 0 < y < 2, |y/2| < 1, so it converges.

Its sum is x = (y/2)/(1 - y/2) = y/(2 - y).

Then x(2 - y) = y, i.e. 2x - xy = y, so 2x = y(1 + x), giving y = 2x/(1 + x).

y = 2x/(1 + x) is proved

Q7. If x = 3y/2 + 9y^2/4 + 27y^3/8 + ... if 0 < y < 2/3, then prove that y = 2x/[3(1 + x)].

The series is geometric with first term 3y/2 and ratio 3y/2. Since 0 < y < 2/3, |3y/2| < 1, so it converges.

Its sum is x = (3y/2)/(1 - 3y/2) = 3y/(2 - 3y).

Then x(2 - 3y) = 3y, i.e. 2x - 3xy = 3y, so 2x = 3y(1 + x), giving y = 2x/[3(1 + x)].

y = 2x/[3(1 + x)] is proved

Q8. The sum of an infinite geometric series is one third the sum of the squares of its terms. If the sum of its first two terms is 3/4, find the series.

Let the first term be a and the ratio be r, |r| < 1. Sum of the series = a/(1 - r).

The squares form a G.P. with first term a^2 and ratio r^2; their sum = a^2/(1 - r^2).

Given: a/(1 - r) = (1/3) a^2/(1 - r^2) = (1/3) a^2/[(1 - r)(1 + r)]. Cancelling a/(1 - r) (non-zero) gives 1 = a/[3(1 + r)], so a = 3(1 + r).

Also a + ar = a(1 + r) = 3/4. Substituting a: 3(1 + r)^2 = 3/4, so (1 + r)^2 = 1/4, 1 + r = +-1/2.

r = -3/2 is rejected (|r| >= 1, the infinite sum would not exist). Hence r = -1/2 and a = 3(1/2) = 3/2.

The series is 3/2 - 3/4 + 3/8 - ... Check: sum = (3/2)/(3/2) = 1, and sum of squares = (9/4)/(3/4) = 3 = 3(1).

The series is 3/2 - 3/4 + 3/8 - ... (a = 3/2, r = -1/2)

Exercise 3.7

Q1. Find the n-th term of the H.P. (i) 1/3, 1/7, 1/11, ... (ii) 9/5, 9/13, 9/21, ...

For an H.P., the reciprocals of the terms form an A.P.; find the n-th term of that A.P. and take its reciprocal.

(i) Reciprocals: 3, 7, 11, ... with a = 3, d = 4. The n-th reciprocal is 3 + (n - 1)(4) = 4n - 1. Hence the n-th term of the H.P. is 1/(4n - 1).

(ii) Reciprocals: 5/9, 13/9, 21/9, ... with a = 5/9, d = 8/9. The n-th reciprocal is 5/9 + (n - 1)(8/9) = (8n - 3)/9. Hence the n-th term of the H.P. is 9/(8n - 3).

(i) 1/(4n - 1); (ii) 9/(8n - 3)

Q2. Find the 10th terms of the following harmonic sequences. (i) 1/3, 1/8, 1/13, ... (ii) 1/3, 2/9, 1/6, ...

(i) Reciprocals: 3, 8, 13, ... with a = 3, d = 5. The 10th reciprocal is 3 + 9(5) = 48. Hence the 10th term is 1/48.

(ii) Reciprocals: 3, 9/2, 6, ... with a = 3, d = 3/2. The 10th reciprocal is 3 + 9(3/2) = 33/2. Hence the 10th term is 2/33.

(i) 1/48; (ii) 2/33

Q3. If the 7th and 10th terms of a H.P. are 1/3 and 5/21 respectively, find its 20th term.

Work with reciprocals, which form an A.P. The 7th reciprocal is 3 and the 10th reciprocal is 21/5.

Let the A.P. have first term a and difference d: a + 6d = 3 and a + 9d = 21/5.

Subtracting: 3d = 21/5 - 3 = 6/5, so d = 2/5. Then a = 3 - 6(2/5) = 3/5.

The 20th reciprocal is 3/5 + 19(2/5) = 41/5. Hence the 20th term of the H.P. is 5/41.

5/41

Q4. If the numbers 1/u, 1/(2u + 1) and 1/(4u - 1) are in H.P., find u.

The reciprocals u, 2u + 1, 4u - 1 must be in A.P.

So twice the middle term equals the sum of the extremes: 2(2u +

1) = u + (4u - 1).

Thus 4u + 2 = 5u - 1, giving u = 3.

u = 3

Q5. If the sum of first and sixth terms of H.P. is 31/116, find the sequence if the first term is 1/4.

Let the reciprocals form an A.P. with first term a and difference d.

The first term of the H.P. is 1/a = 1/4, so a = 4.

The sixth term of the H.P. is 1/(a + 5d) = 1/(4 + 5d). Given: 1/4 + 1/(4 + 5d) = 31/116.

So 1/(4 + 5d) = 31/116 - 29/116 = 2/116 = 1/58, giving 4 + 5d = 58 and d = 54/5.

The reciprocals are 4, 74/5, 128/5, 182/5, 236/5, 58, ... Hence the H.P. is 1/4, 5/74, 5/128, 5/182, 5/236, 1/58, ... with n-th term 1/(4 + (n - 1)(54/5)) = 5/(54n - 34).

1/4, 5/74, 5/128, 5/182, 5/236, 1/58, ... (n-th term 5/(54n - 34))

Q6. In a harmonic progression, 5th and 9th terms are 7/30 and 7/54 respectively, find the 15th term.

Work with reciprocals: the 5th reciprocal is 30/7 and the 9th reciprocal is 54/7.

Let the A.P. have first term a and difference d: a + 4d = 30/7 and a + 8d = 54/7.

Subtracting: 4d = 24/7, so d = 6/7. Then a = 30/7 - 4(6/7) = 6/7.

The 15th reciprocal is 6/7 + 14(6/7) = 90/7. Hence the 15th term of the H.P. is 7/90.

7/90

Exercise 3.8

Q1. Find H.Ms. between the following numbers. (i) 4 and 9; (ii) 3 + sqrt(2) and 3 - sqrt(2); (iii) 2 + 3i and 2 - 3i; (iv) 1/7 and 1/11.

The harmonic mean between a and b is H = 2ab/(a + b).

(i) 2(4)(9)/(4 + 9) = 72/13.

(ii) ab = (3 + sqrt(2))(3 - sqrt(2)) = 7, a + b = 6. H = 2(7)/6 = 7/3.

(iii) ab = (2 + 3i)(2 - 3i) = 13, a + b = 4. H = 2(13)/4 = 13/2.

(iv) ab = 1/77, a + b = 18/77. H = 2(1/77)/(18/77) = 2/18 = 1/9.

(i) 72/13; (ii) 7/3; (iii) 13/2; (iv) 1/9

Q2. Insert four H.Ms. between 1/9 and 1/81.

Work with reciprocals: they form an A.P. with 6 terms, first term 9, sixth term 81.

Let the difference be d: 9 + 5d = 81, so d = 72/5.

The reciprocals of the four means are 9 + 72/5 = 117/5, 189/5, 261/5, 333/5.

Taking reciprocals, the four H.Ms. are 5/117, 5/189, 5/261, 5/333.

5/117, 5/189, 5/261, 5/333

Q3. Insert six H.Ms. between 1/12 and 1/48.

Work with reciprocals: they form an A.P. with 8 terms, first term 12, eighth term 48.

Let the difference be d: 12 + 7d = 48, so d = 36/7.

The reciprocals of the six means are 120/7, 156/7, 192/7, 228/7, 264/7, 300/7.

Taking reciprocals, the six H.Ms. are 7/120, 7/156, 7/192, 7/228, 7/264, 7/300.

7/120, 7/156, 7/192, 7/228, 7/264, 7/300

Q4. Find n so that (a^{n-5} + b^{n-5})/(a^{n-6} + b^{n-6}) may be H.M. between a and b.

The H.M. between a and b is 2ab/(a + b). Put m = n - 6, so the expression is (a^{m+1} + b^{m+1})/(a^m + b^m).

Set it equal to 2ab/(a + b) and cross-multiply: (a + b)(a^{m+1} + b^{m+1}) = 2ab(a^m + b^m).

Expand: a^{m+2} + a b^{m+1} + a^{m+1} b + b^{m+2} = 2a^{m+1} b + 2a b^{m+1}.

Rearrange: a^{m+2} - a^{m+1} b - a b^{m+1} + b^{m+2} = (a - b)(a^{m+1} - b^{m+1}) = 0.

For a different from b this holds identically only when m + 1 = 0, i.e. m = -1. Hence n = 5.

Check: for n = 5 the expression is (1 + 1)/(1/a + 1/b) = 2ab/(a + b).

n = 5

Q5. Find A, G and H and show that G^2 = A x H, if (i) a = 12, b = 3; (ii) a = 2/5, b = 8/5.

Use A = (a + b)/2, G = sqrt(ab), H = 2ab/(a + b).

(i) A = (12 + 3)/2 = 15/2; G = sqrt(36) = 6; H = 2(36)/15 = 24/5. Then A x H = (15/2)(24/5) = 36 = G^2.

(ii) A = (2/5 + 8/5)/2 = 1; G = sqrt(16/25) = 4/5; H = 2(16/25)/(10/5) = 16/25. Then A x H = 16/25 = G^2.

(i) A = 15/2, G = 6, H = 24/5, and G^2 = A x H = 36; (ii) A = 1, G =

4/5, H = 16/25, and G^2 = A x H = 16/25

Review Exercise 3

Q1(i). If the difference between two consecutive terms is same then the sequence is called;

- ✓ arithmetic
- geometric
- harmonic
- none of these

Q1(ii). 3, 6, 12, ... are G.P. then its 6th term is;

- 24
- 48
- ✓ 96
- 192

Q1(iii). G.M. between 4 and 16 is;

- 2
- 4
- 96
- ✓ 8

Q1(iv). If the reciprocal of the terms of sequence are in arithmetic progression then sequence is known as;

- arithmetic
- geometric
- ✓ harmonic
- none of these

Q1(v). n-th term of A.P. is a_n =

✓ a_1 + (n - 1)d

a_1 r^{n-1}

1/(a_1 + (n - 1)d)

(a_1 + (n - 1)d)/(a_1 r^{n-1})

Q1(vi). If a = 9, b = 11 then A.M. =

5

✓ 10

15

20

Q1(vii). If (2/3)^3, (5/3)^3, (8/3)^3, ... are in A.P. then its nth term is (flag: the wording is sloppy — the cubes themselves are NOT in A.P.; the intended answer is the cube of the A.P. (2/3, 5/3, 8/3, ...) nth term, i.e. option (c)).

((5n - 1)/3)^3

((4n - 1)/3)^3

✓ ((3n - 1)/3)^3

((2n - 1)/3)^3

Q1(viii). If a = 4 - 3i, b = 4 + 3i then G.M. is;

20

15

10

✓ 5

Q1(ix). 1 + 2 + 3 + ... + 100 is;

5000

✓ 5050

5100

5150

Q2. The 8th term of an arithmetic sequence is 75 and 20th term is 39. Find 24th term.

Let the first term be a and the difference be d: a + 7d = 75 and a + 19d = 39.

Subtracting: 12d = -36, so d = -3. Then a = 75 - 7(-3) = 96.

The 24th term is a + 23d = 96 + 23(-3) = 96 - 69 = 27.

27

Q3. There are n A.Ms. between 5 and 32 such that the ratio of the third and 7th means is 7:13, find the value of n.

With n means there are n + 2 terms; the common difference is d = (32 - 5)/(n + 1) = 27/(n + 1).

The 3rd mean is 5 + 3d and the 7th mean is 5 + 7d. Given (5 + 3d)/(5 + 7d) = 7/13.

Cross-multiply: 13(5 + 3d) = 7(5 + 7d), i.e. 65 + 39d = 35 + 49d, so 10d = 30 and d = 3.

Then 27/(n + 1) = 3, so n + 1 = 9 and n = 8.

n = 8

Q4. Find the sum of all the integers between 450 and 750 which are divisible by 7.

The first such integer is 455 = 7 x 65 and the last is 749 = 7 x 107.

The count is 107 - 65 + 1 = 43 terms, in A.P. with difference 7.

Sum = (43/2)(455 + 749) = (43/2)(1204) = 43 x 602 = 25886.

25886

Q5. How many terms are there if a_1 = 4, a_n = 1/256, r = 1/2?

a_n = a_1 r^{n-1} gives (1/2)^{n-1} x 4 = 1/256.

So (1/2)^{n-1} = 1/1024 = (1/2)^{10}, giving n - 1 = 10 and n = 11.

11 terms

Q6. The yearly depreciation of a certain machine is 15% of the value at the beginning of the year. If the original cost of the machine is Rs 4,000, find the value after 4 years.

Each year the value is multiplied by 1 - 0.15 = 0.85, a G.P. with ratio 0.85.

Value after 4 years = 4000 x (0.85)^4 = 4000 x 0.52200625 = 2088.025, i.e. about Rs 2088.03.

Rs 2088.03 (approximately)

Q7. Find r if S_10 = 244 S_5, for a geometric series.
S_10/S_5 = (1 - r^{10})/(1 - r^5) = 1 + r^5 = 244.

So r^5 = 243 = 3^5, giving r = 3.

r = 3

Q8. If A.M. and H.M. between two numbers are 5 and 4 1/5 respectively, find the numbers.
Let the numbers be a and b. Then (a + b)/2 = 5, so a + b = 10.

Also 2ab/(a + b) = 21/5, so 2ab/10 = 21/5, giving ab = 21.

Thus a and b are the roots of t^2 - 10t + 21 = 0, i.e. (t - 3)(t - 7) = 0. The numbers are 3 and 7.

3 and 7

Q9. If G.M. and H.M. between two numbers are 15 and 5 10/13 respectively, find the numbers.
Let the numbers be a and b. Then sqrt(ab) = 15, so ab = 225.

Also 2ab/(a + b) = 75/13, so 450/(a + b) = 75/13, giving a + b =

78.

Thus a and b are the roots of t^2 - 78t + 225 = 0. Discriminant = 78^2 - 4(225) = 5184 = 72^2.

So the roots are (78 +- 72)/2 = 75 and 3. The numbers are 75 and 3.

75 and 3

Q10. The second term of an H.P. is 1/2 and the fifth term is -1/4. Find the 12th term.

Work with reciprocals, which form an A.P. The 2nd reciprocal is 2 and the 5th reciprocal is -4.

Let the A.P. have first term a and difference d: a + d = 2 and a + 4d = -4.

Subtracting: 3d = -6, so d = -2 and a = 4.

The 12th reciprocal is 4 + 11(-2) = -18. Hence the 12th term of the H.P. is -1/18.

-1/18

Unit 4: Miscellaneous Series

Exercise 4.1

Method used below: write the k-th term as a polynomial in k, then use sum k = n(n+1)/2, sum k^2 = n(n+1)(2n+1)/6, sum k^3 = [n(n+1)/2]^2. Each answer is checked for n = 1 (and n = 2 where shown).

Q1. 1 x 3 + 2 x 6 + 3 x 9 + ... (up to n terms).

The k-th term is k x 3k = 3k^2.

Sum = 3 x n(n + 1)(2n + 1)/6 = n(n + 1)(2n + 1)/2. Check n = 1: 1 x 2 x 3/2 = 3.

n(n + 1)(2n + 1)/2

Q2. 5 x 5 + 7 x 9 + 9 x 13 + ... (up to n terms).

The k-th term is (2k + 3)(4k + 1) = 8k^2 + 14k + 3.

Sum = 8 x n(n + 1)(2n + 1)/6 + 14 x n(n + 1)/2 + 3n = n(8n^2 + 33n + 34)/3.

Check n = 1: (8 + 33 + 34)/3 = 25 = 5 x 5; n = 2: 2(32 + 66 + 34)/3 = 88 = 25 + 63.

n(8n^2 + 33n + 34)/3

Q3. 1 x 5 + 2 x 8 + 3 x 11 + ... (up to n terms).

The k-th term is k(3k + 2) = 3k^2 + 2k.

Sum = 3 x n(n + 1)(2n + 1)/6 + 2 x n(n + 1)/2 = n(n + 1)(2n + 1)/2 + n(n + 1) = n(n + 1)(2n + 3)/2.

Check n = 1: 1 x 2 x 5/2 = 5.

n(n + 1)(2n + 3)/2

Q4. 3 x 4 x 7 + 4 x 6 x 10 + 5 x 8 x 13 + ... (up to n terms).

The k-th term is (k + 2)(2k + 2)(3k + 4) = 2(k + 1)(k + 2)(3k + 4) = 6k^3 + 26k^2 + 36k + 16.

Sum = 6[n(n + 1)/2]^2 + 26 x n(n + 1)(2n + 1)/6 + 36 x n(n + 1)/2 + 16n.

= (3/2)n^2(n + 1)^2 + (13/3)n(n + 1)(2n + 1) + 18n(n + 1) + 16n.

Check n = 1: 6 + 26 + 36 + 16 = 84 = 3 x 4 x 7.

(3/2)n^2(n + 1)^2 + (13/3)n(n + 1)(2n + 1) + 18n(n + 1) + 16n

Q5. 1 x 4 x 6 + 4 x 7 x 10 + 7 x 10 x 14 + ... (up to n terms).

The k-th term is (3k - 2)(3k + 1)(4k + 2) = 36k^3 + 6k^2 - 14k - 4.

Sum = 36[n(n + 1)/2]^2 + 6 x n(n + 1)(2n + 1)/6 - 14 x n(n + 1)/2 - 4n.

= 9n^2(n + 1)^2 + n(n + 1)(2n + 1) - 7n(n + 1) - 4n. Check n = 1: 36 + 6 - 14 - 4 = 24 = 1 x 4 x 6.

9n^2(n + 1)^2 + n(n + 1)(2n + 1) - 7n(n + 1) - 4n

Q6. 1^2 + 4^2 + 7^2 + ... (up to n terms).

The k-th term is (3k - 2)^2 = 9k^2 - 12k + 4.

Sum = 9 x n(n + 1)(2n + 1)/6 - 12 x n(n + 1)/2 + 4n = (3/2)n(n + 1)(2n + 1) - 6n(n + 1) + 4n.

Check n = 1: 9 - 12 + 4 = 1.

(3/2)n(n + 1)(2n + 1) - 6n(n + 1) + 4n

Q7. 3^2 + 6^2 + 9^2 + ... (up to n terms).

The k-th term is (3k)^2 = 9k^2.

Sum = 9 x n(n + 1)(2n + 1)/6 = (3/2)n(n + 1)(2n + 1). Check n = 1: (3/2)(2)(3) = 9.

(3/2)n(n + 1)(2n + 1)

Q8. 1^3 + 4^3 + 7^3 + ... (up to n terms).

The k-th term is (3k - 2)^3 = 27k^3 - 54k^2 + 36k - 8.

Sum = 27[n(n + 1)/2]^2 - 54 x n(n + 1)(2n + 1)/6 + 36 x n(n + 1)/2 - 8n.

= (27/4)n^2(n + 1)^2 - 9n(n + 1)(2n + 1) + 18n(n + 1) - 8n. Check n = 1: 27 - 54 + 36 - 8 = 1.

(27/4)n^2(n + 1)^2 - 9n(n + 1)(2n + 1) + 18n(n + 1) - 8n

Q9. 3 x 1^2 + 5 x 2^2 + 7 x 3^2 + ... (up to n terms).
The k-th term is (2k + 1)k^2 = 2k^3 + k^2.

Sum = 2[n(n + 1)/2]^2 + n(n + 1)(2n + 1)/6 = n(n + 1)(3n^2 + 5n + 1)/6. Check n = 1: 1 x 2 x 9/6 = 3.

n(n + 1)(3n^2 + 5n + 1)/6

Q10. 5 x 2^2 + 7 x 3^2 + 9 x 4^2 + ... (up to n terms).
The k-th term is (2k + 3)(k + 1)^2 = 2k^3 + 7k^2 + 8k + 3.

Sum = 2[n(n + 1)/2]^2 + 7 x n(n + 1)(2n + 1)/6 + 8 x n(n + 1)/2 + 3n.

= n^2(n + 1)^2/2 + (7/6)n(n + 1)(2n + 1) + 4n(n + 1) + 3n. Check n = 1: 2 + 7 + 8 + 3 = 20 = 5 x 4.

n^2(n + 1)^2/2 + (7/6)n(n + 1)(2n + 1) + 4n(n + 1) + 3n

Q11. 1 + (1 + 2) + (1 + 2 + 3) + ... (up to n terms).
The k-th term is 1 + 2 + ... + k = k(k + 1)/2.

Sum = (1/2)[sum k^2 + sum k] = (1/2)[n(n + 1)(2n + 1)/6 + n(n + 1)/2] = n(n + 1)(n + 2)/6.

Check n = 2: 2 x 3 x 4/6 = 4 = 1 + 3.

n(n + 1)(n + 2)/6

Q12. 2 + (2 + 5) + (2 + 5 + 8) + ... (up to n terms).

The k-th bracket is the sum of the A.P. 2, 5, 8, ... to k terms: (k/2)[4 + 3(k - 1)] = k(3k + 1)/2.

Sum = (1/2)[3 sum k^2 + sum k] = (1/2)[3 x n(n + 1)(2n + 1)/6 + n(n + 1)/2] = n(n + 1)^2/2.

Check n = 2: 2 x 9/2 = 9 = 2 + 7.

n(n + 1)^2/2

Exercise 4.2

Sum of n terms and infinite terms of the following. For an arithmetico-geometric series with first term a, difference d and ratio r: S_n = a/(1 - r) + dr/(1 - r)^2 - dr^n/(1 - r)^2 - [a + (n - 1)d]r^n/(1 - r); and S_inf = a/(1 - r) + dr/(1 - r)^2 when |r| < 1.

Q1. 1 + 2.3 + 3.3^2 + 4.3^3 + ... (i.e. 1 + 2 x 3 + 3 x 3^2 + 4 x 3^3 + ...). Sum of n terms and infinite terms.

Here a = 1, d = 1, r = 3.

S_n = 1/(1 - 3) + 3/(1 - 3)^2 - 3^n/(1 - 3)^2 - n x 3^n/(1 - 3) = -1/2

+ 3/4 - 3^n/4 + n x 3^n/2 = [1 + (2n - 1)3^n]/4.

Check n = 1: [1 + 3]/4 = 1; n = 2: [1 + 27]/4 = 7 = 1 + 6.

Since |r| = 3 >= 1, the infinite sum does not exist (the series diverges).

S_n = [1 + (2n - 1)3^n]/4; the infinite sum diverges (r = 3)

Q2. 1 + 6x + 11x^2 + 16x^3 + ... Sum of n terms and infinite terms.
Here a = 1, d = 5, r = x.

S_n = 1/(1 - x) + 5x/(1 - x)^2 - 5x^n/(1 - x)^2 - [1 + 5(n - 1)]x^n/(1 - x).

Combining over (1 - x)^2: S_n = [1 + 4x - x^n(5 + (5n - 4)(1 - x))]/(1 - x)^2.

Check n = 1: the bracket is 1 + 4x - x(6 - x) = (1 - x)^2, so S_1 = 1.

For |x| < 1, S_inf = 1/(1 - x) + 5x/(1 - x)^2 = (1 + 4x)/(1 - x)^2.

S_n = [1 + 4x - x^n(5 + (5n - 4)(1 - x))]/(1 - x)^2; S_inf = (1 + 4x)/(1 - x)^2 for |x| < 1

Q3. 1 + 4/5 + 7/25 + 10/125 + ... Sum of n terms and infinite terms.

Here a = 1, d = 3, r = 1/5.

S_inf = 1/(1 - 1/5) + 3(1/5)/(1 - 1/5)^2 = 5/4 + 15/16 = 35/16.

S_n = 35/16 - (1/5)^n[3/(1 - 1/5)^2 + (3n - 2)/(1 - 1/5)] = 35/16 - (1/5)^n(60n + 35)/16 = [35 - (60n + 35)/5^n]/16.

Check n = 1: [35 - 95/5]/16 = 16/16 = 1.

S_n = [35 - (60n + 35)/5^n]/16; S_inf = 35/16

Q4. 1 + 7/4 + 13/4^2 + 19/4^3 + ... Sum of n terms and infinite terms.

Here a = 1, d = 6, r = 1/4.

S_inf = 1/(1 - 1/4) + 6(1/4)/(1 - 1/4)^2 = 4/3 + 8/3 = 4.

S_n = 4 - (1/4)^n[6/(1 - 1/4)^2 + (6n - 5)/(1 - 1/4)] = 4 - (1/4)^n(24n + 12)/3 = 4 - (8n + 4)/4^n.

Check n = 1: 4 - 12/4 = 1.

S_n = 4 - (8n + 4)/4^n; S_inf = 4

Q5. 1 - 6y + 11y^2 - 16y^3 + ... Sum of n terms and infinite terms.

Write the terms as (a + kd)r^k with r = -y: the first term gives a = 1; the second gives (a + d)(-y) = -6y, so d = 5 (check: third term (1 + 10)y^2 = 11y^2; fourth (1 + 15)(-y)^3 = -16y^3).

So a = 1, d = 5, r = -y. Then S_n = 1/(1 + y) + 5(-y)/(1 + y)^2 - 5(-y)^n/(1 + y)^2 - [1 + 5(n - 1)](-y)^n/(1 + y).

For |y| < 1, S_inf = 1/(1 + y) - 5y/(1 + y)^2 = (1 - 4y)/(1 + y)^2.

S_n = 1/(1 + y) - 5y/(1 + y)^2 - 5(-y)^n/(1 + y)^2 - [1 + 5(n - 1)](-y)^n/(1 + y); S_inf = (1 - 4y)/(1 + y)^2 for |y| < 1

Q6. Find S_8 if 1 + 5/2 + 9/4 + 13/8 + ...

Here a = 1, d = 4, r = 1/2.

S_n = 1/(1 - 1/2) + 4(1/2)/(1 - 1/2)^2 - (1/2)^n[4/(1 - 1/2)^2 + (4n - 3)/(1 - 1/2)] = 10 - (8n + 10)/2^n.

Check n = 1: 10 - 18/2 = 1.

Hence S_8 = 10 - 74/256 = 10 - 37/128 = 1243/128.

S_8 = 1243/128

Q7. Find n-th term if 3, 7/3, 11/3^2, 15/3^3, ...

This is arithmetico-geometric with a = 3, d = 4, r = 1/3.

The n-th term is (a + (n - 1)d)r^{n-1} = (3 + 4(n - 1))(1/3)^{n-1} = (4n - 1)/3^{n-1}.

(4n - 1)/3^{n-1}

Q8. Find 15th term if 9, 14/5, 19/5^2, ...

This is arithmetico-geometric with a = 9, d = 5, r = 1/5.

The 15th term is (9 + 14 x 5)(1/5)^{14} = 79/5^{14}.

79/5^14

Exercise 4.3

Q1. A man deposits in a bank Rs 15,000 in the first month; Rs 16,500 in the second month; Rs 18,000 in third month and so on. Find how much he will have deposited in the bank in 12th month.

The monthly deposits form an A.P.: 15000, 16500, 18000, ... with first term 15000 and common difference 1500.

Deposit in the 12th month = 15000 + 11(1500) = 15000 + 16500 = 31500.

(For reference, the total deposited over 12 months is (12/2)[2(15000) + 11(1500)] = 6(46500) = 279000.)

Rs 31,500 deposited in the 12th month (Rs 279,000 deposited in total over 12 months)

Q2. A boy saves Rs 100 at the end of first week and goes on increasing his saving Rs 15 weekly. After some weeks, he saves Rs 1000. In how many weeks he will save Rs 1000?

The weekly savings form an A.P. with first term 100 and common

difference 15.

Let the n-th week's saving be 1000: 100 + (n - 1)(15) = 1000.

Then 15(n - 1) = 900, so n - 1 = 60 and n = 61.

In the 61st week

Q3. An object falling from rest, falls 16 meter during the first second, 32 meter during the next second, 48 meters during the third second and so on. How far will it fall during the 9th second?
The distances per second form an A.P.: 16, 32, 48, ... with first term 16 and common difference 16.

Distance during the 9th second = 16 + 8(16) = 16(9) = 144 metres.

144 metres

Q4. The sum of three numbers in A.P. is 21 and their product is 231. Find the numbers.
Let the numbers be a - d, a, a + d. Their sum is 3a = 21, so a = 7.

Their product is (7 - d)(7)(7 + d) = 7(49 - d^2) = 231, so 49 - d^2 = 33 and d^2 = 16, d = +-4.

The numbers are 3, 7, 11.

3, 7, 11

Q5. Find five numbers in A.P. whose sum is 20 and the sum of whose cubes is 440.

Let the numbers be a - 2d, a - d, a, a + d, a + 2d. Their sum is 5a = 20, so a = 4.

Sum of cubes = (a - 2d)^3 + (a - d)^3 + a^3 + (a + d)^3 + (a + 2d)^3 = 5a^3 + 30a d^2.

With a = 4: 320 + 120 d^2 = 440, so d^2 = 1 and d = +-1.

The numbers are 2, 3, 4, 5, 6.

2, 3, 4, 5, 6

Q6. The sum of Rs 21,000 is distributed among five persons so that each person after the first receives Rs 40 less than the preceding person. How much does each person receive?

The shares form a decreasing A.P. with 5 terms and common difference -40.

Sum = (5/2)[2a + 4(-40)] = 5(a - 80) = 21000, so a - 80 = 4200 and a = 4280.

The shares are 4280, 4240, 4200, 4160, 4120. Check: 5 x 4200 = 21000.

Rs 4280, Rs 4240, Rs 4200, Rs 4160, Rs 4120

Q7. Tickets for a certain show were printed bearing numbers from 1 to 150. Odd number tickets were sold by receiving rupees equal to twice of the number on the ticket while even number tickets were issued by receiving rupees equal to thrice of the number on the ticket. How much amount was received by the issuing agency?
Odd tickets: 1, 3, ..., 149 (75 tickets). Their sum is 75^2 = 5625. Amount received = 2 x 5625 = 11250.

Even tickets: 2, 4, ..., 150 (75 tickets). Their sum is 2(1 + 2 + ... + 75) = 2 x 2850 = 5700. Amount received = 3 x 5700 = 17100.

Total amount = 11250 + 17100 = 28350 rupees.

Rs 28,350

Q8. A man borrows Rs 1,450 and agree to repay with a total interest of Rs 350 in 15 installments, each installment being less than the preceding by Rs 20. What should be his first installment?
Total to repay = 1450 + 350 = 1800, in 15 installments forming an A.P. with common difference -20.

Sum = (15/2)[2a + 14(-20)] = 15(a - 140) = 1800, so a - 140 = 120 and a = 260.

The first installment is Rs 260. Flag: with a = 260 and d = -20, the 15th installment would be 260 - 280 = -20, i.e. negative — the

book's numbers (15 installments, Rs 20 steps, Rs 1800 total) cannot all hold for real installments; the answer follows the book's data as printed.

Rs 260 (flag: the book's data are inconsistent — see steps)

Q9. Find three consecutive numbers in G.P. whose sum is 21 and their product is 216.
Let the numbers be a/r, a, ar. Their product is a^3 = 216, so a = 6.

Their sum is 6/r + 6 + 6r = 21, so 1/r + 1 + r = 7/2. Multiply by 2r: 2 + 2r + 2r^2 = 7r, i.e. 2r^2 - 5r + 2 = 0.

Factor: (2r - 1)(r - 2) = 0, so r = 2 or r = 1/2. The numbers are 3, 6, 12.

3, 6, 12

Q10. The number of bacteria in a culture increased geometrically from 625,000 to 18,75,000 in 7 days. Find the daily rate of increase assuming the rate to be constant.
Let the daily growth factor be r. Over 7 days the number is multiplied by r^7: 625000 x r^7 = 1875000.

So r^7 = 3, giving r = 3^{1/7} = 1.1699... to four places.

The daily rate of increase is r - 1 = 0.1699..., i.e. about 17% per day.

Daily growth factor $3^{1/7}$, i.e. a daily rate of about 17%

Review Exercise 4

Q1(i). A series which is obtained by multiplying corresponding terms of arithmetic and geometric series is known as;

- arithmetic series
- geometric series
- ✓ arithmetic-geometric series
- harmonic series

Q1(ii). n-th term of arithmetic-geometric series is;

- $a_1 + (n - 1)d$
- $a_1 r^{n-1}$
- ✓ $[a_1 + (n - 1)d] r^{n-1}$
- $1/(a_1 + (n - 1)d)$

Q1(iii). $\sum_{k=1}^{4} (k^2 + k)$ is equal to;

- 36
- ✓ 40
- 44
- 48

Q1(iv). Sum of infinite arithmetic geometric series is obtained by the formula.

- $a/(1 - r)$
- $dr/(1 - r)^2$
- ✓ $a/(1 - r) + dr/(1 - r)^2$

a/(1 - r) + dr/(1 - r)^3

Q1(v). Sum_{x=1}^{4} x^2 =

25

✓ 30

35

40

Q1(vi). Sum_{k=1}^{n} k^2 =

n

n(n + 1)/2

✓ n(n + 1)(2n + 1)/6

[n(n + 1)/2]^2

Q1(vii). Sum of infinite term of arithmetic geometric series is

a/(1 - r)

dr/(1 - r)^2

✓ a/(1 - r) + dr/(1 - r)^2

a/(1 - r) + dr/(1 - r)^2 + dr^n/(1 - r)^2

Q1(viii). S_inf = a/(1 - r) + (flag: options (a) and (b) are printed identically, both dr/(1 - r)^2; the correct completion is dr/(1 - r)^2).

dr/(1 - r)^2

✓ dr/(1 - r)^2

dr/(1 - r^2)

d^2 r/(1 + r^2)

Q1(ix). 1/[n(n + 1)] =

1/n + 1/(n + 1)

✓ 1/n - 1/(n + 1)

2/n + 1/(n + 1)

1/n - 3/(n + 1)

Q2. Sum the series up to n terms. (i) 2 x 1^2 + 4 x 2^2 + 6 x 2^3 + ... (ii) 1 + 4/3 + 7/3^2 + ...

(i) Flag: as printed (verified at 300 dpi: 2 x 1^2 + 4 x 2^2 + 6 x 2^3 + ...), the terms 2, 16, 48, ... do not form a standard arithmetico-geometric series. The sum below follows the evident continuation 2 x 1^2 + 4 x 2^2 + 6 x 2^3 + 8 x 2^4 + ..., i.e. T_k = 2k x 2^k for k >= 2 with T_1 = 2.

Sum = 2 + sum_{k=2}^{n} k x 2^{k+1} = 2 + 2[(n - 1)2^{n+1} + 2] - 4 = (n - 1)2^{n+2} + 2. Check n = 2: 2^4 + 2 = 18 = 2 + 16; n = 3: 2 x 2^5 + 2 = 66 = 2 + 16 + 48.

(ii) Here a = 1, d = 3, r = 1/3. S_inf = 1/(2/3) + 3(1/3)/(4/9) = 15/4.

S_n = 15/4 - (1/3)^n[3/(4/9) + (3n - 2)/(2/3)] = 15/4 - (1/3)^n(18n + 15)/4 = [15 - (18n + 15)/3^n]/4. Check n = 1: [15 - 33/3]/4 = 1.

(i) S_n = (n - 1)2^{n+2} + 2, for the series as printed (flagged — see steps); (ii) S_n = [15 - (18n + 15)/3^n]/4

Unit 5: Permutation and Combination and Probability

Exercise 5.1

Q1. A student must choose 1 subject from each of 4 categories: Maths (4 options), Science (3 options), Humanities (2 options) and Art (5 options). How many different schedules are possible?

By the fundamental counting principle, multiply the numbers of options: 4 x 3 x 2 x 5 = 120.

120 schedules

Q2. P, Q and R are the vertices of a triangle. In how many ways, can we write the triangles?

The three vertices can be ordered in 3! = 6 ways (PQR, PRQ, QPR, QRP, RPQ, RQP). This is the natural reading of the question as printed.

6 ways (flag: read as the number of ways to name/order the three vertices)

Q3. Evaluate the following. (i) 8! x 6!/(7! x 5!); (ii) 12!/(7! x 8!); (iii) (6!)^2/((4!)^3 x 3!).

(i) 8! x 6!/(7! x 5!) = (8 x 7! x 6 x 5!)/(7! x 5!) = 8 x 6 = 48.

(ii) 12!/(7! x 8!) = (12 x 11 x 10 x 9)/(7 x 6 x 5 x 4 x 3 x 2 x 1) = 11880/5040 = 33/14.

(iii) (6!)^2/((4!)^3 x 3!) = 720^2/(24^3 x 6) = 518400/82944 = 25/4.

(i) 48; (ii) 33/14; (iii) 25/4

Q4. Express the following in a single fraction. (i) (n - 1)!/r! + (n + 1)!/(r - 1)!; (ii) n!/r! + (n - 1)!/(r + 1)! - (n - 2)!/(r - 1)!

(i) Write the second term over r!: (n + 1)!/(r - 1)! = r(n + 1)!/r!. Sum = [(n - 1)! + r(n + 1)!]/r!. Since (n + 1)! = (n + 1)n(n - 1)!, this is (n - 1)![1 + r n(n + 1)]/r!.

(ii) Use the common denominator (r + 1)!: n!/r! = n!(r + 1)/(r + 1)! and (n - 2)!/(r - 1)! = (n - 2)!r(r + 1)/(r + 1)!. Sum = [n!(r + 1) + (n - 1)! - (n - 2)!r(r + 1)]/(r + 1)! = (n - 2)![n(n - 1)(r + 1) + (n - 1) - r(r + 1)]/(r + 1)!.

(i) (n - 1)![1 + r n(n + 1)]/r!; (ii) (n - 2)![n(n - 1)(r + 1) + (n - 1) - r(r + 1)]/(r + 1)!

Q5. In how many ways different batting orders are possible for a cricket team consisting of 11 players?

A batting order is an arrangement of 11 distinct players: 11! = 39,916,800.

39,916,800

Q6. In how many ways can five English books, two chemistry books

and four mathematics books arranged on a shelf so that all books of same kind are together?

Treat each kind as one block: the 3 blocks can be arranged in 3! = 6 ways.

Within the blocks: English books 5! = 120 ways, chemistry books 2! = 2 ways, mathematics books 4! = 24 ways.

Total = 6 x 120 x 2 x 24 = 34,560.

34,560

Q7. How many different arrangement can be formed of the word 'NOSHKI' using all letters when no letter is to be repeated?

NOSHKI has 6 distinct letters. Arrangements = 6! = 720.

720

Q8. How many signals can be given by 7 flags of different colours when any number of them are used at a time?

Signals using k flags: P(7, k). Sum over k = 1 to 7.

P(7,1) = 7; P(7,2) = 42; P(7,3) = 210; P(7,4) = 840; P(7,5) = 2520; P(7,6) = 5040; P(7,7) = 5040.

Total = 7 + 42 + 210 + 840 + 2520 + 5040 + 5040 = 13,699.

13,699

Q9. Find the value of n in each of the following. (i) nP2 = 30; (ii) nP3 = 60; (iii) 8Pn = 8.7.6; (iv) (n+2)P4 : (n+1)P3 = 5 : 1.

(i) n(n - 1) = 30, i.e. n^2 - n - 30 = 0, i.e. (n - 6)(n + 5) = 0. So n = 6.

(ii) n(n - 1)(n - 2) = 60. Trying n = 5: 5 x 4 x 3 = 60. So n = 5.

(iii) 8Pn = 8 x 7 x 6 = 336, so 8!/(8 - n)! = 336, giving (8 - n)! = 40320/336 = 120 = 5!. Hence 8 - n = 5 and n = 3.

(iv) (n+2)P4/(n+1)P3 = (n + 2)(n + 1)n(n - 1)/((n + 1)n(n - 1)) = n + 2 = 5/1. So n = 3.

(i) n = 6; (ii) n = 5; (iii) n = 3; (iv) n = 3

Q10. How many 5-digit numbers can be formed, without repeating any digit from the digits 2, 3, 4, 5, 7, 8?

Choose and arrange 5 out of 6 distinct digits: P(6, 5) = 6 x 5 x 4 x 3 x 2 = 720.

720

Q11. How many words can be formed out of the letters of the following words, all letters taken together? (i) STATISTICS; (ii) ASSASSINATION; (iii) MATHEMATICS; (iv) ASSISTANCE.

(i) STATISTICS: 10 letters with S x 3, T x 3, I x 2. Count = 10!/(3! x 3! x 2!) = 3628800/72 = 50,400.

(ii) ASSASSINATION: 13 letters with A x 3, S x 4, I x 2, N x 2. Count = 13!/(3! x 4! x 2! x 2!) = 6227020800/576 = 10,810,800.

(iii) MATHEMATICS: 11 letters with M x 2, A x 2, T x 2. Count = 11!/(2! x 2! x 2!) = 39916800/8 = 4,989,600.

(iv) ASSISTANCE: 10 letters with A x 2, S x 3. Count = 10!/(2! x 3!) = 3628800/12 = 302,400.

(i) 50,400; (ii) 10,810,800; (iii) 4,989,600; (iv) 302,400

Q12. How many 8 digit numbers can be formed from the digits 1, 1, 1, 3, 3, 3, 5, 6? How many among these are greater than 50000000?

Total: 8 digits with 1 repeated 3 times and 3 repeated 3 times: 8!/(3! x 3!) = 40320/36 = 1120.

Greater than 50,000,000 means the first digit is 5 or 6. First digit 5: remaining digits 1, 1, 1, 3, 3, 3, 6 give 7!/(3! x 3!) = 140. First digit 6: remaining digits 1, 1, 1, 3, 3, 3, 5 give 140. Total greater = 280.

1120 numbers in total; 280 of them are greater than 50,000,000

Q13. How many distinct permutations of letters of the word 'ESSENTIAL' are possible? How many will have the two S's (i) together (ii) separate. (Flag: ESSENTIAL has three S's; 'two S's' is read as the S's as a group — all three together, no two adjacent,

and exactly two together are all given.)

ESSENTIAL: 9 letters with E x 2, S x 3. Total = 9!/(2! x 3!) = 362880/12 = 30,240.

(i) All three S's together: treat SSS as one block; 7 objects with E x 2 give 7!/2! = 2,520.

(ii) No two S's adjacent: arrange the other 6 letters (E x 2, N, T, I, A, L) in 6!/2! = 360 ways; the 3 S's go into 3 of the 7 gaps: C(7, 3) = 35. Count = 360 x 35 = 12,600.

Exactly two S's together (for completeness) = 30240 - 2520 - 12600 = 15,120.

30,240 in total; all three S's together: 2,520; no two S's together: 12,600; exactly two S's together: 15,120

Q14. 14 members of a club form 3 committees of 6, 5, 3 members so that no member is a member of more than one committee. Find the number of committees.

Choose 6 for the first committee, then 5 of the remaining 8 for the second, then 3 of the remaining 3: C(14, 6) x C(8, 5) x C(3, 3) = 3003 x 56 x 1 = 168,168. (Equivalently 14!/(6! x 5! x 3!).)

168,168 ways

Q15. In how many ways can 7 persons be seated at a round table?

Circular arrangements of 7 distinct persons: (7 - 1)! = 720.

720

Q16. In how many ways 6 keys be arranged on a circular key ring?
On a key ring, arrangements that differ only by flipping the ring over are the same, so divide the circular count by 2: (6 - 1)!/2 = 120/2 = 60. (If flipping were not identified, the count would be (6 - 1)! = 120.)

60 (flag: key-ring convention — see steps)

Exercise 5.2

Q1. Find the value of n when (i) nC2 = 21; (ii) nC12 = nC8; (iii) nC15 = (17 x 16)/2!.
(i) n(n - 1)/2 = 21, so n^2 - n - 42 = 0, i.e. (n - 7)(n + 6) = 0. Hence n = 7.

(ii) By symmetry nCr = nC(n - r): n - 12 = 8 gives n = 20.

(iii) (17 x 16)/2! = 136. Now nC15 = 136 with n = 17 gives 17C15 = 17C2 = 136. Hence n = 17.

(i) n = 7; (ii) n = 20; (iii) n = 17

Q2. Find the values of n and r, when (i) nPr = 840, nCr = 35; (ii) (n-1)C(r-1) : nCr : (n+1)C(r+1) = 3 : 6 : 11.
(i) nPr/nCr = r! = 840/35 = 24, so r = 4. Then nP4 = n(n - 1)(n - 2)(n - 3) = 840; trying n = 7: 7 x 6 x 5 x 4 = 840. So n = 7, r = 4.

(ii) (n-1)C(r-1)/nCr = r/n = 3/6 = 1/2, so n = 2r. Also (n+1)C(r+1)/nCr = (n + 1)/(r + 1) = 11/6. Substituting n = 2r: (2r + 1)/(r + 1) = 11/6, so 12r + 6 = 11r + 11, giving r = 5 and n = 10.

(i) n = 7, r = 4; (ii) n = 10, r = 5

Q3. The members of a club are 10 boys and 6 girls. In how many ways can a committee of 4 boys and 3 girls be formed?

Choose 4 boys from 10: C(10, 4) = 210. Choose 3 girls from 6: C(6, 3) = 20.

Total = 210 x 20 = 4,200.

4,200

Q4. A box contains 8 red balls and 5 black balls. In how many ways can 5 balls be selected such that exactly 3 are black?

Choose 3 black from 5: C(5, 3) = 10. Choose the other 2 from the 8 red: C(8, 2) = 28.

Total = 10 x 28 = 280.

280

Q5. How many diagonals can be drawn in a plane figure of 10 sides?

Joining 2 of the 10 vertices gives C(10, 2) = 45 segments; 10 of

these are the sides.

Diagonals = 45 - 10 = 35.

35

Q6. Show that: 9C6 + 9C5 = 10C6.

By Pascal's identity nCr + nC(r - 1) = (n + 1)Cr with n = 9, r = 6:

9C6 + 9C5 = 10C6.

Numerically: 84 + 126 = 210 = 10C6.

9C6 + 9C5 = 10C6 is proved

Q7. There are 10 men and 7 women members of a club. How many committees of 6 members can be formed, having: (i) Exactly three men; (ii) at most three women; (iii) at least five women.

(i) Exactly 3 men (and 3 women): C(10, 3) x C(7, 3) = 120 x 35 = 4,200.

(ii) At most 3 women: sum over 0, 1, 2, 3 women: C(10, 6)C(7, 0) + C(10, 5)C(7, 1) + C(10, 4)C(7, 2) + C(10, 3)C(7, 3) = 210 + 1764 + 4410 + 4200 = 10,584.

(iii) At least 5 women: 5 women + 1 man, or 6 women: C(10, 1)C(7, 5) + C(10, 0)C(7, 6) = 210 + 7 = 217.

(i) 4,200; (ii) 10,584; (iii) 217

Q8. A secure system uses passwords that are 4 characters long, using uppercase English letters (A-Z). No character repeats and order matters. How many different passwords are possible?

Order matters and letters do not repeat: P(26, 4) = 26 x 25 x 24 x 23 = 358,800.

358,800

Q9. A lottery game, 6 numbers are drawn from a set of 49. The order does not matter. What is are odds of choosing the correct 6-number combination?

The number of possible combinations is C(49, 6) = 49!/(6! x 43!) = 13,983,816.

Only one combination is correct, so the odds are 1 in 13,983,816.

1 in 13,983,816

Q10. A DJ has 15 songs and wan to play 4 during a set. The order in which songs are played matters. How many different play lists are possible?

Order matters: P(15, 4) = 15 x 14 x 13 x 12 = 32,760.

32,760

Exercise 5.3

Q1. A bag contains 30 balls out of which 6 are green, 10 are red

and the remaining are black. A ball is drawn out of them. Find the probability of getting: (i) the ball is green or black; (ii) the ball is red or black; (iii) the ball is green or red.

Black balls = 30 - 6 - 10 = 14. Total = 30.

(i) Green or black: (6 + 14)/30 = 20/30 = 2/3.

(ii) Red or black: (10 + 14)/30 = 24/30 = 4/5.

(iii) Green or red: (6 + 10)/30 = 16/30 = 8/15.

(i) 2/3; (ii) 4/5; (iii) 8/15

Q2. Two fair dice are rolled simultaneously. Find the probability that the sum of dots are: (i) 4 or 7; (ii) less than 5 or greater than 9.

There are 36 equally likely outcomes.

(i) Sum 4: (1,3), (2,2), (3,1) — 3 outcomes. Sum 7: 6 outcomes. Total 9. Probability = 9/36 = 1/4.

(ii) Sums less than 5: 2, 3, 4 — 1 + 2 + 3 = 6 outcomes. Sums greater than 9: 10, 11, 12 — 3 + 2 + 1 = 6 outcomes. Total 12. Probability = 12/36 = 1/3.

(i) 1/4; (ii) 1/3

Q3. In a three child family, what is the probability (using sample space): (i) three girls; (ii) exactly two boys; (iii) at least one girl?

The sample space has 8 outcomes: BBB, BBG, BGB, BGG, GBB, GBG, GGB, GGG.

(i) Three girls: GGG only — 1/8.

(ii) Exactly two boys: BBG, BGB, GBB — 3/8.

(iii) At least one girl: all except BBB — 7/8.

(i) 1/8; (ii) 3/8; (iii) 7/8

Q4. A die is rolled. Find the probability (using Venn diagram): (i) an odd number; (ii) a number greater than 3; (iii) a number which is odd and greater than 3.

Sample space: {1, 2, 3, 4, 5, 6}.

(i) Odd numbers {1, 3, 5}: 3/6 = 1/2.

(ii) Numbers greater than 3, {4, 5, 6}: 3/6 = 1/2.

(iii) Odd and greater than 3: {5} only: 1/6.

(i) 1/2; (ii) 1/2; (iii) 1/6

Q5. In a single throw of two fair dice, find the probability (using possibility diagram) that the product of the number on dice is: (i) between 3 and 12 (both inclusive); (ii) divisible by 3.

(i) Count products from 3 to 12 in the 6 x 6 table: first die 1 gives 4

(3, 4, 5, 6); 2 gives 5 (4, 6, 8, 10, 12); 3 gives 4 (3, 6, 9, 12); 4 gives 3 (4, 8, 12); 5 gives 2 (5, 10); 6 gives 2 (6, 12). Total 20. Probability = 20/36 = 5/9.

(ii) The product is divisible by 3 unless neither die shows 3 or 6: 1 - (4/6)^2 = 1 - 16/36 = 20/36 = 5/9.

(i) 5/9; (ii) 5/9

Q6. A marble is drawn at random from a box containing 30 red, 10 white, 20 orange and 15 blue marbles. Find the probability that it is: (i) orange or red; (ii) blue or red.
Total marbles = 30 + 10 + 20 + 15 = 75.

(i) Orange or red: (20 + 30)/75 = 50/75 = 2/3.

(ii) Blue or red: (15 + 30)/75 = 45/75 = 3/5.

(i) 2/3; (ii) 3/5

Q7. A card is drawn from a standard deck. What is the probability of drawing a king or a queen?
Kings: 4, queens: 4, total cards 52. Probability = (4 + 4)/52 = 8/52 = 2/13.

2/13

Q8. There is a 30% chance of rain and 20% chance of snow on a

certain day, and both cannot happen together. What is the probability of either rain or snow?

The events cannot happen together (mutually exclusive), so P(rain or snow) = P(rain) + P(snow) = 0.30 + 0.20 = 0.50 = 1/2.

1/2

Q9. A fast-food restaurant offers 3 meal type: burger 40%, pizza 30% or sandwich 30%. What is the probability a customer order burger or pizza?

The choices are mutually exclusive: P(burger or pizza) = 40% + 30% = 70%.

70%

Q10. A bag has 5 red and 3 black marbles. Find the probability of drawing a red then black, with replacement.

With replacement the draws are independent: P(red) = 5/8, P(black) = 3/8.

P(red then black) = (5/8)(3/8) = 15/64.

15/64

Q11. I have 4 red and 6 blue beads. What is the probability of choosing a red bead then a blue bead without replacement?

P(red first) = 4/10. After removing a red bead, 9 beads remain with 6 blue: P(blue second) = 6/9.

P(red then blue) = (4/10)(6/9) = 24/90 = 4/15.

4/15

Review Exercise 5

Q1(i). If 'n' is the total objects and 'r' are the selected objects then nCr can be calculated by;

- n!/(n - r)!
- n!/r!
- ✓ n!/(r!(n - r)!)
- none of these

Q1(ii). A number between 0 and 1 that is used to measure uncertainty is called

- random variable
- trial
- simple event
- ✓ probability

Q1(iii). A set of all possible outcomes of an experiment is called

- sample point
- ✓ sample space
- combination
- compound event

Q1(iv). An arrangement of objects regarding to a definite order is called

multiplication rule

combination

✓ permutation

sample point

Q1(v). The value of P(n, n - 1) is

n

✓ n!

2n

2n!

Q1(vi). The number of ways in which 8 students can be seated in a line is

5040

50400

40230

✓ 40320

Q1(vii). If nP3 = 12 x (n-1)P3, the value of n is (flag: as printed this equation has no integer solution — n(n-1)(n-2) = 12(n-1)(n-2)(n-3) gives n = 12(n-3), i.e. n = 36/11. The intended equation is presumably nP3 = 12 x (n-1)P2, which gives n = 12, option (c).)

6

10

✓ 12

16

Q1(viii). The number of ways 4 boys and 3 girls can be seated in a

row so that they are alternate is;

12

104

✓ 144

256

Q1(ix). In how many ways 8 distinct toys can be distributed among 5 children?

8P5

5P8

✓ 5^8

8^5

Q2. Evaluate the following. (i) (6C3 x 4C3)/10C4; (ii) (4C1 x 48C9)/52C4.

(i) 6C3 = 20, 4C3 = 4, 10C4 = 210. Value = (20 x 4)/210 = 80/210 = 8/21.

(ii) 4C1 = 4, 48C9 = 1677106640, 52C4 = 270725. Value = (4 x 1677106640)/270725 = 1341685312/54145, about 24779.49. (Note: this is the value of the expression as printed; it is not a probability.)

(i) 8/21; (ii) 1341685312/54145 (about 24779.49)

Q3. There are six seats available in a compartment. In how many ways can six persons be seated?

Six distinct persons in six distinct seats: 6! = 720.

720

Unit 6: Mathematical Induction and Binomial Theorem

Exercise 6.1

Q1. Use principle of mathematical induction to prove for positive integer n: 1 + 4 + 7 + ... + (3n - 2) = n(3n - 1)/2.

Base n = 1: LHS = 1, RHS = 1(3 - 1)/2 = 1. True.

Assume true for n = k: 1 + 4 + ... + (3k - 2) = k(3k - 1)/2.

For n = k + 1, add the next term (3k + 1): k(3k - 1)/2 + (3k + 1) = (3k^2 - k + 6k + 2)/2 = (3k^2 + 5k + 2)/2 = (k + 1)(3k + 2)/2 = (k + 1)(3(k + 1) - 1)/2. This is the formula for n = k + 1. Hence proved.

Proved by induction

Q2. Prove by induction: 2 + 4 + 6 + ... + 2n = n(n + 1).

Base n = 1: LHS = 2, RHS = 1 x 2 = 2. True.

Assume true for n = k: sum = k(k + 1).

For n = k + 1, add 2(k + 1): k(k + 1) + 2(k + 1) = (k + 1)(k + 2). This is the formula for n = k + 1. Hence proved.

Proved by induction

Q3. Prove by induction: 1 + 3 + 9 + ... + 3^(n-1) = (3^n - 1)/2.

Base n = 1: LHS = 1, RHS = (3 - 1)/2 = 1. True.

Assume true for n = k: sum = (3^k - 1)/2.

For n = k + 1, add 3^k: (3^k - 1)/2 + 3^k = (3 x 3^k - 1)/2 = (3^(k+1) - 1)/2. Hence proved.

Proved by induction

Q4. Prove by induction: 5 + 15 + 45 + ... + 5 x 3^(n-1) = 5(3^n - 1)/2.

Base n = 1: LHS = 5, RHS = 5(3 - 1)/2 = 5. True.

Assume true for n = k: sum = 5(3^k - 1)/2.

For n = k + 1, add 5 x 3^k: 5(3^k - 1)/2 + 5 x 3^k = 5(3 x 3^k - 1)/2 = 5(3^(k+1) - 1)/2. Hence proved.

Proved by induction

Q5. Prove by induction: 1 x 3 + 2 x 5 + 3 x 7 + ... + n(2n + 1) = n(n + 1)(4n + 5)/6.

Base n = 1: LHS = 3, RHS = 1 x 2 x 9/6 = 3. True.

Assume true for n = k. For n = k + 1, add (k + 1)(2k + 3): [k(k + 1)(4k + 5) + 6(k + 1)(2k + 3)]/6 = (k + 1)[k(4k + 5) + 6(2k + 3)]/6 = (k + 1)(4k^2 + 17k + 18)/6 = (k + 1)(k + 2)(4k + 9)/6 = (k +

1)(k + 2)(4(k + 1) + 5)/6. Hence proved.

Proved by induction

Q6. Prove by induction: 1 x 2 + 3 x 4 + 5 x 6 + ... + 2n(2n - 1) = n(n + 1)(4n - 1)/3.
Base n = 1: LHS = 2, RHS = 1 x 2 x 3/3 = 2. True.

Assume true for n = k. For n = k + 1, add 2(k + 1)(2k + 1): [k(k + 1)(4k - 1) + 6(k + 1)(2k + 1)]/3 = (k + 1)[k(4k - 1) + 6(2k + 1)]/3 = (k + 1)(4k^2 + 11k + 6)/3 = (k + 1)(k + 2)(4k + 3)/3 = (k + 1)(k + 2)(4(k + 1) - 1)/3. Hence proved.

Proved by induction

Q7. Prove by induction: 1/(1 x 2) + 1/(2 x 3) + 1/(3 x 4) + ... + 1/(n(n + 1)) = 1 - 1/(n + 1).
Base n = 1: LHS = 1/2, RHS = 1 - 1/2 = 1/2. True.

Assume true for n = k. For n = k + 1, add 1/((k + 1)(k + 2)): 1 - 1/(k + 1) + 1/((k + 1)(k + 2)) = 1 - [(k + 2) - 1]/((k + 1)(k + 2)) = 1 - 1/(k + 2). Hence proved.

Proved by induction

Q8. Prove by induction: 1^2 + 2^2 + 3^2 + ... + n^2 = n(n + 1)(2n + 1)/6.
Base n = 1: LHS = 1, RHS = 1 x 2 x 3/6 = 1. True.

Assume true for n = k. For n = k + 1, add (k + 1)^2: [k(k + 1)(2k + 1) + 6(k + 1)^2]/6 = (k + 1)[k(2k + 1) + 6(k + 1)]/6 = (k + 1)(2k^2 + 7k + 6)/6 = (k + 1)(k + 2)(2k + 3)/6 = (k + 1)(k + 2)(2(k + 1) + 1)/6. Hence proved.

Proved by induction

Q9. Prove by induction: 1/(1 x 2 x 3) + 1/(2 x 3 x 4) + 1/(3 x 4 x 5) + ... + 1/(n(n + 1)(n + 2)) = n(n + 3)/(4(n + 1)(n + 2)).
Base n = 1: LHS = 1/6, RHS = 1 x 4/(4 x 2 x 3) = 1/6. True.

Assume true for n = k. For n = k + 1, add 1/((k + 1)(k + 2)(k + 3)): k(k + 3)/(4(k + 1)(k + 2)) + 1/((k + 1)(k + 2)(k + 3)) = [k(k + 3)^2 + 4]/(4(k + 1)(k + 2)(k + 3)) = (k^3 + 6k^2 + 9k + 4)/(4(k + 1)(k + 2)(k + 3)) = (k + 1)^2(k + 4)/(4(k + 1)(k + 2)(k + 3)) = (k + 1)(k + 4)/(4(k + 2)(k + 3)) = (k + 1)((k + 1) + 3)/(4((k + 1) + 1)((k + 1) + 2)). Hence proved.

Proved by induction

Q10. Prove by induction: r + r^2 + r^3 + ... + r^n = r(1 - r^n)/(1 - r), (r not equal to 1).
Base n = 1: LHS = r, RHS = r(1 - r)/(1 - r) = r. True.

Assume true for n = k. For n = k + 1, add r^(k+1): r(1 - r^k)/(1 - r) + r^(k+1) = (r - r^(k+1) + r^(k+1) - r^(k+2))/(1 - r) = r(1 - r^(k+1))/(1 - r). Hence proved.

Proved by induction (r not equal to 1)

Q11. Prove by induction: a + (a + d) + (a + 2d) + ... + (a + (n - 1)d) = n/2 [2a + (n - 1)d].
Base n = 1: LHS = a, RHS = 1/2 x 2a = a. True.

Assume true for n = k. For n = k + 1, add a + kd: k/2[2a + (k - 1)d] + a + kd = [2ak + k(k - 1)d + 2a + 2kd]/2 = [(k + 1)2a + (k^2 + k)d]/2 = (k + 1)/2[2a + kd]. Hence proved.

Proved by induction

Q12. Prove by induction: 1 x 1! + 2 x 2! + 3 x 3! + ... + n x n! = (n + 1)! - 1.
Base n = 1: LHS = 1, RHS = 2! - 1 = 1. True.

Assume true for n = k. For n = k + 1, add (k + 1)(k + 1)!: (k + 1)! - 1 + (k + 1)(k + 1)! = (k + 2)(k + 1)! - 1 = (k + 2)! - 1. Hence proved.

Proved by induction

Q13. Prove by induction: a_n = a_1 + (n - 1)d where a_1, a_1 + d, a_1 + 2d, ... is an A.P.
Base n = 1: a_1 = a_1 + 0 x d. True.

Assume a_k = a_1 + (k - 1)d. The next term adds the common

difference: a_(k+1) = a_k + d = a_1 + (k - 1)d + d = a_1 + kd. Hence proved.

Proved by induction

Q14. Prove by induction: a_n = a_1 r^(n-1) where a_1, a_1 r, a_1 r^2, ... is a G.P.

Base n = 1: a_1 = a_1 r^0. True.

Assume a_k = a_1 r^(k-1). The next term multiplies by r: a_(k+1) = a_k x r = a_1 r^(k-1) x r = a_1 r^k. Hence proved.

Proved by induction

Q15. Prove by induction: C(6,6) + C(7,6) + C(8,6) + ... + C(n+5,6) = C(n+6,7).

Base n = 1: LHS = C(6, 6) = 1, RHS = C(7, 7) = 1. True.

Assume true for n = k. For n = k + 1, add C(k + 6, 6): C(k + 6, 7) + C(k + 6, 6) = C(k + 7, 7) by Pascal's identity. Hence proved.

Proved by induction (hockey-stick identity)

Q16. Prove by mathematical induction for all positive integer n. (i) 5^n - 1 is divisible by 4; (ii) 2^(2n) - 1 is divisible by 3; (iii) 3^(2n) - 1 is divisible by 8; (iv) 8 x 10^n - 2 is divisible by 6; (v) 2^(n+1) + 3^(2n+1) is divisible by 5; (vi) 10^(2n) - 1 is divisible by 9. (Flag: (v) as printed is false — at n = 1 it gives 4 + 27 = 31, not divisible

by 5. The true statement of this shape, 2^(2n+1) + 3^(2n+1) divisible by 5, is proved instead; the book's exponent 2^(n+1) appears to be a misprint for 2^(2n+1).)

(i) Base n = 1: 4 divisible by 4. Step: 5^(k+1) - 1 = 5(5^k - 1) + 4, a sum of multiples of 4.

(ii) 2^(2n) - 1 = 4^n - 1. Base n = 1: 3 divisible by 3. Step: 4^(k+1) - 1 = 4(4^k - 1) + 3.

(iii) 3^(2n) - 1 = 9^n - 1. Base n = 1: 8 divisible by 8. Step: 9^(k+1) - 1 = 9(9^k - 1) + 8.

(iv) Base n = 1: 80 - 2 = 78 = 6 x 13. Step: 8 x 10^(k+1) - 2 = 10(8 x 10^k - 2) + 18, a sum of multiples of 6.

(v) Corrected statement: 2^(2n+1) + 3^(2n+1) is divisible by 5. Base n = 1: 8 + 27 = 35 = 5 x 7. Step: 2^(2k+3) + 3^(2k+3) = 4(2^(2k+1) + 3^(2k+1)) - 3^(2k+1)(4 - 9) = 4(2^(2k+1) + 3^(2k+1)) + 5 x 3^(2k+1), a sum of multiples of 5. (Equivalently, odd powers: a^(2n+1) + b^(2n+1) is divisible by a + b = 5.)

(vi) 10^(2n) - 1 = 100^n - 1. Base n = 1: 99 divisible by 9. Step: 100^(k+1) - 1 = 100(100^k - 1) + 99.

(i)-(iv) and (vi) proved by induction; (v) as printed is FALSE — see flag

Q17. Using mathematical induction, prove the following summation

formulae. (i) sum k^2 = n(n + 1)(2n + 1)/6; (ii) sum k^3 = [n(n + 1)/2]^2; (iii) sum (2k - 1) = n^2; (iv) sum k(k + 1) = n(n + 1)(n + 2)/3. (Flag: the book prints (iv) as n(n + 1)(n + 1)/3, verified at 300 dpi — this is a misprint, since at n = 1 the left side is 2 while n(n + 1)(n + 1)/3 = 8/3. The correct factor is (n + 2).)
(i) Base n = 1: 1 = 1 x 2 x 3/6. Step: add (k + 1)^2: [k(k + 1)(2k + 1) + 6(k + 1)^2]/6 = (k + 1)(k + 2)(2k + 3)/6.

(ii) Base n = 1: 1 = [1 x 2/2]^2. Step: add (k + 1)^3: [k(k + 1)/2]^2 + (k + 1)^3 = (k + 1)^2[k^2/4 + (k + 1)] = (k + 1)^2(k + 2)^2/4 = [(k + 1)(k + 2)/2]^2.

(iii) Base n = 1: 1 = 1^2. Step: add (2k + 1): k^2 + 2k + 1 = (k + 1)^2.

(iv) Base n = 1: LHS = 2, RHS = 1 x 2 x 3/3 = 2. Step: add (k + 1)(k + 2): k(k + 1)(k + 2)/3 + (k + 1)(k + 2) = (k + 1)(k + 2)(k/3 + 1) = (k + 1)(k + 2)(k + 3)/3.

(i)-(iii) proved; (iv) proved with the corrected factor (n + 2) — see flag

Q18. By applying mathematical induction, show the (i) a + 1 is a factor of a^(2n) - 1, (a not equal to -1); (ii) x - y is a factor of x^n - y^n, (x not equal to y); (iii) a + b is a factor of a^(2n-1) + b^(2n-1), (a not equal to -b).
(i) Base n = 1: a^2 - 1 = (a - 1)(a + 1), divisible by a + 1. Step: a^(2k+2) - 1 = a^2(a^(2k) - 1) + (a^2 - 1); both terms are

divisible by a + 1.

(ii) Base n = 1: x - y divides itself. Step: x^(k+1) - y^(k+1) = x(x^k - y^k) + y^k(x - y); both terms are divisible by x - y.

(iii) Base n = 1: a + b divides a + b. Step: a^(2k+1) + b^(2k+1) = a^2(a^(2k-1) + b^(2k-1)) + b^(2k-1)(b^2 - a^2) = a^2(a^(2k-1) + b^(2k-1)) + b^(2k-1)(b - a)(b + a); both terms are divisible by a + b.

Proved by induction

Q19. Use mathematical induction to prove that: 3 + 3 x 5 + 3 x 5^2 + ... + 3 x 5^n = 3(5^(n+1) - 1)/4.
Base n = 1: LHS = 3 + 15 = 18, RHS = 3(25 - 1)/4 = 18. True.

Assume true for n = k. For n = k + 1, add 3 x 5^(k+1): 3(5^(k+1) - 1)/4 + 3 x 5^(k+1) = 3[5^(k+1) - 1 + 4 x 5^(k+1)]/4 = 3(5 x 5^(k+1) - 1)/4 = 3(5^(k+2) - 1)/4. Hence proved.

Proved by induction

Q20. By mathematical induction prove that: ln(x^n) = n ln(x) for any integer n >= 0 and x is positive number.
Base n = 0: ln(x^0) = ln 1 = 0 = 0 x ln x. True.

Assume ln(x^k) = k ln x. Then ln(x^(k+1)) = ln(x^k x x) = ln(x^k) + ln x = k ln x + ln x = (k + 1) ln x. Hence proved.

Proved by induction

Q21. Use principle of mathematical induction prove that: (i) n! > 2^n - 1, for all n >= 4; (ii) n^2 > n + 2, for all n >= 3; (iii) 4^n > 3^n + 2^(n-1), for all n >= 2; (iv) n! > n^2, for all n >= 4; (v) 3^n < n!, for all n > 6; (vi) 1 + nx <= (1 + x)^n for all n >= 2 and x > -1.

(i) Base n = 4: 24 > 15. Step: (k + 1)! = (k + 1)k! > 5(2^k - 1) = 5 x 2^k - 5 > 2^(k+1) - 1, since 3 x 2^k > 4 for k >= 4.

(ii) Base n = 3: 9 > 5. Step: (k + 1)^2 = k^2 + 2k + 1 > (k + 2) + 2k + 1 = 3k + 3 >= (k + 1) + 2.

(iii) Base n = 2: 16 > 9 + 2 = 11. Step: 4^(k+1) = 4 x 4^k > 4(3^k + 2^(k-1)) = 4 x 3^k + 2^(k+1) > 3^(k+1) + 2^k, since 4 x 3^k > 3^(k+1) and 2^(k+1) > 2^k.

(iv) Base n = 4: 24 > 16. Step: (k + 1)! = (k + 1)k! > (k + 1)k^2 > (k + 1)^2, since k^2 > k + 1 for k >= 2.

(v) Base n = 7: 2187 < 5040. Step: 3^(k+1) = 3 x 3^k < 3k! < (k + 1)k! = (k + 1)! for k + 1 > 3.

(vi) Base n = 2: 1 + 2x <= 1 + 2x + x^2. Step: (1 + x)^(k+1) = (1 + x)^k(1 + x) >= (1 + kx)(1 + x) (using 1 + x > 0) = 1 + (k + 1)x + kx^2 >= 1 + (k + 1)x.

Proved by induction

Exercise 6.2

Q1. Using binomial theorem, expand the following. (i) (1 + 3q)^6; (ii) (2m - 3n)^5; (iii) (3x + 2/y)^4; (iv) (a/x - 3x/(2a))^7; (v) (2x + 1)^8; (vi) (3 + 1/(2x))^5.

(i) Coefficients C(6, r) with powers of 3q: 1 + 6(3q) + 15(9q^2) + 20(27q^3) + 15(81q^4) + 6(243q^5) + 729q^6 = 1 + 18q + 135q^2 + 540q^3 + 1215q^4 + 1458q^5 + 729q^6.

(ii) (2m)^5 + 5(2m)^4(-3n) + 10(2m)^3(9n^2) + 10(2m)^2(-27n^3) + 5(2m)(81n^4) + (-243n^5) = 32m^5 - 240m^4n + 720m^3n^2 - 1080m^2n^3 + 810mn^4 - 243n^5.

(iii) (3x)^4 + 4(3x)^3(2/y) + 6(3x)^2(4/y^2) + 4(3x)(8/y^3) + 16/y^4 = 81x^4 + 216x^3/y + 216x^2/y^2 + 96x/y^3 + 16/y^4.

(iv) General term C(7, r)(a/x)^(7-r)(-3x/(2a))^r. For r = 0 to 7: a^7/x^7; -21a^5/(2x^5); 189a^3/(4x^3); -945a/(8x); 2835x/(16a); -5103x^3/(32a^3); 5103x^5/(64a^5); -2187x^7/(128a^7).

(v) C(8, r)(2x)^(8-r): 256x^8 + 1024x^7 + 1792x^6 + 1792x^5 + 1120x^4 + 448x^3 + 112x^2 + 16x + 1.

(vi) C(5, r)3^(5-r)(1/(2x))^r: 243 + 405/(2x) + 135/(2x^2) + 45/(4x^3) + 15/(16x^4) + 1/(32x^5).

(i) 1 + 18q + 135q^2 + 540q^3 + 1215q^4 + 1458q^5 +

729q^6; (ii) 32m^5 - 240m^4n + 720m^3n^2 - 1080m^2n^3 + 810mn^4 - 243n^5; (iii) 81x^4 + 216x^3/y + 216x^2/y^2 + 96x/y^3 + 16/y^4; (iv) a^7/x^7 - 21a^5/(2x^5) + 189a^3/(4x^3) - 945a/(8x) + 2835x/(16a) - 5103x^3/(32a^3) + 5103x^5/(64a^5) - 2187x^7/(128a^7); (v) 256x^8 + 1024x^7 + 1792x^6 + 1792x^5 + 1120x^4 + 448x^3 + 112x^2 + 16x + 1; (vi) 243 + 405/(2x) + 135/(2x^2) + 45/(4x^3) + 15/(16x^4) + 1/(32x^5)

Q2. Expand and simplify the following. (i) (x + y)^5 - (x - y)^5; (ii) (3 + i)^4 - (3 - i)^4.

(i) The even-power terms cancel and the odd-power terms double: 2[5x^4y + 10x^2y^3 + y^5] = 10x^4y + 20x^2y^3 + 2y^5.

(ii) (3 + i)^4 = 28 + 96i and (3 - i)^4 = 28 - 96i, so the difference is 192i. (Directly: 2[4 x 27 x i + 4 x 3 x i^3] = 2[108i - 12i] = 192i.)

(i) 10x^4y + 20x^2y^3 + 2y^5; (ii) 192i

Q3. Find the term involving: (i) x^3 in the expansion of (3 + 2x)^7; (ii) x^6y^6 in the expansion of (2x + y)^12; (iii) a^3 in the expansion of (2/y + a)^9; (iv) x^5 in the expansion of (3 - x)^8.

(i) T_(r+1) = C(7, r)3^(7-r)(2x)^r. For x^3, r = 3: C(7, 3) x 3^4 x 8 = 35 x 81 x 8 = 22680. Term = 22680x^3.

(ii) T_(r+1) = C(12, r)(2x)^(12-r)y^r. For x^6y^6, r = 6: C(12, 6) x 2^6 = 924 x 64 = 59136. Term = 59136x^6y^6.

(iii) T_(r+1) = C(9, r)(2/y)^(9-r)a^r. For a^3, r = 3: C(9, 3) x

(2/y)^6 = 84 x 64/y^6 = 5376/y^6. Term = 5376a^3/y^6.

(iv) T_(r+1) = C(8, r)3^(8-r)(-x)^r. For x^5, r = 5: C(8, 5) x 3^3 x (-1)^5 = 56 x 27 x (-1) = -1512. Term = -1512x^5.

(i) 22680x^3; (ii) 59136x^6y^6; (iii) 5376a^3/y^6; (iv) -1512x^5

Q4. Find the term independent of y in the following. (i) (y - 2/y)^10; (ii) (y^2 - 1/y)^9; (iii) (sqrt(y) + 1/y) with no exponent printed. (Flag: (iii) is printed with NO exponent, verified at 300 dpi — the question as printed cannot be completed. Method: for (sqrt(y) + 1/y)^n, the general term is C(n, r)y^(n/2 - 3r/2), so the independent term needs r = n/3; for the common case n = 6 it is C(6, 2) = 15.)
(i) T_(r+1) = C(10, r)(-2)^r y^(10-2r). Set 10 - 2r = 0, so r = 5: C(10, 5)(-2)^5 = 252 x (-32) = -8064.

(ii) T_(r+1) = C(9, r)(-1)^r y^(18-3r). Set 18 - 3r = 0, so r = 6: C(9, 6)(+1) = 84.

(iii) As printed there is no exponent, so no single numerical term exists. In general, T_(r+1) = C(n, r)(sqrt(y))^(n-r)(1/y)^r = C(n, r)y^(n/2 - 3r/2); the independent term requires n/2 - 3r/2 = 0, i.e. r = n/3 (so n must be a multiple of 3). For n = 6: r = 2 and the term is C(6, 2) = 15.

(i) -8064; (ii) 84; (iii) exponent missing as printed — see flag

Q5. Find the middle term (middle terms) in the following. (i) (a/3 + 8b)^8; (ii) (3x - 7y)^11; (iii) (2u - 5v)^6.

(i) n = 8 is even: one middle term, the 5th. T_5 = C(8, 4)(a/3)^4(8b)^4 = 70 x (a^4/81) x 4096b^4 = 286720a^4b^4/81.

(ii) n = 11 is odd: two middle terms, the 6th and 7th. T_6 = C(11, 5)(3x)^6(-7y)^5 = 462 x 729 x (-16807)x^6y^5 = -5660563986x^6y^5. T_7 = C(11, 6)(3x)^5(-7y)^6 = 462 x 243 x 117649x^5y^6 = 13207982634x^5y^6.

(iii) n = 6 is even: one middle term, the 4th. T_4 = C(6, 3)(2u)^3(-5v)^3 = 20 x 8 x (-125)u^3v^3 = -20000u^3v^3.

(i) 286720a^4b^4/81; (ii) -5660563986x^6y^5 and 13207982634x^5y^6; (iii) -20000u^3v^3

Q6. Expand (1.03)^4 upto four places of decimal by using binomial theorem.

(1.03)^4 = (1 + 0.03)^4 = 1 + 4(0.03) + 6(0.03)^2 + 4(0.03)^3 + (0.03)^4 = 1 + 0.12 + 0.0054 + 0.000108 + 0.00000081 = 1.12550881.

To four decimal places: 1.1255.

1.1255 (to four decimal places)

Q7. Show that sum of the odd places coefficients is equal to the sum of even places coefficient in the binomial expansion (x + y)^n

and each of them is equal to 2^(n-1).

Put x = y = 1 in (x + y)^n: the sum of ALL coefficients is 2^n, i.e. S_odd + S_even = 2^n.

Put x = 1, y = -1: the alternating sum is 0 (for n >= 1), i.e. S_odd - S_even = 0.

Hence S_odd = S_even, and each equals 2^n/2 = 2^(n-1).

Both sums equal 2^(n-1)

Q8. Show that the sum of the binomial coefficients in the expansion of (x + y)^n is 2^n, n in N. Also show that the sum of the binomial coefficient in the expansion of (x + y)^8 is 256.

Put x = y = 1 in (x + y)^n = sum C(n, r)x^(n-r)y^r: the sum of the coefficients is sum C(n, r) = (1 + 1)^n = 2^n.

For n = 8: 2^8 = 256.

2^n, and 256 for n = 8

Q9. Show that: C(n,1) + 2C(n,2) + 3C(n,3) + ... + nC(n,n) = n x 2^(n-1).

Use r x C(n, r) = n x C(n - 1, r - 1). Then the sum is n x [C(n - 1, 0) + C(n - 1, 1) + ... + C(n - 1, n - 1)] = n x 2^(n-1), since the bracket is (1 + 1)^(n-1).

n x 2^(n-1)

Q10. Given that the coefficient of x in the expansion of (1 + ax)(1 + 5x)^10 is 207, find the value of a.

Coefficient of x in (1 + 5x)^10 is C(10, 1) x 5 = 50, and the constant term is 1.

In (1 + ax)(1 + 5x)^10, the x coefficient is 1 x 50 + a x 1 = 50 + a. Set 50 + a = 207, so a = 157.

a = 157

Q11. Given that (1 + ax)^n = 1 + 36x + 576x^2 + ... + a^n x^n, find the values of a and n.

Coefficient of x: na = 36. Coefficient of x^2: C(n, 2)a^2 = n(n - 1)a^2/2 = 576.

From na = 36, a = 36/n. Substitute: n(n - 1)(36/n)^2/2 = 576, i.e. 648(n - 1)/n = 576, i.e. 648n - 648 = 576n, so 72n = 648 and n = 9. Then a = 36/9 = 4.

Check: C(9, 2) x 16 = 36 x 16 = 576. Correct.

a = 4, n = 9

Exercise 6.3

Q1. If |x| < 1 in each of the following questions, find the general term in the expansions of; (i) (1 - x)^(-2); (ii) (1 - x)^(-3); (iii) (1 + x)^(-2).

(i) (1 - x)^(-2) = 1 + 2x + 3x^2 + 4x^3 + ... The general term (r = 0, 1, 2, ...) is (r + 1)x^r.

(ii) (1 - x)^(-3) = 1 + 3x + 6x^2 + 10x^3 + ... The general term is C(r + 2, 2)x^r = (r + 1)(r + 2)/2 x^r.

(iii) (1 + x)^(-2) = 1 - 2x + 3x^2 - 4x^3 + ... The general term is (-1)^r (r + 1)x^r.

(i) (r + 1)x^r; (ii) (r + 1)(r + 2)/2 x^r; (iii) (-1)^r (r + 1)x^r (r = 0, 1, 2, ...)

Q2. Find the first four terms in the following expansions. (i) sqrt(1 + 6x); (ii) cubert(8 - 16x); (iii) 3/(3 + x); (iv) 3x/sqrt(9 + x^3); (v) 8/(1 + x/3)^3; (vi) sqrt(4 + 3x^2); (vii) (1 + 3x)^(-1/3).
(i) (1 + 6x)^(1/2) = 1 + (1/2)(6x) + (1/2)(-1/2)/2 (6x)^2 + (1/2)(-1/2)(-3/2)/6 (6x)^3 = 1 + 3x - 9x^2/2 + 27x^3/2.

(ii) (8 - 16x)^(1/3) = 2(1 - 2x)^(1/3) = 2[1 - 2x/3 - 4x^2/9 - 40x^3/81] = 2 - 4x/3 - 8x^2/9 - 80x^3/81.

(iii) 3/(3 + x) = (1 + x/3)^(-1) = 1 - x/3 + x^2/9 - x^3/27.

(iv) 3x/sqrt(9 + x^3) = x(1 + x^3/9)^(-1/2) = x[1 - x^3/18 + x^6/216 - 5x^9/11664] = x - x^4/18 + x^7/216 - 5x^10/11664.

(v) 8(1 + x/3)^(-3) = 8[1 - x + 2x^2/3 - 10x^3/27] = 8 - 8x + 16x^2/3 - 80x^3/27.

(vi) (4 + 3x^2)^(1/2) = 2(1 + 3x^2/4)^(1/2) = 2[1 + 3x^2/8 - 9x^4/128 + 27x^6/1024] = 2 + 3x^2/4 - 9x^4/64 + 27x^6/512.

(vii) (1 + 3x)^(-1/3) = 1 - x + 2x^2 - 14x^3/3.

(i) 1 + 3x - 9x^2/2 + 27x^3/2; (ii) 2 - 4x/3 - 8x^2/9 - 80x^3/81; (iii) 1 - x/3 + x^2/9 - x^3/27; (iv) x - x^4/18 + x^7/216 - 5x^10/11664; (v) 8 - 8x + 16x^2/3 - 80x^3/27; (vi) 2 + 3x^2/4 - 9x^4/64 + 27x^6/512; (vii) 1 - x + 2x^2 - 14x^3/3

Q3. If x is so small that its square and higher powers may be neglected, then show that; (i) (5 + x)/(2 - x + x^2) approx 5/2 + 7x/4; (ii) sqrt(1 + x/4) approx 1 + x/8; (iii) (1 + 8x)^(1/4) approx 1 + 2x; (iv) sqrt((1 + 2x)/(1 - 4x)) approx 1 + 3x.

(i) Neglecting x^2: (5 + x)/(2 - x) = (5 + x)(1/2)(1 - x/2)^(-1) approx (5 + x)(1/2)(1 + x/2) approx 5/2 + 5x/4 + x/2 = 5/2 + 7x/4.

(ii) (1 + x/4)^(1/2) approx 1 + (1/2)(x/4) = 1 + x/8.

(iii) (1 + 8x)^(1/4) approx 1 + (1/4)(8x) = 1 + 2x.

(iv) (1 + 2x)^(1/2)(1 - 4x)^(-1/2) approx (1 + x)(1 + 2x) approx 1 + 3x.

(i) 5/2 + 7x/4; (ii) 1 + x/8; (iii) 1 + 2x; (iv) 1 + 3x

Q4. If x is so small that its cube and higher powers may be

neglected, then show that; (i) 1/(1 - 3x/2)^2 approx 1 + 3x + 27x^2/4; (ii) 1/(1 - x - x^2) approx 1 + x + 2x^2.
(i) (1 - 3x/2)^(-2) = 1 + 2(3x/2) + 3(3x/2)^2 = 1 + 3x + 27x^2/4, neglecting x^3 and higher.

(ii) (1 - (x + x^2))^(-1) approx 1 + (x + x^2) + (x + x^2)^2 approx 1 + x + x^2 + x^2 = 1 + x + 2x^2, neglecting x^3 and higher.

(i) 1 + 3x + 27x^2/4; (ii) 1 + x + 2x^2

Q5. Using binomial series, find the value of the following to three places of decimals; (i) sqrt(24); (ii) cubert(28); (iii) (0.945)^4; (iv) 5th root of 241.
(i) sqrt(24) = 5(1 - 1/25)^(1/2) = 5[1 - 1/50 - 1/10000 - ...] approx 5(0.979792) = 4.89896 approx 4.899.

(ii) cubert(28) = 3(1 + 1/27)^(1/3) = 3[1 + 1/81 - 1/6561 x ...] approx 3(1.012193) = 3.03658 approx 3.037.

(iii) (0.945)^4 = (1 - 0.055)^4 = 1 - 0.22 + 6(0.055)^2 - 4(0.055)^3 + (0.055)^4 = 1 - 0.22 + 0.01815 - 0.0006655 + 0.0000092 = 0.7974937 approx 0.797.

(iv) 5th root of 241 = 3(1 - 2/243)^(1/5) = 3[1 - (1/5)(2/243) - ...] approx 3(0.998354) = 2.99506 approx 2.995.

(i) 4.899; (ii) 3.037; (iii) 0.797; (iv) 2.995

Q6. Use binomial theorem, to find the remainder when, (i) 5^105 is divided by 13; (ii) 8^95 is divided by 9; (iii) 41^7 is divided by 21; (iv) 93^8 is divided by 23.

(i) 5^2 = 25 = 2 x 13 - 1, so 5^2 is congruent to -1 mod 13. Then 5^104 is congruent to 1 and 5^105 is congruent to 5. Remainder 5.

(ii) 8 is congruent to -1 mod 9, so 8^95 is congruent to (-1)^95 = -1, i.e. 8 mod 9. Remainder 8.

(iii) 41 = 42 - 1 is congruent to -1 mod 21, so 41^7 is congruent to (-1)^7 = -1, i.e. 20 mod 21. Remainder 20.

(iv) 93 = 92 + 1 is congruent to 1 mod 23, so 93^8 is congruent to 1. Remainder 1.

(i) 5; (ii) 8; (iii) 20; (iv) 1

Q7. Use binomial theorem, to find the last digit of a number. (i) When 9^6 is divided by 8; (ii) 52^7 is divided by 17. (Flag: the question mixes 'last digit' with 'divided by'; both are answered.)

(i) 9 = 8 + 1, so 9^6 = (8 + 1)^6 is congruent to 1 mod 8: remainder 1. Also 9^6 = 531441, whose last digit is 1.

(ii) 52 = 51 + 1, so 52^7 = (51 + 1)^7 is congruent to 1 mod 17: remainder 1. The last digit of 52^7 is the last digit of 2^7 = 128, i.e. 8.

(i) remainder 1, last digit 1; (ii) remainder 1, last digit 8 (flag: Q7 is worded confusingly — see steps)

Q8. If u = 1/2 x 1/16 - 1/(2! x 4) x 1/16^2 + 1/(3! x 8) x 1/16^3 - ... then show that 16u(u + 2) = 1. (Flag: the third term is typeset cramped; the series is the binomial expansion of (1 + 1/16)^(1/2) - 1, whose third term is (1/16)(1/16^3), i.e. 1 x 3/(2^3 x 3!) x (1/16)^3. With the literal 1/(3! x 8) = 1/48 the identity would only be approximate.)
The series is (1 + x)^(1/2) - 1 with x = 1/16: u = (1/2)(1/16) - (1/8)(1/16^2) + (1/16)(1/16^3) - ... = (1 + 1/16)^(1/2) - 1.

So u + 1 = (17/16)^(1/2), giving (u + 1)^2 = 17/16, i.e. u^2 + 2u = 1/16, i.e. 16u(u + 2) = 1.

16u(u + 2) = 1 is proved (series identified as (1 + 1/16)^(1/2) - 1; flag on the cramped third term)

Q9. If 1/y = 2/5 + (1 x 3)/2! (2/5)^2 + (1 x 3 x 5)/3! (2/5)^3 + ..., then show that 4y^2 - 2y - 1 = 0.
The series is sum_{n>=1} C(2n, n)(1/5)^n, since (2n - 1)!!/n! x (2/5)^n = C(2n, n)(1/5)^n.

Now sum_{n>=0} C(2n, n)t^n = (1 - 4t)^(-1/2). With t = 1/5: 1/y = (1 - 4/5)^(-1/2) - 1 = sqrt(5) - 1.

Hence y = 1/(sqrt(5) - 1) = (sqrt(5) + 1)/4. Then 4y^2 - 2y - 1 = (6

+ 2sqrt(5))/4 - (sqrt(5) + 1)/2 - 1 = 0.

4y^2 - 2y - 1 = 0 is proved

Q10. If z/2 = 1/2 (4/9) + (1 x 3)/(2^2 x 2!) (4/9)^2 + (1 x 3 x 5)/(2^3 x 3!) (4/9)^3 + ... then show that 5z^2 + 20z - 16 = 0.
The series is (1 - t)^(-1/2) - 1 with t = 4/9: the coefficients are 1/2, 3/8, 5/16, ... So z/2 = (1 - 4/9)^(-1/2) - 1 = (5/9)^(-1/2) - 1 = 3/sqrt(5) - 1.

Hence z = 6/sqrt(5) - 2. Then 5z^2 + 20z - 16 = 5(56/5 - 24/sqrt(5)) + 120/sqrt(5) - 40 - 16 = 56 - 24sqrt(5) + 24sqrt(5) - 56 = 0.

5z^2 + 20z - 16 = 0 is proved

Q11. A company hires new employees at a starting salary of Rs.30,000 per year, with a 2% annual raise. What will be the salary of an employee after 5 years? Use mathematical induction.
By induction, the salary after n years is 30000 x (1.02)^n: true for n = 0, and if true for year k, year k + 1 multiplies by 1.02.

After 5 years: 30000 x (1.02)^5 = 30000 x 1.1040808 = 33122.42. So about Rs 33,122.

Rs 33,122.42 (about Rs 33,122)

Q12. A population of rabbits doubles every year. If there are 10 rabbits initially, how many rabbits will there be after 5 years? Use

mathematical induction.
By induction, the population after n years is 10 x 2^n: true for n = 0, and each year doubles the previous count.

After 5 years: 10 x 2^5 = 10 x 32 = 320 rabbits.

320 rabbits

Q13. A company is forecasting the demand for its product over the next 5 years. The demand in the first year is expected to be 100 units. If the demand in the nth year is qn units, where q is a constant, find the demand in the 5th year using mathematical induction and the binomial theorem. (Flag: read literally, qn means q times n. Year 1 gives q = 100, so year 5 is 500. If q^n (geometric growth) were meant, the 5th-year demand would be 100q^4, but q is not numerically given.)
Let D_n be the demand in year n, with D_n = qn. For n = 1: D_1 = q = 100, so q = 100.

By induction D_n = 100n for every year n. For n = 5: D_5 = 100 x 5 = 500 units.

500 units (flag: literal reading qn = q x n — see steps)

Q14. In a puzzle game, a player starts with 1 tile and doubles the number of tiles in each subsequent level. For example, in level 1 there is 1 tile, in level 2 there are 2 tiles, in level 3 there are 4 tiles, and so on. If the player reaches level n, what is the total number of

tiles in all levels combined using mathematical induction and the binomial theorem? (Flag: the target level is blurred in the print ('reaches lev?l'); it is read as level n, giving the general total.)
Level k has 2^(k-1) tiles. By induction, the total through level n is 1 + 2 + 4 + ... + 2^(n-1) = 2^n - 1: true for n = 1 (1 = 2 - 1), and adding level n + 1 with 2^n tiles gives (2^n - 1) + 2^n = 2^(n+1) - 1.

2^n - 1 tiles through level n (flag: target level blurred in print)

Q15. In a domino effect, a chain reaction occurs when a domino falls and knocks over the next domino, which then knocks over the next domino, and so on until the entire line of dominos has fallen. If there are n dominos in a line, how many dominos will fall in total using mathematical induction and the binomial theorem?
By induction: with 1 domino, 1 falls. Assume k dominoes in a line all fall; adding one more domino, the kth domino knocks it over, so k + 1 fall.

Hence all n dominoes fall: n in total.

n dominoes

Review Exercise 6

Q1(i). Number of terms in the expansion (x + 3y)^6 are;

6

✓ 7

8

9

Q1(ii). Third term in the expansion (x + y)^6 is;

✓ 15x^4y^2

20x^3y^2

20x^4y^3

15x^3y^3

Q1(iii). Middle term in the expansion (3a + 7b)^6 is;

3rd

✓ 4th

5th

6th

Q1(iv). Third term in the expansion (1 + x)^(-1) is;

x^(-1)

1

x

✓ x^2

Q1(v). Sum of binomial coefficients in the expansion (x + y)^5 is;

16

✓ 32

64

128

Q1(vi). In the expansion (x + 1/x)^6 the middle term is;

15

✓ 20

25

30

Q1(vii). In the expansion (x + 1/2)^7, how many middle terms are there?

one

✓ two

three

four

Q1(viii). In (y + 1/y)^8, what is the sum of binomial coefficient?

64

128

✓ 256

512

Q1(ix). In the expansion (a - 1/(3a))^6 the fifth term is

✓ 5/(27a^2)

10/(81a^3)

15/(243a^4)

20/(729a^5)

Q1(x). If (1 + ax/3)^(-5) = 1 - 5x + 15x^2 + ... then the value of a is

12

9

6

✓ 3

Q2. Prove by principle of mathematical induction for all n in Z+, 8 x 10^n - 2 is divisible by 6.

Base n = 1: 80 - 2 = 78 = 6 x 13. True.

Assume 8 x 10^k - 2 is divisible by 6. Then 8 x 10^(k+1) - 2 = 10(8 x 10^k - 2) + 18, a sum of multiples of 6. Hence proved.

Proved by induction

Q3. Prove that n^2 > n + 3 for integral values of n >= 3.

Base n = 3: 9 > 6. True.

Assume k^2 > k + 3. Then (k + 1)^2 = k^2 + 2k + 1 > (k + 3) + 2k + 1 = 3k + 4 > k + 4 = (k + 1) + 3. Hence proved.

Proved by induction

Q4. Show that; C(n,0) + C(n,2) + C(n,4) + ... + C(n,n) = 2^(n-1). (Flag: the printed last term C(n,n) fits only for even n; the statement is the sum of the EVEN-index coefficients, whose last term is C(n,n) for even n and C(n,n-1) for odd n.)

Put x = 1 and x = -1 in (1 + x)^n: (1 + 1)^n = 2^n is the sum of all coefficients, and (1 - 1)^n = 0 is the alternating sum (even-index minus odd-index).

Adding: twice the even-index sum equals 2^n, so the even-index

sum is 2^(n-1).

Sum of even-index coefficients = 2^(n-1)

Q5. Given that the coefficient of x^2 in the expansion 1/(1 + ax)^2 is 96, find a. (Flag: the equation 3a^2 = 96 has two roots; the book's single-answer format hides the negative root.)
(1 + ax)^(-2) = 1 - 2ax + 3a^2x^2 - ... So the x^2 coefficient is 3a^2 = 96, giving a^2 = 32 and a = 4sqrt(2) or a = -4sqrt(2).

a = 4sqrt(2) or a = -4sqrt(2) (flag: two values)

Q6. Give that terms involving x^4 and higher powers may be neglected and that; 1/(a + bx)^3 - 1/(1 + 3x)^4 = cx^2 + dx^3, find the values of a, b, c and d. (Flag: the right side has no constant or x term, so those coefficients on the left must vanish — this fixes a and b.)
Expand: 1/(a + bx)^3 = (1/a^3)[1 - 3(b/a)x + 6(b/a)^2x^2 - 10(b/a)^3x^3 + ...] and 1/(1 + 3x)^4 = 1 - 12x + 90x^2 - 540x^3 + ...

Constant term: 1/a^3 - 1 = 0, so a = 1. x coefficient: -3b/a^4 + 12 = 0, so b = 4.

x^2 coefficient: 6b^2/a^5 - 90 = 96 - 90 = 6, so c = 6. x^3 coefficient: -10b^3/a^6 + 540 = -640 + 540 = -100, so d = -100.

a = 1, b = 4, c = 6, d = -100

Unit 7: Division of Polynomial

Exercise 7.1

Q1. Divide a polynomial by a linear and quadratic polynomial to identify quotient and remainder. (i) p(x) = 2x^3 + 4x^2 + 5x + 1, g(x) = x + 1; (ii) p(x) = 16x^3 + x^2 - x + 1, g(x) = 2x - 1; (iii) p(x) = 8x^4 + 4x^2 + 4x + 4, g(x) = x^2 - x + 1; (iv) p(x) = 6x^4 + 3x^2 + 2, g(x) = x^2 + 1.

(i) Synthetic division by -1 on 2, 4, 5, 1: bring 2; multiply -2, add 2; multiply -2, add 3; multiply -3, add -2. Quotient 2x^2 + 2x + 3, remainder -2.

(ii) Synthetic division by 1/2 on 16, 1, -1, 1: bring 16; multiply 8, add 9; multiply 9/2, add 7/2; multiply 7/4, add 11/4. Quotient 16x^2 + 9x + 7/2, remainder 11/4.

(iii) Long division: 8x^4/(x^2) = 8x^2, leaving 8x^3 - 4x^2 + 4x; next +8x, leaving 4x^2 - 4x + 4; next +4, leaving 0. Quotient 8x^2 + 8x + 4, remainder 0.

(iv) Long division: 6x^4/(x^2) = 6x^2, leaving -3x^2 + 2; next -3, leaving 5. Quotient 6x^2 - 3, remainder 5.

(i) Quotient 2x^2 + 2x + 3, remainder -2; (ii) quotient 16x^2 + 9x + 7/2, remainder 11/4; (iii) quotient 8x^2 + 8x + 4, remainder 0; (iv) quotient 6x^2 - 3, remainder 5

Q2. Find the remainder by means of remainder theorem when. (i) (2x - 1)^2 + 6(3 + 4x) - 10 is divided by 2x - 1; (ii) 2x^3 + 7x^2 - 3x + 3 is divided by x - 2; (iii) x^4 + 4x^3 - 3x^2 + 5x + 8 is divided by x + 1.

(i) The zero of 2x - 1 is x = 1/2. f(1/2) = 0 + 6(3 + 2) - 10 = 30 - 10 = 20.

(ii) f(2) = 16 + 28 - 6 + 3 = 41.

(iii) f(-1) = 1 - 4 - 3 - 5 + 8 = -3.

(i) 20; (ii) 41; (iii) -3

Q3. Find K if x^3 + 3x^2 - Kx + 4 leaves a remainder of 8 when divided by x - 2.

By the remainder theorem, f(2) = 8 + 12 - 2K + 4 = 24 - 2K. Set 24 - 2K = 8, so 2K = 16 and K = 8.

K = 8

Q4. The expression 8x^3 + ax^2 + bx - 9 leaves remainder -95 and 3 when divided by x + 2 and 2x - 3 respectively. Calculate the value of a and b.

f(-2) = -64 + 4a - 2b - 9 = -95, so 4a - 2b = -22, i.e. 2a - b = -11.

f(3/2) = 27 + 9a/4 + 3b/2 - 9 = 3, so 9a/4 + 3b/2 = -15, i.e. 3a + 2b = -20.

From b = 2a + 11: 3a + 4a + 22 = -20, so a = -6 and b = -1.

a = -6, b = -1

Q5. Determine whether x - 2 and x - 3 are the factors of x^3 - 10x^2 + 31x - 30.

f(2) = 8 - 40 + 62 - 30 = 0, so x - 2 is a factor.

f(3) = 27 - 90 + 93 - 30 = 0, so x - 3 is a factor.

Yes, both are factors

Q6. Given that 4x^3 - 6x^2 + 1 = (x - 2)(x + 1)Q(x) + ax + b where Q(x) is a polynomial. Find a and b.

Put x = 2: 32 - 24 + 1 = 9 = 2a + b. Put x = -1: -4 - 6 + 1 = -9 = -a + b.

Subtracting: 9 - (-9) = 3a, so a = 6. Then b = 9 - 12 = -3.

a = 6, b = -3

Q7. -4 is the root of x^4 + ax^3 - 19x^2 - 46x + 120. What is the value of a?

f(-4) = 256 - 64a - 304 + 184 + 120 = 256 - 64a. Setting f(-4) = 0 gives 64a = 256, so a = 4.

a = 4

Q8. What is the remainder when 2x^2 - 3x + 5 is divided by 2x - 1?

The zero of 2x - 1 is x = 1/2. f(1/2) = 2(1/4) - 3/2 + 5 = 1/2 - 3/2 + 5 = 4.

4

Q9. If (x - 7) and (x - 4) are the factors of px^3 + qx^2 - 5x + 84. Find the values of p and q.

f(7) = 343p + 49q - 35 + 84 = 0, i.e. 343p + 49q + 49 = 0, i.e. 7p + q + 1 = 0.

f(4) = 64p + 16q - 20 + 84 = 0, i.e. 64p + 16q + 64 = 0, i.e. 4p + q + 4 = 0.

Subtracting: 3p - 3 = 0, so p = 1. Then q = -4 - 4 = -8.

p = 1, q = -8

Exercise 7.2

Q1. Factorize: 3x^3 - 10x^2 + 9x - 2.

Try x = 1: 3 - 10 + 9 - 2 = 0, so (x - 1) is a factor.

Divide: 3x^3 - 10x^2 + 9x - 2 = (x - 1)(3x^2 - 7x + 2) = (x - 1)(3x - 1)(x - 2).

(x - 1)(3x - 1)(x - 2)

Q2. Factorize: x^3 - 11x^2 + 36x - 36.

Try x = 2: 8 - 44 + 72 - 36 = 0, so (x - 2) is a factor.

Divide: x^3 - 11x^2 + 36x - 36 = (x - 2)(x^2 - 9x + 18) = (x - 2)(x - 3)(x - 6).

(x - 2)(x - 3)(x - 6)

Q3. Factorize: x(x + 3)(x - 1) - x - 8.

Expand: x(x^2 + 2x - 3) - x - 8 = x^3 + 2x^2 - 4x - 8.

Try x = 2: 8 + 8 - 8 - 8 = 0, so (x - 2) is a factor. Divide: (x - 2)(x^2 + 4x + 4) = (x - 2)(x + 2)^2.

(x - 2)(x + 2)^2

Q4. Factorize: x^3 + x^2 - 10x + 8.

Try x = 1: 1 + 1 - 10 + 8 = 0, so (x - 1) is a factor.

Divide: (x - 1)(x^2 + 2x - 8) = (x - 1)(x + 4)(x - 2).

(x - 1)(x + 4)(x - 2)

Q5. Factorize: x^3 - 9x^2 + 5x + 3.

Try x = 1: 1 - 9 + 5 + 3 = 0, so (x - 1) is a factor.

Divide: (x - 1)(x^2 - 8x - 3). The quadratic has discriminant 64 + 12 = 76, not a perfect square, so it does not factor further over the

rationals.

(x - 1)(x^2 - 8x - 3) (quadratic irreducible over the rationals)

Q6. Factorize: 4x^3 + 5x^2 - x - 5/4.

Try x = 1/2: 4(1/8) + 5(1/4) - 1/2 - 5/4 = 1/2 + 5/4 - 1/2 - 5/4 = 0, so (x - 1/2) is a factor.

Divide: (x - 1/2)(4x^2 + 7x + 5/2). The quadratic factors as (2x + 1)(2x + 5/2). Hence (x - 1/2)(2x + 1)(2x + 5/2) = (2x - 1)(2x + 1)(4x + 5)/4.

(2x - 1)(2x + 1)(4x + 5)/4, i.e. (x - 1/2)(2x + 1)(2x + 5/2)

Q7. Factorize: x^3 - 5x^2 + 7x - 3.

Try x = 1: 1 - 5 + 7 - 3 = 0, so (x - 1) is a factor.

Divide: (x - 1)(x^2 - 4x + 3) = (x - 1)(x - 1)(x - 3) = (x - 1)^2(x - 3).

(x - 1)^2 (x - 3)

Q8. Find the factors of x^3 + 7x^2 + 8x + 2.

Try x = -1: -1 + 7 - 8 + 2 = 0, so (x + 1) is a factor.

Divide: (x + 1)(x^2 + 6x + 2). The quadratic has discriminant 36 - 8 = 28, not a perfect square, so it does not factor further over the rationals.

(x + 1)(x^2 + 6x + 2) (quadratic irreducible over the rationals)

Q9. Given x^3 + px^2 - 2x + 4sqrt(3) has a factor x + sqrt(2). Find value of p.

Since x + sqrt(2) is a factor, f(-sqrt(2)) = 0: -2sqrt(2) + 2p + 2sqrt(2) + 4sqrt(3) = 0, i.e. 2p + 4sqrt(3) = 0, so p = -2sqrt(3).

p = -2sqrt(3)

Q10. Determine whether -3/2 is a zero (root) of the function f(x) = 2x^3 + 3x^2 - 8x - 12.

f(-3/2) = 2(-27/8) + 3(9/4) - 8(-3/2) - 12 = -27/4 + 27/4 + 12 - 12 = 0.

Hence -3/2 is a zero of f.

Yes, -3/2 is a zero

Review Exercise 7

Q1(i). Factors of x^2 - 16 are

✓ (x + 4)(x - 4)

(x + 4)^2

(x + 4)(x + 5)

(x - 4)^2

Q1(ii). Factors of x^2 + 8x + 15 are

(x + 5)(x - 3)

✓ (x + 5)(x + 3)

(x + 6)(x + 8)

(x - 3)(x - 5)

Q1(iii). What should be added to 3x^2 + 8y^2 to complete the square.

✓ 2sqrt(24)xy

24xy

3sqrt(24)xy

sqrt(24)xy

Q1(iv). Factors of 64/x^3 - x^3 are

✓ (4/x - x)(16/x^2 + 4 + x^2)

(4/x - x)(16/x^2 - x)(16/x^2 - 4 + x^2)

(x - 4/x)(x^2 + 4 + 16/x^2)

(4/x + x)(16/x^2 + 4 - x^2)

Q1(v). Factor of x^3 - 4x^2 + 3x + 2 is

✓ x - 2

x + 2

x + 3

x - 3

Q2. Show that (x - 1) is a factor of 2x^3 + x^2 - 21x + 18.

f(1) = 2 + 1 - 21 + 18 = 0. By the factor theorem, (x - 1) is a factor.

(x - 1) is a factor

Q3. Find the values of a and b if x^2 - 5x + 6 is a factor of 3x^3 +

ax^2 + bx + 12.

x^2 - 5x + 6 = (x - 2)(x - 3), so f(2) = 0 and f(3) = 0.

f(2) = 24 + 4a + 2b + 12 = 0, i.e. 2a + b = -18. f(3) = 81 + 9a + 3b + 12 = 0, i.e. 3a + b = -31.

Subtracting: a = -13. Then b = -18 - 2(-13) = 8.

a = -13, b = 8

Q4. Suppose r and R are the remainders when the polynomials x^3 + 2x^2 - 5kx - 7 and x^3 + kx^2 - 12x + 6 are divided by x + 1 and x - 2, respectively. Find the value of k if 2r + R = 6.

r = f(-1) = -1 + 2 + 5k - 7 = 5k - 6. R = g(2) = 8 + 4k - 24 + 6 = 4k - 10.

2r + R = 2(5k - 6) + (4k - 10) = 14k - 22 = 6, so 14k = 28 and k = 2.

k = 2

Unit 8: Vectors in Space

Exercise 8.1

Q1. If a = 5i - j - 3k and b = i + 3j - 5k, then show that the vectors a + b and a - b are perpendicular.

a + b = 6i + 2j - 8k. a - b = 4i - 4j + 2k.

(a + b) . (a - b) = 24 - 8 - 16 = 0. Since the dot product is zero, the vectors are perpendicular.

(a + b) . (a - b) = 0, so the vectors are perpendicular

Q2. Whether following vectors are orthogonal are not; (i) a = 2i - 3j + 2k, b = 5i + 8j + 7k; (ii) a = i + 3j - 4k, b = -2i + j + 3k.

(i) a . b = 10 - 24 + 14 = 0, so the vectors are orthogonal.

(ii) a . b = -2 + 3 - 12 = -11, not zero, so the vectors are not orthogonal.

(i) orthogonal; (ii) not orthogonal

Q3. Prove the commutative law of a and b and distributive laws of scalar product for the following vectors. (i) a = i - j + 2k, b = 2i + 3j - k, c = 5i - j + 2k; (ii) a = 3i - 5j + 7k, b = -4i + j + k, c = 6i + j - 2k.

(i) Commutative: a . b = 2 - 3 - 2 = -3 and b . a = -3. Distributive: b

+ c = 7i + 2j + k, so a . (b + c) = 7 - 2 + 2 = 7; also a . b + a . c = -3 + (5 + 1 + 4) = 7. Equal.

(ii) Commutative: a . b = -12 - 5 + 7 = -10 and b . a = -10. Distributive: b + c = 2i + 2j - k, so a . (b + c) = 6 - 10 - 7 = -11; also a . b + a . c = -10 + (18 - 5 - 14) = -11. Equal.

Commutative and distributive laws verified in both parts

Q4. Find angle between vectors; (i) i - 2j + 3k and 3i - 2j + k; (ii) 3i + 5j - k and i - 4j + k.

(i) Dot product = 3 + 4 + 3 = 10. |a| = sqrt(14), |b| = sqrt(14). cos(theta) = 10/14 = 5/7, so theta = cos^(-1)(5/7), about 44.4 degrees.

(ii) Dot product = 3 - 20 - 1 = -18. |a| = sqrt(35), |b| = sqrt(18). cos(theta) = -18/sqrt(630) = -6/sqrt(70), about -0.7171, so theta is about 135.8 degrees.

(i) cos^(-1)(5/7), about 44.4 degrees; (ii) about 135.8 degrees

Q5. Prove that cos(alpha + beta) = cos(alpha)cos(beta) - sin(alpha)sin(beta).

Take unit vectors u = (cos alpha, sin alpha, 0) and v = (cos beta, -sin beta, 0). The angle between them is alpha + beta.

Their dot product is u . v = cos(alpha)cos(beta) - sin(alpha)sin(beta). But also u . v = |u||v|cos(alpha + beta) =

cos(alpha + beta). Hence cos(alpha + beta) = cos(alpha)cos(beta) - sin(alpha)sin(beta).

cos(alpha + beta) = cos(alpha)cos(beta) - sin(alpha)sin(beta) is proved

Q6. Prove that in any triangle ABC. (i) b = c cos A + a cos C; (ii) c = a cos B + b cos A; (iii) b^2 = c^2 + a^2 - 2ca cos B; (iv) c^2 = a^2 + b^2 - 2ab cos C.
Let vectors AB, BC, CA have lengths c, a, b with AC = AB + BC as vectors.

(i) Take the dot product of AC = AB + BC with the unit vector along AC: b = c cos A + a cos C, since the projections of AB and BC onto AC are c cos A and a cos C.

(ii) Similarly, projecting onto AB gives c = a cos B + b cos A.

(iii) b^2 = |AB + BC|^2 = c^2 + a^2 + 2(AB . BC). The angle between AB and BC vectors is 180 degrees - B, so AB . BC = -ca cos B. Hence b^2 = c^2 + a^2 - 2ca cos B.

(iv) By the same argument with the other sides, c^2 = a^2 + b^2 - 2ab cos C.

Projection formulae and cosine rules proved

Q7. Find the work done by the force F = 7i + 9j - 11k in moving an

object along a straight line from (4, 2, 7) to (6, 4, 9).

Displacement d = (6 - 4)i + (4 - 2)j + (9 - 7)k = 2i + 2j + 2k.

Work W = F . d = 14 + 18 - 22 = 10 J.

10 J

Q8. Calculate the projection of a along b and projection of b along a when; (i) a = i - 2j + 3k, b = 3i - j - 5k; (ii) a = 2i + 3j + 2k, b = i + 2j + k.

(i) a . b = 3 + 2 - 15 = -10. Projection of a along b: scalar (a . b)/|b| = -10/sqrt(35); vector (a . b)/|b|^2 b = (-10/35)(3i - j - 5k) = (-6/7)i + (2/7)j + (10/7)k. Projection of b along a: scalar -10/sqrt(14); vector (-10/14)(i - 2j + 3k) = (-5/7)i + (10/7)j - (15/7)k.

(ii) a . b = 2 + 6 + 2 = 10. Projection of a along b: scalar 10/sqrt(6); vector (10/6)(i + 2j + k) = (5/3)(i + 2j + k). Projection of b along a: scalar 10/sqrt(17); vector (10/17)(2i + 3j + 2k).

(i) proj of a along b: scalar -10/sqrt(35), vector (-6/7)i + (2/7)j + (10/7)k; proj of b along a: scalar -10/sqrt(14), vector (-5/7)i + (10/7)j - (15/7)k. (ii) proj of a along b: scalar 10/sqrt(6), vector (5/3)(i + 2j + k); proj of b along a: scalar 10/sqrt(17), vector (10/17)(2i + 3j + 2k).

Exercise 8.2

Q1. Find the scalar triple product of vectors i + 2j + 3k, -i - j + k and i + j + k.

The scalar triple product is the determinant with rows (1, 2, 3), (-1, -1, 1), (1, 1, 1).

= 1(-1 - 1) - 2(-1 - 1) + 3(-1 + 1) = -2 + 4 + 0 = 2.

2

Q2. Find lambda if the vectors i + j + 2k, lambda i - j + k and 3i - 2j - k are coplanar.

Coplanar vectors have zero scalar triple product: determinant with rows (1, 1, 2), (lambda, -1, 1), (3, -2, -1) = 0.

= 1(1 + 2) - 1(-lambda - 3) + 2(-2lambda + 3) = 3 + lambda + 3 - 4lambda + 6 = 12 - 3lambda = 0, so lambda = 4.

lambda = 4

Q3. Show that the vectors 4i - j + k, 3i - 2j - k and i + j + 2k are coplanar.

Scalar triple product = determinant with rows (4, -1, 1), (3, -2, -1), (1, 1, 2) = 4(-4 + 1) + 1(6 + 1) + 1(3 + 2) = -12 + 7 + 5 = 0.

Since the scalar triple product is zero, the vectors are coplanar.

Coplanar (scalar triple product = 0)

Q4. If a = i + 2j + k, b = 3i - j - k, c = i + 8j + k then show that a x (b + c) = a x b + a x c.

b + c = 4i + 7j. a x (b + c) = determinant with rows i, j, k; 1, 2, 1; 4, 7, 0 = -7i + 4j - k.

a x b = determinant with rows i, j, k; 1, 2, 1; 3, -1, -1 = -i + 4j - 7k.
a x c = determinant with rows i, j, k; 1, 2, 1; 1, 8, 1 = -6i + 6k.

Sum: (-i + 4j - 7k) + (-6i + 6k) = -7i + 4j - k. Equal.

a x (b + c) = a x b + a x c = -7i + 4j - k

Q5. Find the volume of parallelepiped whose edges are represented by the vectors: (i) 3i + 2j - k, i - 2j + k, i + 2j - 4k; (ii) i - 2j + 3k, 2i - j - k, i + 2j - 4k.
(i) Scalar triple product = determinant with rows (3, 2, -1), (1, -2, 1), (1, 2, -4) = 3(8 - 2) - 2(-4 - 1) - 1(2 + 2) = 18 + 10 - 4 = 24.
Volume = 24 cubic units.

(ii) Scalar triple product = determinant with rows (1, -2, 3), (2, -1, -1), (1, 2, -4) = 1(4 + 2) + 2(-8 + 1) + 3(4 + 1) = 6 - 14 + 15 = 7.
Volume = 7 cubic units.

(i) 24 cubic units; (ii) 7 cubic units

Q6. Compute [i + j, i, i - j + k].
The box product is the determinant with rows (1, 1, 0), (1, 0, 0), (1, -1, 1) = 1(0 - 0) - 1(1 - 0) + 0 = -1.

-1

Q7. Find the volume of the tetrahedron whose vertices are; A(2, 1, 8), B(3, 2, 9), C(2, 1, 4) and D(3, 3, 10).

AB = (1, 1, 1), AC = (0, 0, -4), AD = (1, 2, 2).

Scalar triple product = determinant with rows (1, 1, 1), (0, 0, -4), (1, 2, 2) = 1(0 + 8) - 1(0 + 4) + 0 = 4.

Volume = (1/6) x |4| = 2/3 cubic units.

2/3 cubic units

Q8. A boat is moving at a velocity of 8 m/s in a direction 45 degrees east of north. If there is a current flowing south at 2 m/s, determine the resultant velocity of the boat.

Boat velocity: east component 8 sin 45 = 4sqrt(2), about 5.66 m/s; north component 8 cos 45 = 4sqrt(2), about 5.66 m/s.

Current: (0, -2). Resultant = (5.66, 3.66) m/s.

Speed = sqrt(32 + 13.31) = sqrt(45.31), about 6.73 m/s. Direction: tan^(-1)(5.66/3.66), about 57.1 degrees east of north.

About 6.73 m/s at about 57.1 degrees east of north

Q9. Given the parametric equations x = 2t, y = 3t^2, z = 4t - 1, sketch the curve in 3D space. (Flag: a sketch question — the curve is described exactly rather than drawn; any accurate sketch follows

this description.)

Eliminate t: t = x/2, so y = 3x^2/4 and z = 2x - 1.

The curve is the parabola y = 3x^2/4 lying in the plane z = 2x - 1. Sample points: t = -1 gives (-2, 3, -5); t = 0 gives (0, 0, -1); t = 1 gives (2, 3, 3). The vertex is at (0, 0, -1) and the parabola opens in the +y direction within its plane.

Parabola y = 3x^2/4 in the plane z = 2x - 1 (sketch described, not drawn to a unique scale)

Q10. Find the image of triangle PQR with vertices P(2, -3), Q(0, 1) and R(-1, -2) under the translation vector [5, -1]. Sketch the original and image graph. (Flag: a sketch question — both triangles are given exactly; any accurate sketch follows these coordinates.)

Add (5, -1) to each vertex: P' = (2 + 5, -3 - 1) = (7, -4); Q' = (0 + 5, 1 - 1) = (5, 0); R' = (-1 + 5, -2 - 1) = (4, -3).

The image triangle is the original shifted 5 units right and 1 unit down; corresponding sides are equal and parallel.

Image triangle P'(7, -4), Q'(5, 0), R'(4, -3) (sketch described)

Q11. A cargo delivery company needs to transport goods from point A(2, 4, 6) to point B(8, 12, 10). The cargo plane has maximum speed vector of [3, 5, 2] units per minute. Determine the optimal navigation path from point A to point B and calculate the time it takes to deliver the cargo. (Flag: the book's 'maximum speed

vector' is ambiguous; the reading used is the straight-line displacement at the vector's speed.)
Displacement AB = (6, 8, 4). The optimal (shortest) path is the straight line from A to B, of length sqrt(36 + 64 + 16) = sqrt(116), about 10.77 units.

Speed from the vector [3, 5, 2] is sqrt(9 + 25 + 4) = sqrt(38), about 6.16 units per minute.

Time = distance/speed = sqrt(116)/sqrt(38), about 1.75 minutes.

Displacement (6, 8, 4); distance sqrt(116), about 10.77 units; speed sqrt(38), about 6.16 units/min; time about 1.75 min (flag: model reading — see steps)

Q12. An airplane travels from point A(1, 2, 3) to B(4, 7, 2). Determine the direction vector of its flight path.
Direction vector = B - A = (4 - 1, 7 - 2, 2 - 3) = (3, 5, -1).

(3, 5, -1)

Q13. A ship is moving at a velocity of [2, -1, 4] m/s. Calculate its speed and direction.
Speed = sqrt(4 + 1 + 16) = sqrt(21), about 4.58 m/s.

Direction: unit vector (2i - j + 4k)/sqrt(21), i.e. direction cosines (2/sqrt(21), -1/sqrt(21), 4/sqrt(21)).

Speed sqrt(21), about 4.58 m/s; direction (2i - j + 4k)/sqrt(21)

Q14. Two cars start at the origin and travel along vectors [2, 3, 1] and [-1, 2, 4]. Find the angle between their path.

Dot product = -2 + 6 + 4 = 8. Magnitudes: sqrt(14) and sqrt(21).

cos(theta) = 8/sqrt(294) = 8/(7sqrt(6)), about 0.4667, so theta is about 62.2 degrees.

About 62.2 degrees

Review Exercise 8

Q1(i). Two vectors are said to be equal if their corresponding components are;

- ✓ equal
- unequal
- proportional
- none of these

Q1(ii). Unit vector of 2i + 2j + k is;

- (2i + 2j + k)/5
- ✓ (2i + 2j + k)/3
- (2i - 2j + k)/3
- (2i + 2j - k)/5

Q1(iii). Angle between two vectors a and b is;

- theta = cos^(-1)((|a|.|b|)/(a.b))
- ✓ theta = cos^(-1)((a.b)/(|a|.|b|))

theta = tan^(-1)((a.b)/(|a|))

theta = cosec^(-1)((a.b)/(|b|))

Q1(iv). If a = 3i + 4j + k then its magnitude is

3

✓ sqrt(26)

5

4

Q1(v). Vector in Cartesian plane is

✓ xi + yj

x + yj

xi + yj + zk

xi - yj

Q1(vi). Sum of two vectors a = 5i + 6j + k and b = 7i - 4j - 9k is

i + 2j - 8k

2i + 2j - 8k

12i + j - 8k

✓ 12i + 2j - 8k

Q1(vii). If a = 3i + 5k and b = 5i - 2j + 3k then |a - b| is

sqrt(8)

sqrt(15)

2sqrt(6)

✓ 2sqrt(3)

Q1(viii). If a = 3i - j + 2k, b = 5i + 7j + k then a.b is; (flag: the book

prints the first vector as '3i - i + 2k'; it is read as 3i - j + 2k, which gives dot product 15 - 7 + 2 = 10, option (a).)

✓ 10

7

-10

20

Q1(ix). Condition for orthogonality of two vectors is

✓ a.b = 0

a.b = cos(theta)

a.b = 1

a.b = |a||b|cos(theta)

Q1(x). Distributive law for dot product a.(b + c) is

a.b + c

a + b + c

✓ a.b + a.c

a.b - a.c

Q2. Find the vector parallel to vector -i + 7j + 3k and has magnitude 12 unit.

| -i + 7j + 3k | = sqrt(1 + 49 + 9) = sqrt(59). The unit vector is (-i + 7j + 3k)/sqrt(59).

A parallel vector of magnitude 12 is 12(-i + 7j + 3k)/sqrt(59) (or its negative, in the opposite direction).

12(-i + 7j + 3k)/sqrt(59) or its negative

Q3. Show that A(1, 2, 7), B(2, 6, 3) and C(3, 10, -1) are collinear.
AB = (1, 4, -4). AC = (2, 8, -8) = 2 AB.

Since AC is a scalar multiple of AB, the three points lie on one line.

Collinear (AC = 2 AB)

Q4. Show that the vectors 1/7(2i + 3j + 6k), 1/7(3i - 6j + 2k) and 1/7(6i + 2j + 3k) are mutually perpendicular. (Flag: as printed, verified at 300 dpi (crop zoom-r8q4crop.png), this is FALSE: with u = (2, 3, 6)/7, v = (3, -6, 2)/7, w = (6, 2, 3)/7, u.v = 0 but u.w = 36/49 and v.w = 12/49, not zero. The intended third vector is (6i + 2j - 3k)/7, for which u.w = (12 + 6 - 18)/49 = 0 and v.w = (18 - 12 - 6)/49 = 0. The proof below uses the corrected vector.)
u.v = (6 - 18 + 12)/49 = 0.

With the corrected w = (6, 2, -3)/7: u.w = (12 + 6 - 18)/49 = 0 and v.w = (18 - 12 - 6)/49 = 0.

Hence the three vectors are mutually perpendicular (and each has magnitude 1).

As printed the claim is FALSE — see flag; with the corrected third vector it holds

Q5. An airplane is heading south with a groundspeed of 300 knots. If there is a crosswind from the east at 40 knots, determine the

component of the wind that affects the airplane's northward speed. (Flag: model reading — the east wind blows along the east-west axis, perpendicular to the north-south axis.)
The wind blows from the east, i.e. along the east-west axis. Its component along the north-south axis is 40 x cos(90 degrees) = 0.

So the component of the wind affecting the northward (or southward) speed is 0 knots; the crosswind only pushes the airplane sideways (westward drift), it does not change its northward speed.

0 knots (flag: model reading — see steps)

Unit 9: Fundamental Law of Trigonometry

Exercise 9.1

Q1. Show that; (cos(alpha + beta) + cos(alpha - beta))/(sin(alpha + beta) + sin(alpha - beta)) = cot(alpha).

Numerator: cos(alpha + beta) + cos(alpha - beta) = 2cos(alpha)cos(beta).

Denominator: sin(alpha + beta) + sin(alpha - beta) = 2sin(alpha)cos(beta).

Ratio = cos(alpha)/sin(alpha) = cot(alpha).

cot(alpha) is proved

Q2. Show that; sin(alpha + beta)sin(alpha - beta) = sin^2(alpha) - sin^2(beta).

sin(alpha + beta)sin(alpha - beta) = (sin(alpha)cos(beta) + cos(alpha)sin(beta))(sin(alpha)cos(beta) - cos(alpha)sin(beta)) = sin^2(alpha)cos^2(beta) - cos^2(alpha)sin^2(beta).

= sin^2(alpha)(1 - sin^2(beta)) - (1 - sin^2(alpha))sin^2(beta) = sin^2(alpha) - sin^2(beta).

sin^2(alpha) - sin^2(beta) is proved

Q3. Prove that; (i) sin(pi + theta) = -sin(theta); (ii) sin(2pi + theta) = sin(theta); (iii) sin(3pi/2 + theta) = -cos(theta); (iv) tan(2pi + theta) = tan(theta); (v) tan(2pi - theta) = -tan(theta); (vi) tan(3pi/2 + theta) = -cot(theta).

(i) sin(pi + theta) = sin(pi)cos(theta) + cos(pi)sin(theta) = -sin(theta).

(ii) sin(2pi + theta) = sin(2pi)cos(theta) + cos(2pi)sin(theta) = sin(theta).

(iii) sin(3pi/2 + theta) = sin(3pi/2)cos(theta) + cos(3pi/2)sin(theta) = -cos(theta).

(iv) tan(2pi + theta) = tan(theta) since tan has period pi (hence also 2pi).

(v) tan(2pi - theta) = tan(-theta) = -tan(theta).

(vi) tan(3pi/2 + theta) = sin(3pi/2 + theta)/cos(3pi/2 + theta) = -cos(theta)/sin(theta) = -cot(theta).

All six allied-angle identities proved

Q4. Show that; sin(90 degrees - theta)/cosec(90 degrees - theta) + cos(90 degrees - theta)/sec(90 degrees - theta) = 1.

sin(90 - theta) = cos(theta) and cosec(90 - theta) = sec(theta) = 1/cos(theta), so the first term is cos^2(theta).

cos(90 - theta) = sin(theta) and sec(90 - theta) = cosec(theta) = 1/sin(theta), so the second term is sin^2(theta).

Sum = cos^2(theta) + sin^2(theta) = 1.

1 is proved

Q5. Evaluate the following without using calculator. (i) sin(765 degrees); (ii) tan(5pi/4); (iii) cosec(2040 degrees); (iv) sec70 degrees/cosec20 degrees + sin59 degrees/cos31 degrees; (v) 2 sin43 degrees/cos47 degrees - cot30 degrees/tan60 degrees - sqrt(2) sin45 degrees.
(i) sin(765) = sin(720 + 45) = sin45 = sqrt(2)/2.

(ii) tan(5pi/4) = tan(pi + pi/4) = tan(pi/4) = 1.

(iii) 2040 = 5 x 360 + 240, so cosec(2040) = cosec(240) = 1/sin240 = 1/(-sqrt(3)/2) = -2/sqrt(3).

(iv) sec70 = 1/cos70 = 1/sin20 = cosec20, so the first term is 1. cos31 = sin59, so the second term is 1. Total 2.

(v) cos47 = sin43, so 2sin43/cos47 = 2. cot30/tan60 = sqrt(3)/sqrt(3) = 1. sqrt(2)sin45 = 1. Total 2 - 1 - 1 = 0.

(i) sqrt(2)/2; (ii) 1; (iii) -2/sqrt(3); (iv) 2; (v) 0

Q6. Prove that; (i) sin(45 degrees - theta) = 1/sqrt(2) (cos(theta) -

sin(theta)); (ii) (cos11 degrees - sin11 degrees)/(cos11 degrees + sin11 degrees) = tan34 degrees.
(i) sin(45 - theta) = sin45 cos(theta) - cos45 sin(theta) = (1/sqrt(2))(cos(theta) - sin(theta)).

(ii) Divide numerator and denominator by cos11: (1 - tan11)/(1 + tan11) = tan(45 - 11) = tan34.

Both identities proved

Q7. Express the following in the form r sin(theta + phi) where theta and phi are in the first quadrant. (i) 8sin(theta) + 6cos(theta); (ii) 5sin(theta) + 7cos(theta); (iii) sin(theta) + cos(theta); (iv) sqrt(3)/2 cos(theta) + 1/2 sin(theta).
(i) r = sqrt(64 + 36) = 10, phi = tan^(-1)(6/8) = tan^(-1)(3/4), about 36.87 degrees, in the first quadrant. So 10sin(theta + phi).

(ii) r = sqrt(25 + 49) = sqrt(74), phi = tan^(-1)(7/5), about 54.46 degrees. So sqrt(74)sin(theta + phi).

(iii) r = sqrt(2), phi = 45 degrees. So sqrt(2)sin(theta + 45).

(iv) (1/2)sin(theta) + (sqrt(3)/2)cos(theta) = sin(theta)cos60 + cos(theta)sin60 = sin(theta + 60). So r = 1, phi = 60 degrees.

(i) 10sin(theta + phi), phi = tan^(-1)(3/4); (ii) sqrt(74)sin(theta + phi), phi = tan^(-1)(7/5); (iii) sqrt(2)sin(theta + 45 degrees); (iv) sin(theta + 60 degrees)

Q8. If sin(alpha) = 3/5 and sin(beta) = 12/13 where 0 < alpha < pi/2 and 0 < beta < pi/2 then find (i) sin(alpha + beta); (ii) tan(alpha - beta); (iii) sin(alpha - beta); (iv) cos(alpha + beta); (v) cos(alpha - beta); (vi) tan(alpha + beta). In which quadrant (alpha + beta) and (alpha - beta) lie?

cos(alpha) = 4/5, cos(beta) = 5/13, tan(alpha) = 3/4, tan(beta) = 12/5.

(i) sin(alpha + beta) = (3/5)(5/13) + (4/5)(12/13) = 63/65.

(ii) tan(alpha - beta) = (3/4 - 12/5)/(1 + 36/20) = -33/56.

(iii) sin(alpha - beta) = (3/5)(5/13) - (4/5)(12/13) = -33/65.

(iv) cos(alpha + beta) = (4/5)(5/13) - (3/5)(12/13) = -16/65.

(v) cos(alpha - beta) = (4/5)(5/13) + (3/5)(12/13) = 56/65.

(vi) tan(alpha + beta) = (3/4 + 12/5)/(1 - 36/20) = -63/16.

Quadrants: (alpha + beta) has sin > 0, cos < 0, so Q2. (alpha - beta) has sin < 0, cos > 0, so Q4.

(i) 63/65; (ii) -33/56; (iii) -33/65; (iv) -16/65; (v) 56/65; (vi) -63/16; (alpha + beta) in Q2, (alpha - beta) in Q4

Q9. If sin(alpha) = 4/5 and cos(beta) = -3/5 where pi/2 < alpha < pi

and pi < beta < 3pi/2 then find; (i) sin(alpha + beta); (ii) cos(alpha - beta); (iii) tan(alpha - beta). In which quadrant (alpha - beta) lie?
cos(alpha) = -3/5, tan(alpha) = -4/3. sin(beta) = -4/5, tan(beta) = 4/3.

(i) sin(alpha + beta) = (4/5)(-3/5) + (-3/5)(-4/5) = -12/25 + 12/25 = 0.

(ii) cos(alpha - beta) = (-3/5)(-3/5) + (4/5)(-4/5) = 9/25 - 16/25 = -7/25.

(iii) tan(alpha - beta) = (-4/3 - 4/3)/(1 - 16/9) = (-8/3)/(-7/9) = 24/7.

Quadrant: sin(alpha - beta) = (4/5)(-3/5) - (-3/5)(-4/5) = -24/25 < 0 and cos(alpha - beta) = -7/25 < 0, so (alpha - beta) lies in Q3.

(i) 0; (ii) -7/25; (iii) 24/7; (alpha - beta) in Q3

Q10. If tan(alpha) = 5/12 and sin(beta) = -4/5 where pi < alpha < 3pi/2 and 3pi/2 < beta < 2pi then find; (i) sin(alpha + beta); (ii) cos(alpha + beta); (iii) tan(alpha + beta). In which quadrant (alpha + beta) lie?
sin(alpha) = -5/13, cos(alpha) = -12/13. cos(beta) = 3/5, tan(beta) = -4/3.

(i) sin(alpha + beta) = (-5/13)(3/5) + (-12/13)(-4/5) = -15/65 + 48/65 = 33/65.

(ii) cos(alpha + beta) = (-12/13)(3/5) - (-5/13)(-4/5) = -36/65 - 20/65 = -56/65.

(iii) tan(alpha + beta) = (5/12 - 4/3)/(1 - 20/36) = (-11/12)/(16/36) = -33/16.

Quadrant: sin > 0, cos < 0, so (alpha + beta) lies in Q2.

(i) 33/65; (ii) -56/65; (iii) -33/16; (alpha + beta) in Q2

Exercise 9.2

Q1. Prove: sin(2theta)/sin(theta) - cos(2theta)/cos(theta) = sec(theta).

sin(2theta)/sin(theta) = 2cos(theta). cos(2theta)/cos(theta) = (2cos^2(theta) - 1)/cos(theta).

Difference = [2cos^2(theta) - 2cos^2(theta) + 1]/cos(theta) = 1/cos(theta) = sec(theta).

sec(theta) is proved

Q2. Prove: tan(2theta) = 2/(cot(theta) - tan(theta)).

tan(2theta) = 2tan(theta)/(1 - tan^2(theta)). Divide numerator and denominator by tan(theta): = 2/(cot(theta) - tan(theta)).

tan(2theta) = 2/(cot(theta) - tan(theta)) is proved

Q3. Prove: tan^2(theta/2) = (tan(theta) - sin(theta))/(tan(theta) +

sin(theta)).

tan(theta) - sin(theta) = sin(theta)(1 - cos(theta))/cos(theta).

tan(theta) + sin(theta) = sin(theta)(1 + cos(theta))/cos(theta).

Ratio = (1 - cos(theta))/(1 + cos(theta)) = tan^2(theta/2).

tan^2(theta/2) = (tan(theta) - sin(theta))/(tan(theta) + sin(theta)) is proved

Q4. Prove: sec^2(theta/2) = 2sec(theta)/(sec(theta) + 1).

sec^2(theta/2) = 2/(1 + cos(theta)). Multiply numerator and denominator by sec(theta): = 2sec(theta)/(sec(theta) + 1).

sec^2(theta/2) = 2sec(theta)/(sec(theta) + 1) is proved

Q5. Prove: cos(2x) = (1 - tan^2(x))/(1 + tan^2(x)).

cos(2x) = cos^2(x) - sin^2(x). Divide numerator and denominator by cos^2(x): = (1 - tan^2(x))/(1 + tan^2(x)), since 1/cos^2(x) = 1 + tan^2(x) in the denominator after dividing the identity cos^2 + sin^2 = 1.

cos(2x) = (1 - tan^2(x))/(1 + tan^2(x)) is proved

Q6. Prove: (sin(theta/2) - cos(theta/2))^2 = 1 - sin(theta).

Expand: sin^2(theta/2) + cos^2(theta/2) - 2sin(theta/2)cos(theta/2) = 1 - sin(theta), since 2sin(theta/2)cos(theta/2) = sin(theta).

(sin(theta/2) - cos(theta/2))^2 = 1 - sin(theta) is proved

Q7. Prove: tan(theta/2) = cosec(theta) - cot(theta).

cosec(theta) - cot(theta) = (1 - cos(theta))/sin(theta). Multiply numerator and denominator by (1 + cos(theta)): = sin(theta)/(1 + cos(theta)) = tan(theta/2).

tan(theta/2) = cosec(theta) - cot(theta) is proved

Q8. Prove: sin(2theta) = 2tan(theta)/(1 + tan^2(theta)).

sin(2theta) = 2sin(theta)cos(theta). Divide numerator and denominator by cos^2(theta): = 2tan(theta)/(1 + tan^2(theta)), since 1/cos^2(theta) = 1 + tan^2(theta).

sin(2theta) = 2tan(theta)/(1 + tan^2(theta)) is proved

Q9. Prove: cos(2beta) = (cot^2(beta) - 1)/(cot^2(beta) + 1).

cos(2beta) = cos^2(beta) - sin^2(beta). Divide numerator and denominator by sin^2(beta): = (cot^2(beta) - 1)/(cot^2(beta) + 1).

cos(2beta) = (cot^2(beta) - 1)/(cot^2(beta) + 1) is proved

Q10. Prove: tan(4x) = 4tan(x)(1 - tan^2(x))/(tan^4(x) - 6tan^2(x) + 1).

Let t = tan(x). Then tan(2x) = 2t/(1 - t^2) and tan(4x) = 2tan(2x)/(1 - tan^2(2x)).

Numerator: 2tan(2x) = 4t/(1 - t^2). Denominator: 1 - 4t^2/(1 -

t^2)^2 = (1 - 6t^2 + t^4)/(1 - t^2)^2.

Ratio = 4t(1 - t^2)/(t^4 - 6t^2 + 1).

tan(4x) = 4tan(x)(1 - tan^2(x))/(tan^4(x) - 6tan^2(x) + 1) is proved

Q11. Prove: cos(5alpha) = 16cos^5(alpha) - 20cos^3(alpha) + 5cos(alpha).
cos(5alpha) = cos(3alpha + 2alpha) = cos(3alpha)cos(2alpha) - sin(3alpha)sin(2alpha).

= (4c^3 - 3c)(2c^2 - 1) - (3s - 4s^3)(2sc), where c = cos(alpha), s = sin(alpha).

= 8c^5 - 10c^3 + 3c - 6s^2c + 8s^4c. With s^2 = 1 - c^2 and s^4 = 1 - 2c^2 + c^4, this simplifies to 16c^5 - 20c^3 + 5c.

cos(5alpha) = 16cos^5(alpha) - 20cos^3(alpha) + 5cos(alpha) is proved

Q12. Prove: sin(4x) = 4sin(x)(2cos^3(x) - cos(x)).
sin(4x) = 2sin(2x)cos(2x) = 4sin(x)cos(x)cos(2x) = 4sin(x)cos(x)(2cos^2(x) - 1) = 4sin(x)(2cos^3(x) - cos(x)).

sin(4x) = 4sin(x)(2cos^3(x) - cos(x)) is proved

Q13. If cos(theta) = -4/5 and pi/2 < theta < pi then find the

following: (i) sin(theta/2); (ii) cos(theta/2); (iii) tan(theta/2); (iv) sin(2theta); (v) cos(2theta); (vi) tan(2theta); and also find in which quadrant theta/2 and 2theta lies.

sin(theta) = 3/5. Since pi/2 < theta < pi, pi/4 < theta/2 < pi/2, so theta/2 is in Q1 with positive sin and cos.

(i) sin(theta/2) = sqrt((1 - cos(theta))/2) = sqrt((1 + 4/5)/2) = 3/sqrt(10).

(ii) cos(theta/2) = sqrt((1 + cos(theta))/2) = sqrt((1/5)/2) = 1/sqrt(10).

(iii) tan(theta/2) = 3.

(iv) sin(2theta) = 2(3/5)(-4/5) = -24/25.

(v) cos(2theta) = 2(16/25) - 1 = 7/25.

(vi) tan(2theta) = -24/7.

2theta lies in (pi, 2pi) with sin < 0, cos > 0, so in Q4.

(i) 3/sqrt(10); (ii) 1/sqrt(10); (iii) 3; (iv) -24/25; (v) 7/25; (vi) -24/7; theta/2 in Q1, 2theta in Q4

Exercise 9.3

Q1. Express the following products as sums or differences; (i) 2sin(8theta)cos(4theta); (ii) cos60 degrees sin30 degrees; (iii)

2sin(x + y)cos(x - y); (iv) -2sin((R + S)/2)sin((R - S)/2); (v) sin(x + 45 degrees)sin(x - 45 degrees); (vi) 2cos(2x + 30 degrees)cos(2x - 30 degrees).

(i) 2sin(8theta)cos(4theta) = sin(12theta) + sin(4theta).

(ii) cos60 sin30 = (1/2)[sin90 - sin30] = (1/2)(1 - 1/2) = 1/4.

(iii) 2sin(x + y)cos(x - y) = sin(2x) + sin(2y).

(iv) -2sin((R + S)/2)sin((R - S)/2) = -(cos(S) - cos(R)) = cos(R) - cos(S).

(v) sin(x + 45)sin(x - 45) = (1/2)[cos90 - cos(2x)] = -(1/2)cos(2x).

(vi) 2cos(2x + 30)cos(2x - 30) = cos(4x) + cos60 = cos(4x) + 1/2.

(i) sin(12theta) + sin(4theta); (ii) 1/4; (iii) sin(2x) + sin(2y); (iv) cos(R) - cos(S); (v) -(1/2)cos(2x); (vi) cos(4x) + 1/2

Q2. Express the following sums or difference as products. (i) sin(2y) - sin(2z); (ii) cos(3theta/2) + cos(theta/2); (iii) cos12 degrees + cos48 degrees; (iv) cos(x + 30 degrees) - cos(x - 30 degrees); (v) 1 + sin(2x) (Hint: sin90 degrees = 1); (vi) sin2(x + 40 degrees) + sin2(x - 40 degrees).

(i) sin(2y) - sin(2z) = 2cos(y + z)sin(y - z).

(ii) cos(3theta/2) + cos(theta/2) = 2cos(theta)cos(theta/2).

(iii) cos12 + cos48 = 2cos30 cos18 = sqrt(3)cos18.

(iv) cos(x + 30) - cos(x - 30) = -2sin(x)sin30 = -sin(x).

(v) 1 + sin(2x) = sin90 + sin(2x) = 2sin(45 + x)cos(45 - x).

(vi) sin2(x + 40) + sin2(x - 40) = 2sin(2x)cos80.

(i) 2cos(y + z)sin(y - z); (ii) 2cos(theta)cos(theta/2); (iii) sqrt(3)cos18 degrees; (iv) -sin(x); (v) 2sin(45 degrees + x)cos(45 degrees - x); (vi) 2sin(2x)cos80 degrees

Q3. Prove the following identities. (i) (cos(alpha) + cos(beta))/(sin(alpha) - sin(beta)) = cot((alpha - beta)/2); (ii) (sin(A) - sin(B))/(sin(A) + sin(B)) = cot((A + B)/2)tan((A - B)/2); (iii) (sin(8x) + sin(2x))/(cos(8x) + cos(2x)) = tan(5x); (iv) (sin(theta) + sin(3theta) + sin(5theta) + sin(7theta))/(cos(theta) + cos(3theta) + cos(5theta) + cos(7theta)) = tan(4theta).
(i) Numerator = 2cos((alpha + beta)/2)cos((alpha - beta)/2); denominator = 2cos((alpha + beta)/2)sin((alpha - beta)/2). Ratio = cot((alpha - beta)/2).

(ii) Numerator = 2cos((A + B)/2)sin((A - B)/2); denominator = 2sin((A + B)/2)cos((A - B)/2). Ratio = cot((A + B)/2)tan((A - B)/2).

(iii) Numerator = 2sin(5x)cos(3x); denominator = 2cos(5x)cos(3x). Ratio = tan(5x).

(iv) Group: (sin(theta) + sin(7theta)) + (sin(3theta) + sin(5theta)) = 2sin(4theta)cos(3theta) + 2sin(4theta)cos(theta) = 2sin(4theta)(cos(3theta) + cos(theta)) = 4sin(4theta)cos(2theta)cos(theta). Similarly the denominator is 4cos(4theta)cos(2theta)cos(theta). Ratio = tan(4theta).

All four identities proved

Q4. Prove that; (i) sin10 degrees sin30 degrees sin50 degrees sin70 degrees = 1/16; (ii) sin(pi/9)sin(2pi/9)sin(pi/3)sin(4pi/9) = 3/16; (iii) cos20 degrees cos40 degrees cos80 degrees = 1/8.
(i) sin10 sin50 sin70 = (1/2)[cos40 - cos60]sin70? Better: sin50 sin70 = (1/2)[cos20 + 1/2]; times sin10 gives (1/2)sin10 cos20 + (1/4)sin10 = (1/4)[sin30 - sin10] + (1/4)sin10 = 1/8. Multiply by sin30 = 1/2: 1/16.

(ii) sin20 sin40 sin80 = (1/2)[cos20 - 1/2]sin80 = (1/2)sin80 cos20 - (1/4)sin80 = (1/4)[sin100 + sin60] - (1/4)sin80 = sqrt(3)/8, since sin100 = sin80. Multiply by sin60 = sqrt(3)/2: 3/16.

(iii) Multiply by sin20: sin20 cos20 cos40 cos80 = (1/2)sin40 cos40 cos80 = (1/4)sin80 cos80 = (1/8)sin160 = (1/8)sin20. Divide by sin20: 1/8.

(i) 1/16; (ii) 3/16; (iii) 1/8

Review Exercise 9

Q1(i). Distance between two points A(x1, y1) and B(x2, y2) can be

determined by the formula; (flag: NO option is exactly correct — the true formula is sqrt((x2 - x1)^2 + (y2 - y1)^2). Option (b) is the intended one with the square on (y2 - y1) omitted in print.)

- sqrt((x2 - x1)^2 + (y2 + y1)^2)
- ✓ sqrt((x2 - x1)^2 + (y2 - y1))
- sqrt((x2 - x1) + (y2 - y1)^2)
- sqrt((x2 + x1)^2 + (y2 + y1)^2)

Q1(ii). sin(alpha - beta) is equal to;

- sin(alpha)sin(beta) - cos(alpha)cos(beta)
- sin(alpha)sin(beta) + cos(alpha)cos(beta)
- sin(alpha)cos(beta) + cos(alpha)sin(beta)
- ✓ sin(alpha)cos(beta) - cos(alpha)sin(beta)

Q1(iii). sin(2alpha) is equal to;

- sin(alpha)cos(alpha)
- ✓ 2sin(alpha)cos(alpha)
- (1 + cos(alpha))/2
- (1 - cos(alpha))/2

Q1(iv). tan(pi + theta) =

- cot(theta)
- ✓ tan(theta)
- -tan(theta)
- sin(theta)

Q1(v). 3sin(alpha) - 4sin^3(alpha) is equal to

- ✓ sin(3alpha)

-sin(3alpha)

cos(3alpha)

3sin(alpha)

Q1(vi). 2sin((alpha + beta)/2)cos((alpha - beta)/2) is equal to

sin(alpha) - sin(beta)

sin(alpha) + cos(beta)

sin(alpha)cos(beta)

✓ sin(alpha) + sin(beta)

Q1(vii). cos(2alpha) is equal to;

✓ 2cos^2(alpha) - 1

2sin(10alpha)cos(6alpha)

sin(10alpha)cos(6alpha)

cos(10alpha)sin(6alpha)

Q1(viii). tan(2x) is equal to;

sin(x)cos(x)

2tan(x)/(1 + tan^2(x))

tan(x)/(1 + tan^2(x))

✓ 2tan(x)/(1 - tan^2(x))

Q1(ix). cos(3theta) is equal to; (flag: options (a) and (d) are the same expression, 4cos^3(theta) - 3cos(theta), written two ways.)

✓ 4cos^3(theta) - 3cos(theta)

cos^3(theta) - 3cos(theta)

4cos^3(theta) - cos(theta)

4cos(theta)^3 - 3cos(theta)

Q1(x). sin(2pi - x) is equal to;

✓ -sin(x)

sin(x)

cos(x)

-cos(x)

Q2. Prove the following; (i) cos(pi/4 - x)cos(pi/4 - y) - sin(pi/4 - x)sin(pi/4 - y) = sin(x + y); (ii) cos(3pi/4 + x) - cos(3pi/4 - x) = -sqrt(2)sin(x); (iii) sin^2(6x) - sin^2(4x) = 4sin(5x)cos(5x)sin(x)cos(x); (iv) (sin(5x) + sin(3x))/(cos(5x) + cos(3x)) = tan(4x); (v) (cos(4x) + cos(3x) + cos(2x))/(sin(4x) + sin(3x) + sin(2x)) = cot(3x); (vi) cot(alpha - beta) = (cot(alpha)cot(beta) + 1)/(cot(beta) - cot(alpha)); (vii) cos(2x)cos(x/2) - cos(3x)cos(9x/2) = sin(10x)sin(5x). (Flag: (vii) as printed is FALSE — at x = pi/10 the left side is about 0.7071 while sin(10x)sin(5x) = 0. The correct identity is cos(2x)cos(x/2) - cos(3x)cos(9x/2) = sin(5x)sin(5x/2), proved below.)
(i) Left side = cos((pi/4 - x) + (pi/4 - y)) = cos(pi/2 - (x + y)) = sin(x + y).

(ii) cos(3pi/4 + x) - cos(3pi/4 - x) = -2sin(3pi/4)sin(x) = -2(sqrt(2)/2)sin(x) = -sqrt(2)sin(x).

(iii) sin^2(6x) - sin^2(4x) = (sin(6x) - sin(4x))(sin(6x) + sin(4x)) = (2cos(5x)sin(x))(2sin(5x)cos(x)) = 4sin(5x)cos(5x)sin(x)cos(x).

(iv) Numerator = 2sin(4x)cos(x); denominator = 2cos(4x)cos(x).

Ratio = tan(4x).

(v) Numerator = (cos(4x) + cos(2x)) + cos(3x) = 2cos(3x)cos(x) + cos(3x) = cos(3x)(2cos(x) + 1). Denominator = (sin(4x) + sin(2x)) + sin(3x) = 2sin(3x)cos(x) + sin(3x) = sin(3x)(2cos(x) + 1). Ratio = cot(3x).

(vi) cot(alpha - beta) = 1/tan(alpha - beta) = (1 + tan(alpha)tan(beta))/(tan(alpha) - tan(beta)). Dividing numerator and denominator by tan(alpha)tan(beta) gives (cot(alpha)cot(beta) + 1)/(cot(beta) - cot(alpha)).

(vii) Corrected: cos(2x)cos(x/2) = (1/2)[cos(5x/2) + cos(3x/2)] and cos(3x)cos(9x/2) = (1/2)[cos(15x/2) + cos(3x/2)]. Difference = (1/2)[cos(5x/2) - cos(15x/2)] = (1/2)[-2sin(5x)sin(-5x/2)] = sin(5x)sin(5x/2).

(i)-(vi) proved; (vii) corrected — see flag

Unit 10: Trigonometric Functions and Their Graph

Exercise 10.1

Q1. Find the domain and range of each of the following function: (i) cos(2x); (ii) 5tan(x); (iii) 5sin(5x); (iv) cosec(3x); (v) 3cot(2x); (vi) sin(pi x); (vii) 3sin(2x); (viii) sec(2 pi x + 5).

(i) cos(2x) is defined for all real x and takes values in [-1, 1].

(ii) tan(x) is undefined at pi/2 + k pi; elsewhere 5tan(x) takes all real values.

(iii) sin(5x) is defined everywhere with range [-1, 1], so 5sin(5x) has range [-5, 5].

(iv) cosec(3x) is undefined where sin(3x) = 0, i.e. x = k pi/3; its range is (-inf, -1] union [1, inf).

(v) cot(2x) is undefined where sin(2x) = 0, i.e. x = k pi/2; its range is all reals.

(vi) sin(pi x) is defined everywhere with range [-1, 1].

(vii) 3sin(2x) is defined everywhere with range [-3, 3].

(viii) sec(2 pi x + 5) is undefined where 2 pi x + 5 = pi/2 + k pi, i.e. x = 1/4 + k/2 - 5/(2 pi); its range is (-inf, -1] union [1, inf).

(i) domain all reals, range [-1, 1]; (ii) domain x not equal to pi/2 + k pi, range all reals; (iii) domain all reals, range [-5, 5]; (iv) domain x not equal to k pi/3, range (-inf, -1] union [1, inf); (v) domain x not equal to k pi/2, range all reals; (vi) domain all reals, range [-1, 1]; (vii) domain all reals, range [-3, 3]; (viii) domain 2 pi x + 5 not equal to pi/2 + k pi, range (-inf, -1] union [1, inf)

Q2. Find the values of the following functions. (i) cos(-pi/6); (ii) sec(-pi/4); (iii) tan(-pi/3); (iv) cosec(-pi/2); (v) sin(-pi/3); (vi) cot(-pi/4).

(i) cos is even: cos(-pi/6) = cos(pi/6) = sqrt(3)/2.

(ii) sec(-pi/4) = sec(pi/4) = sqrt(2).

(iii) tan is odd: tan(-pi/3) = -tan(pi/3) = -sqrt(3).

(iv) cosec(-pi/2) = -cosec(pi/2) = -1.

(v) sin(-pi/3) = -sin(pi/3) = -sqrt(3)/2.

(vi) cot(-pi/4) = -cot(pi/4) = -1.

(i) sqrt(3)/2; (ii) sqrt(2); (iii) -sqrt(3); (iv) -1; (v) -sqrt(3)/2; (vi) -1

Q3. Find the period of the following functions. (i) sin(3x); (ii) cos(4x); (iii) cot(x/3); (iv) cosec(x/5); (v) sec(8x); (vi) tan(x/6); (vii) sqrt(5)sin(3x/2); (viii) 3tan(sqrt(2)x); (ix) 3cos(x/3); (x) 3sin(x/4).

The period of sin(bx), cos(bx), sec(bx), cosec(bx) is 2pi/|b|; of tan(bx), cot(bx) it is pi/|b|. Amplitude does not change the period.

(i) 2pi/3. (ii) 2pi/4 = pi/2. (iii) pi/(1/3) = 3pi. (iv) 2pi/(1/5) = 10pi. (v) 2pi/8 = pi/4. (vi) pi/(1/6) = 6pi. (vii) 2pi/(3/2) = 4pi/3. (viii) pi/sqrt(2). (ix) 2pi/(1/3) = 6pi. (x) 2pi/(1/4) = 8pi.

(i) 2pi/3; (ii) pi/2; (iii) 3pi; (iv) 10pi; (v) pi/4; (vi) 6pi; (vii) 4pi/3; (viii) pi/sqrt(2); (ix) 6pi; (x) 8pi

Q4. Find the maximum and minimum value of each of the following functions. (i) y = 3 + 4sin(theta); (ii) y = 1/3 - 5cos(theta); (iii) y = 4 - 1/2 sin(2theta + 3); (iv) y = -3 + 4cos(theta - 15); (v) y = 1/(15 - 10sin(2theta - 3)); (vi) y = 1/(9 + 5cos(4theta - 1)).
(i) sin ranges over [-1, 1]: max = 3 + 4 = 7, min = 3 - 4 = -1.

(ii) cos ranges over [-1, 1]: max = 1/3 + 5 = 16/3, min = 1/3 - 5 = -14/3.

(iii) sin ranges over [-1, 1]: max = 4 + 1/2 = 9/2, min = 4 - 1/2 = 7/2.

(iv) cos ranges over [-1, 1]: max = -3 + 4 = 1, min = -3 - 4 = -7.

(v) The denominator 15 - 10sin ranges over [5, 25], so y ranges over [1/25, 1/5]: max 1/5, min 1/25.

(vi) The denominator 9 + 5cos ranges over [4, 14], so y ranges over

[1/14, 1/4]: max 1/4, min 1/14.

(i) max 7, min -1; (ii) max 16/3, min -14/3; (iii) max 9/2, min 7/2; (iv) max 1, min -7; (v) max 1/5, min 1/25; (vi) max 1/4, min 1/14

Exercise 10.2

Q1-Q8. Draw the graph of the following trigonometric functions for the given interval. 1. y = 2cos(x), 0 <= x <= 2pi; 2. y = -sin(x), 0 <= x <= 2pi; 3. y = sin(2x), 0 <= x <= 2pi; 4. y = tan(x), -pi <= x <= 2pi; 5. y = 2sec(2x), 0 <= x <= 2pi; 6. y = sec(x/2), -pi <= x <= 2pi; 7. y = -2cot(x), -2pi <= x <= 2pi; 8. y = -cosec(x/2), -pi <= x <= 2pi. (Flag: these are sketch questions; each graph is described exactly by its key points, asymptotes and shape, which fully determines an accurate sketch.)

1. y = 2cos(x): amplitude 2, period 2pi. Key points: (0, 2), (pi/2, 0), (pi, -2), (3pi/2, 0), (2pi, 2). One cosine wave.

2. y = -sin(x): reflection of sin(x). Key points: (0, 0), (pi/2, -1), (pi, 0), (3pi/2, 1), (2pi, 0).

3. y = sin(2x): period pi, two full waves on [0, 2pi]. Key points: (0, 0), (pi/4, 1), (pi/2, 0), (3pi/4, -1), (pi, 0), (5pi/4, 1), (3pi/2, 0), (7pi/4, -1), (2pi, 0).

4. y = tan(x): branches with vertical asymptotes at x = -pi/2, pi/2, 3pi/2; zeros at x = -pi, 0, pi, 2pi; each branch rises from -inf to +inf.

5. y = 2sec(2x): period pi; vertical asymptotes at x = pi/4, 3pi/4,

5pi/4, 7pi/4. Key values: (0, 2), (pi/2, -2), (pi, 2), (3pi/2, -2), (2pi, 2); U-shaped branches above 2 and below -2.

6. y = sec(x/2): period 4pi; vertical asymptotes at x = -pi and pi (within the interval). Key values: (0, 1), (2pi, -1); the branch through (0, 1) is U-shaped above 1, the branch through (2pi, -1) is inverted below -1.

7. y = -2cot(x): vertical asymptotes at x = -2pi, -pi, 0, pi, 2pi; zeros at x = -3pi/2, -pi/2, pi/2, 3pi/2; each branch rises from -inf to +inf (the negative of the falling cotangent branches).

8. y = -cosec(x/2): vertical asymptotes at x = 0 and 2pi (where sin(x/2) = 0). Key values: (-pi, 1), (pi, -1); the branch on (-pi, 0) rises above 1, the branch on (0, 2pi) falls below -1.

Graphs described by key points and asymptotes (flag: sketch questions — see steps)

Exercise 10.3

Q1. Without drawing, guess the graph of the following functions. Also find its period. (i) y = cos(2theta); (ii) y = sin(theta/3); (iii) y = cos(4theta); (iv) y = sin(5theta); (v) y = cos(theta/4); (vi) y = sin(4theta). (Flag: the parts are labelled (i), (ii), (iii), (iv), (vi), (vi) in print — the second (vi) is read as (v).)

(i) y = cos(2theta): a cosine wave compressed to period 2pi/2 = pi.

(ii) y = sin(theta/3): a sine wave stretched to period 2pi/(1/3) = 6pi.

(iii) y = cos(4theta): a cosine wave compressed to period 2pi/4 = pi/2.

(iv) y = sin(5theta): a sine wave compressed to period 2pi/5.

(v) y = cos(theta/4): a cosine wave stretched to period 2pi/(1/4) = 8pi.

(vi) y = sin(4theta): a sine wave compressed to period 2pi/4 = pi/2.

(i) cosine wave, period pi; (ii) sine wave, period 6pi; (iii) cosine wave, period pi/2; (iv) sine wave, period 2pi/5; (v) cosine wave, period 8pi; (vi) sine wave, period pi/2 (flag: part numbering misprinted — see steps)

Q2. By using symmetry properties of the sine, cosine and tangent functions to show the following identities. (i) sin(pi/2 + theta) = cos(theta); (ii) cos(pi/2 - theta) = sin(theta); (iii) sin(3pi/2 - theta) = -cos(theta); (iv) cos(3pi/2 + theta) = sin(theta); (v) sin(pi - theta) = cos(theta); (vi) cos(pi + theta) = -cos(theta); (vii) tan(pi - theta) = -tan(theta); (viii) tan(2pi + theta) = tan(theta); (ix) sin(3pi/2 + theta) = -cos(theta); (x) cos(2pi - theta) = cos(theta). (Flag: (v) as printed is FALSE — sin(pi - theta) = sin(theta), not cos(theta). The correct identity is proved below.)

(i) sin(pi/2 + theta) = sin(pi/2)cos(theta) + cos(pi/2)sin(theta) = cos(theta).

(ii) cos(pi/2 - theta) = cos(pi/2)cos(theta) + sin(pi/2)sin(theta) = sin(theta).

(iii) sin(3pi/2 - theta) = sin(3pi/2)cos(theta) - cos(3pi/2)sin(theta) = -cos(theta).

(iv) cos(3pi/2 + theta) = cos(3pi/2)cos(theta) - sin(3pi/2)sin(theta) = sin(theta).

(v) Corrected: sin(pi - theta) = sin(pi)cos(theta) - cos(pi)sin(theta) = sin(theta). (The printed 'cos(theta)' is wrong.)

(vi) cos(pi + theta) = cos(pi)cos(theta) - sin(pi)sin(theta) = -cos(theta).

(vii) tan(pi - theta) = sin(pi - theta)/cos(pi - theta) = sin(theta)/(-cos(theta)) = -tan(theta).

(viii) tan(2pi + theta) = tan(theta) by the period pi of tan.

(ix) sin(3pi/2 + theta) = sin(3pi/2)cos(theta) + cos(3pi/2)sin(theta) = -cos(theta).

(x) cos(2pi - theta) = cos(2pi)cos(theta) + sin(2pi)sin(theta) = cos(theta).

(i)-(iv) and (vi)-(x) proved; (v) as printed is FALSE — see flag

Q3. For any integer k, deduce that; (i) cos(theta + 2k pi) = cos(theta); (ii) tan(theta + k pi) = tan(theta); (iii) sec(theta + 2k pi) = sec(theta); (iv) cosec(theta + 2k pi) = cosec(theta); (v) cot(theta + k pi) = cot(theta).

(i) cos(theta + 2k pi) = cos(theta)cos(2k pi) - sin(theta)sin(2k pi) = cos(theta), since cos(2k pi) = 1 and sin(2k pi) = 0.

(ii) tan has period pi, so tan(theta + k pi) = tan(theta).

(iii) sec(theta + 2k pi) = 1/cos(theta + 2k pi) = 1/cos(theta) = sec(theta).

(iv) cosec(theta + 2k pi) = 1/sin(theta + 2k pi) = 1/sin(theta) = cosec(theta).

(v) cot has period pi, so cot(theta + k pi) = cot(theta).

All five periodicity identities proved

Exercise 10.4

Q1. A ship is 20 km away from a light house. At a certain point, the angle of elevation to the top of the light house is 45 degrees. Find the height of the light house.

Height = 20 x tan(45) = 20 x 1 = 20 km.

20 km

Q2. A Ferris wheel starts from the bottom. If it takes 30 seconds to

reach the top and has a diameter of 40 metres, find the equation of its height as a function of time [h(t) = a sin(bx + c)]. (Flag: the template h(t) = a sin(bt + c) alone cannot give height 0 at t = 0 and 40 at t = 30; the vertical shift +20 is needed, as shown.)
Radius = 20 m. Time to the top is half a period, so the period is 60 s and b = 2pi/60 = pi/30.

Height from the ground: h(t) = 20 - 20cos(pi t/30). Check: h(0) = 0 (bottom), h(30) = 40 (top).

In sine form: h(t) = 20 + 20sin(pi t/30 - pi/2), i.e. a = 20, b = pi/30, c = -pi/2, plus the vertical shift 20.

h(t) = 20 - 20cos(pi t/30) = 20 + 20sin(pi t/30 - pi/2) (flag: the book's template lacks the vertical shift — see steps)

Q3. The temperature follows a sinusoidal pattern with an amplitude of 15 degrees C and a period of 24 hours. Write the equation for temperature. (Flag: the book gives no mean temperature or phase, so the equation is given with the mean as a constant; with mean 0 and starting at the mean rising, T(t) = 15sin(pi t/12).)
Amplitude 15, period 24 hours gives angular frequency 2pi/24 = pi/12.

Equation: T(t) = T_mean + 15sin(pi t/12 + phase). With mean 0 and zero phase: T(t) = 15sin(pi t/12), where t is in hours.

T(t) = T_mean + 15sin(pi t/12) (flag: mean temperature not given)

Q4. An airplane ascends at an angle of 30 degrees to the horizontal. If its altitude increases at a rate of 500m/min, find the horizontal speed.

Vertical speed = v sin30 = 500, so the airspeed v = 500/(1/2) = 1000 m/min.

Horizontal speed = v cos30 = 1000 x sqrt(3)/2 = 500sqrt(3), about 866 m/min.

500sqrt(3), about 866 m/min

Q5. Express f(x) = sin(3x) as a sum of difference of trigonometric functions.

Write 3x = 2x + x and expand: sin(3x) = sin(2x)cos(x) + cos(2x)sin(x).

Equivalently, sin(3x) = 3sin(x) - 4sin^3(x).

sin(3x) = sin(2x)cos(x) + cos(2x)sin(x) = 3sin(x) - 4sin^3(x)

Q6. Determine if g(x) = cos(2x) is an even, odd or neither function.

g(-x) = cos(-2x) = cos(2x) = g(x). Hence g is an even function.

Even

Q7. Find the period of h(x) = tan(4x).

The period of tan(bx) is pi/|b| = pi/4.

pi/4

Q8. Given that P(x) = 3 - 4sin(x), find the extreme values.

sin(x) ranges over [-1, 1]. Maximum at sin(x) = -1: 3 + 4 = 7.

Minimum at sin(x) = 1: 3 - 4 = -1.

Maximum 7, minimum -1

Q9. Express r(x) = cos^2(3x) as a sum or difference of trigonometric functions.

cos^2(3x) = (1 + cos(6x))/2.

(1 + cos(6x))/2

Q10. If S(x) = tan(x), determine whether the function is even, odd or neither even nor odd.

S(-x) = tan(-x) = -tan(x) = -S(x). Hence tan is an odd function.

Odd

Review Exercise 10

Q1(i). The range of both the sine and cosine function is;

- -inf < y < inf
- -1 < y < 1
- ✓ -1 <= y <= 1
- 0 <= y <= 1

Q1(ii). If f(-x) = f(x), then the function is;

- periodic
- ✓ even
- odd
- rational

Q1(iii). If f(x + p) = f(x) then the function is; (flag: printed as f(n + p) = f(x), read as f(x + p) = f(x).)

- linear
- quadratic
- even
- ✓ periodic

Q1(iv). The period of tan(x) is;

- pi/2
- ✓ pi
- 3pi/2
- 2pi

Q1(v). The range of both the tangent and cotangent function is

- 0 < y < inf
- -inf < y < 0
- -1 <= y <= 1
- ✓ -inf < y < inf

Q1(vi). If f(-x) = -f(x) then the function is;

- even
- ✓ odd

algebraic

periodic

Q1(vii). 1 + tan^2(-theta) is;

cosec^2(theta)

✓ sec^2(theta)

cos^2(theta)

sin^2(theta)

Q1(viii). If y = sin(x) then its domain is;

-inf < x < 0

0 < x < inf

✓ -inf < x < inf

none of these

Q1(ix). The period of sec(x) is;

pi

✓ 2pi

3pi

4pi

Q1(x). The minimum value of (a + b cos(theta)) is;

a + |b|

|a| + |b|

✓ a - |b|

|a| - |b|

Q2. Find the maximum and minimum value of the each of the

following functions. (i) y = 5 - 7cos(theta); (ii) y = 4 + 3sin(2theta - 5).

(i) cos ranges over [-1, 1]: max = 5 + 7 = 12, min = 5 - 7 = -2.

(ii) sin ranges over [-1, 1]: max = 4 + 3 = 7, min = 4 - 3 = 1.

(i) max 12, min -2; (ii) max 7, min 1

Q3. Sketch the graph of y = 2sin(x), 0 <= x <= 2pi. (Flag: a sketch question — the graph is described exactly by its key points.)

Amplitude 2, period 2pi. Key points: (0, 0), (pi/2, 2), (pi, 0), (3pi/2, -2), (2pi, 0). Join with a smooth wave through these points.

Sine wave of amplitude 2 on [0, 2pi] (sketch described)

Unit 11: Inverse Trigonometric Function

Exercise 11.1

Q1. Without using tables / calculator, find the value of; (i) Sin^(-1)(1/sqrt(2)); (ii) Sin^(-1)(-1/2); (iii) Cos^(-1)(-1/sqrt(2)); (iv) Cos^(-1)(1/2); (v) Tan^(-1)(1/sqrt(3)); (vi) Tan^(-1)(-1).

Principal values: (i) Sin^(-1)(1/sqrt(2)) = pi/4. (ii) Sin^(-1)(-1/2) = -pi/6. (iii) Cos^(-1)(-1/sqrt(2)) = 3pi/4. (iv) Cos^(-1)(1/2) = pi/3. (v) Tan^(-1)(1/sqrt(3)) = pi/6. (vi) Tan^(-1)(-1) = -pi/4.

(i) pi/4; (ii) -pi/6; (iii) 3pi/4; (iv) pi/3; (v) pi/6; (vi) -pi/4

Q2. Without using tables / calculator, find the value of; (i) Cot^(-1)(1/sqrt(3)); (ii) Cot^(-1)(-1); (iii) Sec^(-1)(2); (iv) Sec^(-1)(1); (v) Cosec^(-1)(-2/sqrt(3)); (vi) Cosec^(-1)(-1).

(i) Cot^(-1)(1/sqrt(3)) = pi/3, since cot(pi/3) = 1/sqrt(3). (ii) Cot^(-1)(-1) = 3pi/4. (iii) Sec^(-1)(2) = pi/3, since sec(pi/3) = 2. (iv) Sec^(-1)(1) = 0. (v) Cosec^(-1)(-2/sqrt(3)) = -pi/3, since sin(-pi/3) = -sqrt(3)/2. (vi) Cosec^(-1)(-1) = -pi/2.

(i) pi/3; (ii) 3pi/4; (iii) pi/3; (iv) 0; (v) -pi/3; (vi) -pi/2

Q3. Find the value of each of the following; (i) Cos(Sin^(-1)(sqrt(3)/2)); (ii) Cosec(Cos^(-1)(1/2)); (iii) Tan(Cos^(-1)(-1)); (iv) Sec(Tan^(-1)(1)); (v) Cot(Tan^(-1)(-sqrt(3))); (vi) Sin(Sec^(-1)(sqrt(2))).

(i) Sin^(-1)(sqrt(3)/2) = pi/3, so Cos(pi/3) = 1/2.

(ii) Cos^(-1)(1/2) = pi/3, so Cosec(pi/3) = 2/sqrt(3).

(iii) Cos^(-1)(-1) = pi, so Tan(pi) = 0.

(iv) Tan^(-1)(1) = pi/4, so Sec(pi/4) = sqrt(2).

(v) Tan^(-1)(-sqrt(3)) = -pi/3, so Cot(-pi/3) = -1/sqrt(3).

(vi) Sec^(-1)(sqrt(2)) = pi/4, so Sin(pi/4) = 1/sqrt(2).

(i) 1/2; (ii) 2/sqrt(3); (iii) 0; (iv) sqrt(2); (v) -1/sqrt(3); (vi) 1/sqrt(2)

Q4. Without using tables / calculator, show that; (i) Tan^(-1)(12/5) = Cot^(-1)(5/12); (ii) Tan^(-1)(sqrt(3)) = Sin^(-1)(sqrt(3)/2); (iii) Cosec^(-1)(2/sqrt(3)) = Cos^(-1)(1/2); (iv) Cos^(-1)(4/9) = Cot^(-1)(4/sqrt(65)).

(i) If tan(theta) = 12/5 with theta in (0, pi/2), then cot(theta) = 5/12, so theta = Cot^(-1)(5/12).

(ii) Tan^(-1)(sqrt(3)) = pi/3 = Sin^(-1)(sqrt(3)/2).

(iii) Cosec^(-1)(2/sqrt(3)) = pi/3 = Cos^(-1)(1/2).

(iv) Let theta = Cos^(-1)(4/9). Then sin(theta) = sqrt(1 - 16/81) = sqrt(65)/9, so cot(theta) = (4/9)/(sqrt(65)/9) = 4/sqrt(65). Hence theta = Cot^(-1)(4/sqrt(65)).

All four identities proved

Q5. Draw the graph of the following functions. (i) y = Sin^(-1)(x) for -1 <= x <= 1; (ii) y = Cot^(-1)(x) for -inf < x < inf; (iii) y = Sec^(-1)(x) for -inf < x < inf. (Flag: sketch questions — each graph is described exactly by its key points and branches.)

(i) y = Sin^(-1)(x): increasing curve through (-1, -pi/2), (-0.5, -pi/6), (0, 0), (0.5, pi/6), (1, pi/2); steep (vertical tangent) at x = +/-1.

(ii) y = Cot^(-1)(x): decreasing curve from pi (as x -> -inf) to 0 (as x -> +inf), through (0, pi/2), (1, pi/4), (-1, 3pi/4).

(iii) y = Sec^(-1)(x): defined for |x| >= 1. Branch on [1, inf): from (1, 0) rising towards pi/2 as x -> inf. Branch on (-inf, -1]: from (-1, pi) falling towards pi/2 as x -> -inf.

Graphs described by key points (flag: sketch questions — see steps)

Exercise 11.2

Q1. Prove: Sin^(-1)(7/25) + Sin^(-1)(5/13) = Cos^(-1)(253/325).

Let A = Sin^(-1)(7/25), B = Sin^(-1)(5/13). Then cos(A) = 24/25, cos(B) = 12/13.

Cos(A + B) = cos(A)cos(B) - sin(A)sin(B) = (24 x 12 - 7 x 5)/(25 x 13) = 253/325.

Since A + B is in (0, pi), A + B = Cos^(-1)(253/325).

Cos^(-1)(253/325) is proved

Q2. Prove: Tan^(-1)(1/6) + Tan^(-1)(1/7) = Tan^(-1)(13/41).

Tan^(-1)(1/6) + Tan^(-1)(1/7) = Tan^(-1)((1/6 + 1/7)/(1 - 1/42)) = Tan^(-1)((13/42)/(41/42)) = Tan^(-1)(13/41). Both angles are small, so their sum is in (0, pi/2) and no adjustment is needed.

Tan^(-1)(13/41) is proved

Q3. Prove: 2Tan^(-1)(2/3) = Cos^(-1)(5/13).

Let theta = Tan^(-1)(2/3). Then cos(theta) = 3/sqrt(13) and cos(2theta) = 2cos^2(theta) - 1 = 18/13 - 1 = 5/13.

Since 2theta is in (0, pi), 2theta = Cos^(-1)(5/13).

Cos^(-1)(5/13) is proved

Q4. Prove: Cos^(-1)(2/sqrt(5)) + Tan^(-1)(1/3) = pi/4.

Let A = Cos^(-1)(2/sqrt(5)). Then sin(A) = 1/sqrt(5) and tan(A) = 1/2.

Tan^(-1)(1/2) + Tan^(-1)(1/3) = Tan^(-1)((1/2 + 1/3)/(1 - 1/6)) = Tan^(-1)(1) = pi/4.

pi/4 is proved

Q5. Prove: Cos^(-1)(3/5) - Cos^(-1)(8/17) = Cos^(-1)(104/85).
(Flag: as printed this is IMPOSSIBLE — 104/85 > 1 is outside the domain of Cos^(-1). The computed value is 84/85: Cos(A - B) = (3 x 8 + 4 x 15)/(5 x 17) = 84/85. Moreover, since A = Cos^(-1)(3/5) is about 53.13 degrees and B = Cos^(-1)(8/17) is about 61.93 degrees, A - B is negative, so strictly Cos^(-1)(3/5) - Cos^(-1)(8/17) = -Cos^(-1)(84/85); the book's equality holds only for |A - B|.)
Let A = Cos^(-1)(3/5) (sin(A) = 4/5) and B = Cos^(-1)(8/17) (sin(B) = 15/17).

Cos(A - B) = cos(A)cos(B) + sin(A)sin(B) = (24 + 60)/85 = 84/85.

Hence |A - B| = Cos^(-1)(84/85); the printed 104/85 cannot occur.

Printed RHS is IMPOSSIBLE; corrected value is Cos^(-1)(84/85), up to sign — see flag

Q6. Prove: Sin^(-1)(77/85) + Sin^(-1)(3/5) = Cos^(-1)(87/425).
(Flag: the printed RHS has the wrong sign. With A = Sin^(-1)(77/85) (about 64.9 degrees) and B = Sin^(-1)(3/5) (about 36.9 degrees), A + B is about 101.8 degrees, whose cosine is negative: Cos(A + B) = (36 x 4 - 77 x 3)/(85 x 5) = -87/425. The correct identity is Sin^(-1)(77/85) + Sin^(-1)(3/5) = Cos^(-1)(-87/425).)
Let A = Sin^(-1)(77/85) (cos(A) = 36/85) and B = Sin^(-1)(3/5) (cos(B) = 4/5).

Cos(A + B) = cos(A)cos(B) - sin(A)sin(B) = (144 - 231)/425 = -87/425.

Since A + B is in (0, pi), A + B = Cos^(-1)(-87/425).

Printed sign is WRONG; correct RHS is Cos^(-1)(-87/425) — see flag

Q7. Prove: Cos^(-1)(63/65) + 2Tan^(-1)(1/5) = Cos^(-1)(4/5).
Let A = Cos^(-1)(63/65) (sin(A) = 16/65). Let 2theta = 2Tan^(-1)(1/5): tan(2theta) = (2/5)/(1 - 1/25) = 5/12, so cos(2theta) = 12/13, sin(2theta) = 5/13.

Cos(A + 2theta) = (63 x 12 - 16 x 5)/(65 x 13) = 676/845 = 4/5.
Since A + 2theta is in (0, pi), the sum is Cos^(-1)(4/5).

Cos^(-1)(4/5) is proved

Q8. Prove: Tan^(-1)(3/4) + Tan^(-1)(3/5) - Tan^(-1)(8/19) = pi/4.
Tan^(-1)(3/4) + Tan^(-1)(3/5) = Tan^(-1)((3/4 + 3/5)/(1 - 9/20)) = Tan^(-1)(27/11).

Tan^(-1)(27/11) - Tan^(-1)(8/19) = Tan^(-1)((27/11 - 8/19)/(1 + 216/209)) = Tan^(-1)(425/425) = Tan^(-1)(1) = pi/4.

pi/4 is proved

Q9. Prove: Sec^(-1)(sqrt(5)/2) + Cot^(-1)(3) = pi/4.
Sec^(-1)(sqrt(5)/2) = Cos^(-1)(2/sqrt(5)) = Tan^(-1)(1/2), since cos = 2/sqrt(5) and sin = 1/sqrt(5).

Cot^(-1)(3) = Tan^(-1)(1/3). Sum = Tan^(-1)(1/2) + Tan^(-1)(1/3) = Tan^(-1)((5/6)/(5/6)) = pi/4.

pi/4 is proved

Q10. Prove: Cos^(-1)(3/5) + Cos^(-1)(12/13) + Cos^(-1)(63/65) = pi/2.
Let A = Cos^(-1)(3/5) (sin 4/5) and B = Cos^(-1)(12/13) (sin 5/13). Then Cos(A + B) = (36 - 20)/65 = 16/65 and Sin(A + B) = (48 + 15)/65 = 63/65.

Let C = Cos^(-1)(63/65) (sin(C) = 16/65). Then A + B = pi/2 - C, since sin(A + B) = cos(C) and cos(A + B) = sin(C). Hence A + B + C = pi/2.

pi/2 is proved

Q11. Prove: 2Tan^(-1)(1/3) + Tan^(-1)(1/7) = pi/4.
2Tan^(-1)(1/3) = Tan^(-1)((2/3)/(1 - 1/9)) = Tan^(-1)(3/4).

Tan^(-1)(3/4) + Tan^(-1)(1/7) = Tan^(-1)((3/4 + 1/7)/(1 - 3/28)) = Tan^(-1)(1) = pi/4.

pi/4 is proved

Q12. Prove: Tan^(-1)(1/3) + Tan^(-1)(1/2) = Tan^(-1)(1/11) + Tan^(-1)(5/6).
Left side = Tan^(-1)((1/3 + 1/2)/(1 - 1/6)) = Tan^(-1)(1) = pi/4.

Right side = Tan^(-1)((1/11 + 5/6)/(1 - 5/66)) = Tan^(-1)(61/61) = Tan^(-1)(1) = pi/4. Equal.

Both sides equal pi/4

Q13. Show that Sin(Cos^(-1)(y)) = sqrt(1 - y^2).
Let theta = Cos^(-1)(y), so theta is in [0, pi] and cos(theta) = y. Then sin(theta) = sqrt(1 - cos^2(theta)) = sqrt(1 - y^2), taking the non-negative root since sin is non-negative on [0, pi].

sqrt(1 - y^2) is proved

Q14. Show that Tan^(-1)(y) + Tan^(-1)(-y) = 0.
Tan^(-1) is an odd function: Tan^(-1)(-y) = -Tan^(-1)(y). Hence the sum is 0.

0 is proved

Q15. Show that Cos^(-1)(y) + Cos^(-1)(-y) = pi.
Cos^(-1)(-y) = pi - Cos^(-1)(y), since cos(pi - theta) = -cos(theta). Hence the sum is pi.

pi is proved

Q16. Show that Tan(Cos^(-1)(y)) = sqrt(1 - y^2)/y.
Let theta = Cos^(-1)(y). Then cos(theta) = y and sin(theta) = sqrt(1 - y^2). So tan(theta) = sin(theta)/cos(theta) = sqrt(1 -

y^2)/y (for y not equal to 0).

sqrt(1 - y^2)/y is proved

Q17. Show that Sin(2Cos^(-1)(y)) = 2y sqrt(1 - y^2).

Let theta = Cos^(-1)(y). Then Sin(2theta) = 2sin(theta)cos(theta) = 2 sqrt(1 - y^2) x y.

2y sqrt(1 - y^2) is proved

Q18. Show that Cos(2Sin^(-1)(y)) = 1 - 2y^2.

Let theta = Sin^(-1)(y). Then Cos(2theta) = 1 - 2sin^2(theta) = 1 - 2y^2.

1 - 2y^2 is proved

Review Exercise 11

Q1(i). Domain of the following y = Sin^(-1)(x) is;

- ✓ [-1, 1]
- (-inf, inf)
- [0, pi]
- [-pi/2, pi/2]

Q1(ii). Range of the function y = Cos^(-1)(x) is

- [pi/2, -pi/2]
- [pi, 0]
- ✓ [0, pi]
- [-pi, pi]

Q1(iii). Domain of y = Tan^(-1)(x) is

[-pi/2, pi/2]

(0, pi)

[-1, 1]

✓ [-inf, inf]

Q1(iv). Which point lies on the graph of y = Sin^(-1)(x)? (flag: TWO options are correct — (b) (0.5, pi/6) since Sin^(-1)(0.5) = pi/6, and (c) (1, pi/2). The tick marks (c).)

(1, pi)

(0.5, pi/6)

✓ (1, pi/2)

(2, pi/2)

Q1(v). Which function has a graph that passes through (0, pi/2)

✓ y = Cos^(-1)(x)

y = Tan^(-1)(x)

y = Sec^(-1)(x)

y = Sin^(-1)(x)

Q1(vi). Sin^(-1)(x) + Cos^(-1)(x) is equal to;

✓ pi/2

pi

0

undefined

Q1(vii). Sin^(-1)(1/2) + Cos^(-1)(1/2) is equal to;

0

pi

✓ pi/2

pi/4

Q1(viii). Tan^(-1)(1) + Tan^(-1)(sqrt(3)) is equal to;

pi/6

pi/3

pi/2

✓ 7pi/12

Q2. Find the value of each of the following. (i) Sin[Cos^(-1)(sqrt(3)/2)]; (ii) Sec[Cosec^(-1)(2)]; (iii) Cosec[Tan^(-1)(1)]; (iv) Sec[Sin^(-1)(1/sqrt(2))].

(i) Cos^(-1)(sqrt(3)/2) = pi/6, so Sin(pi/6) = 1/2.

(ii) Cosec^(-1)(2) = pi/6, so Sec(pi/6) = 2/sqrt(3).

(iii) Tan^(-1)(1) = pi/4, so Cosec(pi/4) = sqrt(2).

(iv) Sin^(-1)(1/sqrt(2)) = pi/4, so Sec(pi/4) = sqrt(2).

(i) 1/2; (ii) 2/sqrt(3); (iii) sqrt(2); (iv) sqrt(2)

Q3. Show that; Tan^(-1)(12/5) = Sin^(-1)(12/13).

Let theta = Tan^(-1)(12/5), with theta in (0, pi/2). Then tan(theta) = 12/5, so sin(theta) = 12/13 (from the 5-12-13 triangle). Hence theta = Sin^(-1)(12/13).

Sin^(-1)(12/13) is proved

Q4. Prove that; Sin^(-1)(1/sqrt(5)) + Cot^(-1)(3) = pi/4.

Sin^(-1)(1/sqrt(5)) = Tan^(-1)(1/2), since sin = 1/sqrt(5) and cos = 2/sqrt(5). Cot^(-1)(3) = Tan^(-1)(1/3).

Sum = Tan^(-1)(1/2) + Tan^(-1)(1/3) = Tan^(-1)((5/6)/(5/6)) = pi/4.

pi/4 is proved

Q5. Prove: Sin^(-1)(4/5) + Sin^(-1)(5/13) + Sin^(-1)(16/65) = pi/2.

Let S = Sin^(-1)(4/5) + Sin^(-1)(5/13). Then sin(S) = (4 x 12 + 3 x 5)/65 = 63/65 and cos(S) = (3 x 12 - 4 x 5)/65 = 16/65.

Let T = Sin^(-1)(16/65), so sin(T) = 16/65, cos(T) = 63/65. Then sin(S + T) = (63 x 63 + 16 x 16)/4225 = 1. Since S + T is in (0, pi), S + T = pi/2.

pi/2 is proved

Unit 12: Solution of Trigonometric Equations

Exercise 12.1

Q1-Q9. Find solutions of the trigonometric equations. 1. sin(theta) = 1/2; 2. cos(theta) = -1/sqrt(2); 3. tan(theta) = sqrt(3); 4. sin(theta) = -sqrt(3)/2; 5. cos(theta) = 1; 6. tan(theta) = -1; 7. tan(theta) = -1/sqrt(3); 8. cos(theta) = 0; 9. tan(theta) = 1.

1. sin(theta) = 1/2: reference angle pi/6, positive in quadrants I and II: theta = pi/6 + 2n pi, 5pi/6 + 2n pi.

2. cos(theta) = -1/sqrt(2): reference angle pi/4, negative in quadrants II and III: theta = 3pi/4 + 2n pi, 5pi/4 + 2n pi.

3. tan(theta) = sqrt(3): period pi: theta = pi/3 + n pi.

4. sin(theta) = -sqrt(3)/2: reference angle pi/3, negative in quadrants III and IV: theta = 4pi/3 + 2n pi, 5pi/3 + 2n pi.

5. cos(theta) = 1: theta = 2n pi.

6. tan(theta) = -1: theta = -pi/4 + n pi (equivalently 3pi/4 + n pi).

7. tan(theta) = -1/sqrt(3): theta = -pi/6 + n pi.

8. cos(theta) = 0: theta = pi/2 + n pi.

9. tan(theta) = 1: theta = pi/4 + n pi.

1. theta = pi/6 + 2n pi, 5pi/6 + 2n pi; 2. theta = 3pi/4 + 2n pi, 5pi/4 + 2n pi; 3. theta = pi/3 + n pi; 4. theta = 4pi/3 + 2n pi, 5pi/3 + 2n pi; 5. theta = 2n pi; 6. theta = -pi/4 + n pi; 7. theta = -pi/6 + n pi; 8. theta = pi/2 + n pi; 9. theta = pi/4 + n pi

Q10-Q13. By using graph, find the solution of the following equations for -pi/2 <= theta <= pi/2. 10. 2sin(theta) - theta = 0; 11. cos(theta) = 1 + theta; 12. sin(theta) = theta; 13. cos(theta) = theta/2. (Flag: Q10-Q12 have only the trivial solution 0 in this interval; Q13 has one solution, approximately 1.03, found graphically/numerically.)

10. 2sin(theta) = theta, i.e. sin(theta) = theta/2. The graphs of y = sin(theta) and y = theta/2 meet at theta = 0 inside [-pi/2, pi/2]; their other crossings, at about +/-1.895, lie outside this interval. So theta = 0 only.

11. cos(theta) = 1 + theta. At theta = 0 both sides are 1. The function cos(theta) - 1 - theta has derivative -sin(theta) - 1 <= 0 on the interval, so it is decreasing and has no other zero. So theta = 0 only.

12. sin(theta) = theta. Since |sin(theta)| < |theta| for theta not equal to 0, the only solution is theta = 0.

13. cos(theta) = theta/2. On the negative side cos(theta) > 0 > theta/2, so no solution there. On (0, pi/2], cos falls from 1 to 0 while

theta/2 rises from 0 to pi/4, so they cross once, at theta about 1.03 (check: cos(1.03) is about 0.515 and 1.03/2 = 0.515).

10. theta = 0 only; 11. theta = 0 only; 12. theta = 0 only; 13. theta about 1.03

Exercise 12.2

Q1-Q6. Solve each equation giving general solution. 1. cosec(2theta) = 1/2; 2. cos(theta) = -1/2; 3. cosec(3theta/2) = -2; 4. sec^2(theta) = 4/3; 5. 4sin^2(theta) - 1 = 0; 6. sec^2(theta) - 4 = 0.

1. cosec(2theta) = 1/2 means sin(2theta) = 2, impossible since |sin| <= 1. NO solution.

2. cos(theta) = -1/2: reference angle pi/3, negative in quadrants II and III: theta = 2pi/3 + 2n pi, 4pi/3 + 2n pi.

3. sin(3theta/2) = -1/2: 3theta/2 = 7pi/6 + 2n pi or 11pi/6 + 2n pi, so theta = 7pi/9 + 4n pi/3, 11pi/9 + 4n pi/3.

4. sec^2(theta) = 4/3 gives cos(theta) = +/-sqrt(3)/2, so theta = +/-pi/6 + n pi.

5. sin^2(theta) = 1/4 gives sin(theta) = +/-1/2, so theta = +/-pi/6 + n pi.

6. sec^2(theta) = 4 gives cos(theta) = +/-1/2, so theta = +/-pi/3 + n pi.

1. NO solution; 2. theta = 2pi/3 + 2n pi, 4pi/3 + 2n pi; 3. theta = 7pi/9 + 4n pi/3, 11pi/9 + 4n pi/3; 4. theta = +/-pi/6 + n pi; 5. theta = +/-pi/6 + n pi; 6. theta = +/-pi/3 + n pi

Q7-Q12. Find the values of theta satisfying the following equations. 7. 3tan^2(theta) + 2sqrt(3)tan(theta) + 1 = 0; 8. cot^2(theta) - cosec(theta) - 1 = 0; 9. 2cos(theta) + sin^2(theta) - 1 = 0; 10. 4cos^2(theta) - 8sin(theta) + 1 = 0; 11. (sec(theta) - 2)(2sin(theta) - 1) = 0; 12. (cosec(theta) + 2)(2cos(theta) - 1) = 0. (Values given in [0, 2pi); general forms in the steps.)

7. (sqrt(3)tan(theta) + 1)^2 = 0, so tan(theta) = -1/sqrt(3): theta = 5pi/6, 11pi/6; generally -pi/6 + n pi.

8. Using cot^2 = cosec^2 - 1: cosec^2(theta) - cosec(theta) - 2 = 0, so cosec(theta) = 2 or -1: theta = pi/6, 5pi/6, 3pi/2.

9. Using sin^2 = 1 - cos^2: cos(theta)(2 - cos(theta)) = 0, so cos(theta) = 0: theta = pi/2, 3pi/2.

10. Using cos^2 = 1 - sin^2: 4sin^2(theta) + 8sin(theta) - 5 = 0, so sin(theta) = 1/2 (the other root -5/2 is impossible): theta = pi/6, 5pi/6.

11. sec(theta) = 2 gives theta = pi/3, 5pi/3; sin(theta) = 1/2 gives theta = pi/6, 5pi/6.

12. cosec(theta) = -2 gives theta = 7pi/6, 11pi/6; cos(theta) = 1/2

gives theta = pi/3, 5pi/3.

7. theta = 5pi/6, 11pi/6 in [0, 2pi); 8. theta = pi/6, 5pi/6, 3pi/2; 9. theta = pi/2, 3pi/2; 10. theta = pi/6, 5pi/6; 11. theta = pi/6, pi/3, 5pi/6, 5pi/3; 12. theta = pi/3, 5pi/3, 7pi/6, 11pi/6

Q13-Q22. Find the solution sets of the following equation. 13. tan(x) = 2sin(x); 14. 4sin^2(theta/2) - 3 = 0; 15. cos(2theta) = cos(theta); 16. sin(4x) - sin(2x) = cos(3x); 17. sin(x) + cos(3x) = cos(5x); 18. cos(x) + cos(3x) + cos(5x) = 0; 19. sin(theta) + sin(3theta) + sin(5theta) + sin(7theta) = 0; 20. cos(theta) + cos(3theta) + cos(5theta) + cos(7theta) = 0; 21. cosec(x) = sqrt(3) + cot(x); 22. 2cos(theta) + sin^2(theta) - 1 = 0. (Flag: Q21 has an extraneous root — see steps. Q22 duplicates Q9 exactly.)
13. sin(x)(1 - 2cos(x)) = 0 with cos(x) not 0: x = n pi or x = +/-pi/3 + 2n pi.

14. sin(theta/2) = +/-sqrt(3)/2, so theta = +/-2pi/3 + 2n pi.

15. 2theta = +/-theta + 2n pi, giving theta = 2n pi/3 (which contains both families).

16. 2cos(3x)sin(x) = cos(3x), so cos(3x)(2sin(x) - 1) = 0: x = pi/6 + n pi/3 or x = pi/6 + 2n pi, 5pi/6 + 2n pi.

17. sin(x) = cos(5x) - cos(3x) = -2sin(4x)sin(x), so sin(x)(1 + 2sin(4x)) = 0: x = n pi or x = 7pi/24 + n pi/2, 11pi/24 + n pi/2.

18. cos(3x)(2cos(2x) + 1) = 0: x = pi/6 + n pi/3 or x = pi/3 + n pi, 2pi/3 + n pi.

19. 4sin(4theta)cos(2theta)cos(theta) = 0. The cos factors give families contained in theta = n pi/4, so theta = n pi/4.

20. 4cos(4theta)cos(2theta)cos(theta) = 0: theta = pi/8 + n pi/4, pi/4 + n pi/2, pi/2 + n pi.

21. Multiplying by sin(x): 1 = sqrt(3)sin(x) + cos(x), i.e. sin(x + pi/6) = 1/2, giving x = 2n pi or x = 2pi/3 + 2n pi. But x = 2n pi makes sin(x) = 0, where the original equation is undefined, so it is extraneous: x = 2pi/3 + 2n pi only.

22. As Q9: cos(theta)(2 - cos(theta)) = 0, so theta = pi/2 + n pi.

13. x = n pi, +/-pi/3 + 2n pi; 14. theta = +/-2pi/3 + 2n pi; 15. theta = 2n pi/3; 16. x = pi/6 + n pi/3, 5pi/6 + 2n pi; 17. x = n pi, 7pi/24 + n pi/2, 11pi/24 + n pi/2; 18. x = pi/6 + n pi/3, pi/3 + n pi, 2pi/3 + n pi; 19. theta = n pi/4; 20. theta = pi/8 + n pi/4, pi/4 + n pi/2, pi/2 + n pi; 21. x = 2pi/3 + 2n pi only (see flag); 22. theta = pi/2 + n pi (duplicates Q9)

Exercise 12.3

Q1-Q8. Applied problems. 1. A hydraulic lift raises a platform 2m off the ground with the arm extending 5m. Find the angle between the arm and the ground. 2. In a mechanical linkage, a bar 10cm long connects to a vertical support, forming a triangle with a horizontal

base of 8cm. Find the angle of inclination. 3. A wheel chair ramp is 9m long and rises to a height of 1.5m. What is the angle of elevation. 4. From a point 30m from the base, an engineer measures an angle of elevation of 60 degrees. Find the height. 5. A boat is 100m from a light house. The angle of elevation to the top is 33 degrees. Find the height. 6. From a height of 200m, the angle of depression to a village is 12 degrees. Find the horizontal distance. 7. A drone at 120m observes a target on the ground at a 35 degrees angle. Find the distance to the target. 8. A model bridge ramp is 12cm long and rises 5cm. Find the angle it makes. (Flag for Q7: 'distance to the target' is ambiguous — both the slant distance and the horizontal distance are given.)

1. sin(angle) = 2/5 = 0.4, so the angle is Sin^(-1)(0.4), about 23.6 degrees.

2. cos(angle) = 8/10 = 0.8, so the angle is Cos^(-1)(0.8), about 36.9 degrees.

3. sin(angle) = 1.5/9 = 1/6, so the angle is about 9.6 degrees.

4. Height = 30 x tan(60) = 30sqrt(3), about 51.96 m.

5. Height = 100 x tan(33), about 64.94 m.

6. Horizontal distance = 200/tan(12), about 940.9 m.

7. Slant distance = 120/sin(35), about 209.2 m; horizontal distance = 120/tan(35), about 171.4 m.

8. sin(angle) = 5/12, so the angle is about 24.6 degrees.

1. about 23.6 degrees; 2. about 36.9 degrees; 3. about 9.6 degrees; 4. about 51.96 m; 5. about 64.94 m; 6. about 940.9 m; 7. slant about 209.2 m, horizontal about 171.4 m (see flag); 8. about 24.6 degrees

Review Exercise 12

Q1(i). If alpha = tan^(-1)(1) and beta = tan^(-1)(1/sqrt(3)) where alpha and beta lie in first quadrant then alpha + beta is;

- pi/6
- pi/4
- pi/3
- ✓ 5pi/12

Q1(ii). If cos(theta) = -sqrt(3)/2 and 0 < theta < pi then theta is;

- pi/6
- pi/3
- 2pi/3
- ✓ 5pi/6

Q1(iii). The general solution of cos^2(theta) = 3/4 is; (flag: NO option is fully correct — cos(theta) = +/-sqrt(3)/2 gives four families pi/6, 5pi/6, 7pi/6, 11pi/6 (+ 2n pi). Option (a) is the intended one: it gives the solutions in [0, pi] but omits 7pi/6 and 11pi/6.)

- ✓ pi/6 + 2n pi, 5pi/6 + 2n pi, n in Z
- pi/3 + 2n pi, 5pi/3 + 2n pi, n in Z

pi/4 + 2n pi, 3pi/4 + 2n pi, n in Z

pi/2 + 2n pi, pi + 2n pi, n in Z

Q1(iv). The general solution of sin^2(theta) - 3sin(theta) + 2 = 0 is;

pi + 2n pi, n in Z

✓ pi/2 + 2n pi, n in Z

pi/6 + 2n pi, n in Z

pi/3 + 2n pi, n in Z

Q1(v). If cos(theta) = -1/2, then theta is;

30 degrees

60 degrees

90 degrees

✓ 120 degrees

Q2. Using graph, find the solution of the following equations for -pi/2 <= theta <= pi/2. (i) 2cos(theta) - theta = 0; (ii) sin(theta) = 1 + theta. (Flag: (ii) has NO solution in this interval — see steps.)

(i) 2cos(theta) = theta is the same equation as Ex 12.1 Q13: cos(theta) = theta/2, crossing once at theta about 1.03.

(ii) sin(theta) - 1 - theta has derivative cos(theta) - 1 <= 0, so it is decreasing, and it is already negative at theta = -pi/2 (about -0.43) and at theta = 0 (-1). Hence it never reaches 0: NO solution in this interval.

(i) theta about 1.03; (ii) NO solution

Q3. Find the solution of the following trigonometric equations.

tan^2(theta) - sec(theta) - 1 = 0.

Using tan^2 = sec^2 - 1: sec^2(theta) - sec(theta) - 2 = 0, so (sec(theta) - 2)(sec(theta) + 1) = 0.

sec(theta) = 2 gives theta = +/-pi/3 + 2n pi; sec(theta) = -1 gives theta = pi + 2n pi.

theta = +/-pi/3 + 2n pi, pi + 2n pi

Q4. 4sin^2(theta) - 8cos(theta) + 1 = 0.

Using sin^2 = 1 - cos^2: 4cos^2(theta) + 8cos(theta) - 5 = 0, so cos(theta) = 1/2 (the other root -5/2 is impossible). Hence theta = +/-pi/3 + 2n pi.

theta = +/-pi/3 + 2n pi

Q5. sec(3theta) = sec(theta). (Flag: the raw families theta = n pi and theta = n pi/2 must be cut by the domain — sec is undefined where cos = 0, which removes the odd multiples of pi/2, leaving theta = n pi.)

sec(3theta) = sec(theta) gives 3theta = +/-theta + 2n pi, i.e. theta = n pi or theta = n pi/2.

But sec(theta) is undefined at theta = pi/2 + n pi, so the odd multiples of pi/2 are excluded, leaving theta = n pi.

theta = n pi (see flag)

Q6. A display ramp is 8cm long with a 2.5cm rise. What is the angle of elevation?

sin(angle) = 2.5/8 = 0.3125, so the angle is about 18.2 degrees.

about 18.2 degrees